Internalizing Environmental Risk: Insurance Design and Firm Behavior in Hazardous Industries

Kaleb K. Javier*

March 16, 2026

Abstract

Public trust funds in thirty-eight U.S. states insure underground storage tank (UST) leaks at a uniform flat fee, even though leak risk varies widely with observable traits such as tank age and wall construction. This paper quantifies how replacing flat-fee pooling with actuarially priced coverage changes firm behavior and environmental outcomes. I exploit Texas's 1999 closure of its trust fund, which forced all owners into a private market that sets premiums according to facility risk. A facility-year panel covering 1990–2021 links EPA inventories with administrative microdata for Texas and eighteen control states that retained public funds. I construct a make-model comparison sample that matches Texas facilities to control-state counterparts on the observable risk characteristics, tank age bin and wall construction type, that private insurers price, ensuring the comparison group is balanced on the dimensions along which the treatment effect is theoretically heterogeneous. Difference-in-differences and event-study estimates show that risk-rated premiums increased annual closure rates by two to three percentage points, shifted closures from market exit to on-site replacement with safer equipment, and reduced the probability of a reported leak by about one percentage point, nearly the entire pre-policy baseline for the high-risk single-wall cohort. The treatment effect is concentrated among older and single-walled facilities, consistent with the premium-gradient mechanism, and is near zero for young double-walled tanks. The findings provide the first causal micro-evidence that cost-based environmental liability insurance can internalize pollution risk more effectively than uniform public pooling, offering clear guidance for states that are reconsidering the structure of UST insurance programs.

*University of California, Berkeley, Department of Agricultural and Resource Economics. Email: `kalebkja@berkeley.edu`. Website: `kalebkjavier.com`. I thank [Insert acknowledgments and funding notes here].

Introduction

Underground Storage Tanks (UST) holding motor fuels and other hazardous liquids sit beneath about half a million U.S. facilities, and leaks from these tanks are the leading documented source of groundwater contamination EPA (2024). Cleanup is costly; median remediation expenses exceed $100 000 per site Marcus (2024), and the probability of a release rises sharply with observable characteristics such as tank age and whether the walls are single or double. Despite these predictable differences, thirty-eight states finance remediation through public trust funds that charge every facility the same premium, diluting incentives to invest in safer technology and crowding out private insurers ASTSWMO (1999).

Economic theory predicts that tying premiums to expected damages should induce firms to internalize environmental risk (Shavell, 2025b; Polinsky, 2025), yet the relevant margin here is not liability per se but the rating regime under which liability is financed. Uniform-premium public funds operate as community-rated pools: every covered facility pays the same fee regardless of its observable risk characteristics, replicating the cross-subsidization and incentive dilution that Pauly (1974) identified in flat-rate insurance and that Rothschild and Stiglitz (1976) showed prevents competitive separating equilibria. Under community rating, two distortions operate simultaneously. Adverse selection pushes high-risk facilities toward the subsidized pool and crowds private risk-rated alternatives out of the market. Moral hazard suppresses prevention investment because the marginal high-risk facility bears no incremental premium from operating leak-prone equipment. Empirical tests of this mechanism for USTs are scarce. Michigan's mid-1990s shift toward private, risk-rated coverage suggested large gains but relied on aggregate counts that could not isolate facility-level responses (Yin et al., 2024b, 2025). Whether actuarial pricing disciplines high-risk owners without simply driving them out of the market therefore remains unresolved.

Texas provides a clean natural experiment. In 1999 the state closed its public fund overnight and required all owners to purchase private coverage whose premiums vary with observable risk. I assemble a new facility-year panel covering 1990 to 2021 that links EPA inventories with administrative microdata from the Texas Commission on Environmental Quality and comparable data from eighteen control states that retained trust funds. The identification strategy rests on a make-model comparison sample that matches Texas facilities to control-state counterparts on the observable risk characteristics, tank age bin and wall construction type, that the private market prices. Because the treatment effect is theoretically heterogeneous along these pricing dimensions (Section 3, Proposition 3), comparability on them is required for valid causal inference; Section 4 shows that matching on make and model is what earns parallel pre-trends.

[REWIRTE RESULTS]

Related Literature

A well-established body of theoretical work suggests that strict liability rules can be an effective policy instrument for environmental risks, particularly when paired with insurance contracts that align a firm's private incentives with social costs (Shavell, 2025b,c,a; Polinsky, 2025; Boyd and Kunreuther, 2025; Shavell, 2024). A crucial condition for this alignment is

the insurer's ability to accurately price the risk of environmental harm. This paper provides the first empirical test of this principle in the context of environmental liability, offering causal estimates of how firms respond to a transition from flat-fee to more efficient, cost-based insurance contracts.

The challenge of pricing environmental risk connects this study to the economics of insurance rating regimes. The pooling equilibrium maintained by uniform-premium state funds is the environmental analog of community rating: premiums are invariant to observable risk, which creates cross-subsidies that suppress precautionary investment and crowd out risk-differentiated private alternatives (Rothschild and Stiglitz, 1976; Pauly, 1974; Cutler and Zeckhauser, 2000). Theory predicts that a shift to experience rating generates two behavioral responses. On the selection margin, high-risk facilities face a cost increase and may exit or upgrade. On the moral hazard margin, facilities that remain now face a direct financial return to risk reduction, inducing investment in safer technology. Identifying these margins separately requires within-facility panel variation that aggregate data cannot provide. The empirical literature on rating regimes confirms both mechanisms, though rarely in the same market. In health insurance, risk adjustment reforms that moved Medicare Advantage toward experience rating reduced favorable selection but generated gaming responses among plans (Brown et al., 2025; McWilliams et al., 2012). In credit markets, moving from flat-rate to risk-differentiated pricing improves screening and changes borrower composition (Einav et al., 2025b, 2024; Nelson, 2025), with efficiency gains depending on whether information asymmetries favor the insurer or the insured (Finkelstein and McGarry, 2006; Liberman et al., 2025; Einav et al., 2025a). In the environmental insurance context, Yin et al. (2025) find that switching from community-rated state funds to private coverage in Michigan reduced leak incidents by over twenty percent. The Texas reform studied here provides a sharper identification environment and facility-level data that allow the selection and moral hazard margins to be separated.

This paper also contributes to the literature on U.S. UST policy. While much of the existing work focuses on estimating the marginal damages of pollution from leaks (Marcus, 2024; Zabel and Guignet, 2024; Guignet, 2025a; Guignet and Martinez-Cruz, 2018; Guignet et al., 2025; Walsh and Mui, 2025; Guignet, 2025b), this study is complementary, examining the design of management policies intended to prevent those damages.

This research is most closely related to studies of a similar insurance market reform in Michigan by Yin et al. (2024b), Yin et al. (2025), and Yin et al. (2024a). That body of work finds that transitioning to risk-based insurance reduced the number of small firms and reported leaks. However, those analyses relied on aggregate or cross-sectional data, which cannot control for unobserved facility-level heterogeneity and may be susceptible to omitted variable bias. This paper advances on prior work by adopting an empirical strategy similar to other modern studies of environmental liability (Boomhower, 2024). By using a rich, facility-level panel dataset and a comparison sample matched on observable risk characteristics, I implement difference-in-differences models that control for unobserved facility characteristics and common time shocks, providing more credible causal estimates of how firms adjust their technology, replacement, and exit decisions in response to cost-based pricing.

Institutional Background

Underground Storage Tanks: Physical Characteristics, Regulatory Lock-in, and Environmental Risk

The U.S. Environmental Protection Agency defines a UST system as any tank and connected underground piping with at least 10% of its volume below the surface (EPA, 2024). The construction technology of a UST system is the primary determinant of its environmental risk profile and the observable margin on which actuarial pricing differentiates. The critical physical distinction within this capital stock is between single-walled systems (bare steel or fiberglass shells with no interstitial containment layer) and double-walled systems (which incorporate a secondary wall and continuous interstitial monitoring space).

The construction choice that creates this risk profile is economically consequential and effectively irreversible. Single-walled construction is cheaper to install: a 1987 GAO survey priced a standard 10,000-gallon single-walled unit at approximately $23,000 versus $39,000 for a comparable double-walled unit, a 67% premium that contemporaneous manufacturer filings confirm has persisted (GAO, 1992; Containment Solutions, 2021; Xerxes, 2018). The savings are locked in at installation. Once a single-walled tank is in the ground, the owner cannot upgrade the wall construction without removing and replacing the tank entirely, a capital expenditure comparable to installing new equipment from scratch. The practical irreversibility of this choice is confirmed directly in the administrative data: in Texas from 1998 to 2021, only 0.003% of active facilities ever retrofitted tank construction prior to terminal closure. Wall type is therefore not a choice variable that responds to price signals at the intensive margin. It is a fixed, observable, and verifiable physical attribute, precisely the conditions required for an actuarially feasible pricing margin and the reason the private insurance market in Texas conditions premiums directly on it.

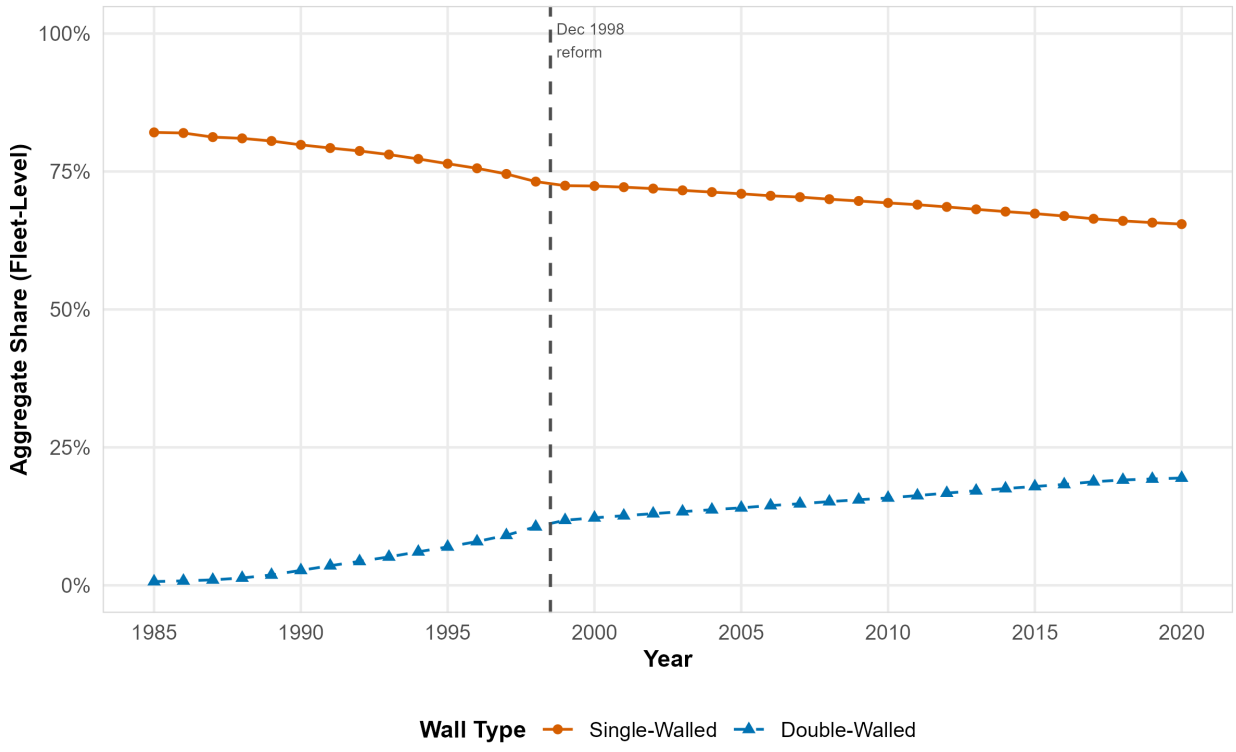
The regulatory framework reinforces this physical lock-in with a legal one. Under 40 CFR Part 280, UST systems are subject to vintage-differentiated technical standards: tanks installed under the 1988 regulations are grandfathered into the compliance classification they earned at the federal deadline, provided they remain unmodified. Any construction activity that alters the wall structure or piping configuration of an existing system, including the partial upgrades that would otherwise be the natural intermediate response to a rising insurance premium, triggers reclassification of the entire facility under the most current technical standards in effect at the time of modification. The 2005 and subsequent EPA technical revisions substantially raised those standards (US EPA, 2024). An owner who initiates a partial upgrade to reduce premium costs therefore faces the prospect of a facility-wide compliance overhaul whose cost dwarfs the incremental premium savings.

This is a direct instance of the mechanism Gruenspecht (2024) identifies in the context of auto emissions and that Stavins (2024) formalizes as a general property of vintage-differentiated regulation: exempting existing capital from new standards inadvertently extends the operating life of the oldest and most hazardous stock by making any modification prohibitively costly. The modification trigger in 40 CFR Part 280 operates exactly as the analogous New Source Review provision does in the Clean Air Act, where Bushnell and Wolfram (2024)

show that the threat of reclassification deterred capital investment at grandfathered coal plants; Heutel (2024) provides structural estimates of the same retardation effect for boiler vintages. The practical effect here is to make the choice set binary: operate the existing system without modification, or close it entirely. As Sallee (2023) and Jacobsen et al. (2024) argue in the context of vehicle fleet decarbonization, when new-capital regulation cannot reach the existing stock, retirement-focused instruments are the appropriate complement. The UST setting illustrates precisely why: incremental investment in risk reduction is not an option the regulatory environment makes available, which is why the extensive margin of closure is the sole behavioral margin the empirical analysis can and should target.

The environmental challenge posed by underground storage tanks is fundamentally a problem of managing an existing stock, not regulating new flows into it. Figure 1 documents the composition of the active fleet from 1985 through 2020. Single-walled tanks accounted for approximately 83% of the active capital stock in the mid-1980s and remained the dominant technology at the federal compliance deadline in December 1998, when they still constituted roughly 73% of active tanks. By 2020, after two decades of post-deadline attrition, their share had fallen only to approximately 63%. Double-walled construction, by contrast, grew from near zero at the start of the study window to just under 20% of the fleet by 2020, reflecting that new installations post-1988 are overwhelmingly double-walled while the legacy stock turns over slowly. The figure makes plain that the trajectory of environmental risk in this market is determined almost entirely by the pace at which the incumbent single-walled stock exits, not by the character of new entrants. The remediation liability sitting beneath the surface today was installed decades ago and cannot be neutralized by tightening standards on new entrants. It can only be reduced by accelerating the exit of the single-walled incumbents. Because the hazardous technology is already in the ground, the relevant policy question is not how to screen entrants but how to price the continued operation of the existing fleet so that owners internalize the expected cost of a release.

Figure 1: Aggregate Composition of Active Underground Storage Tank Fleet by Wall Type, 1989–2020.



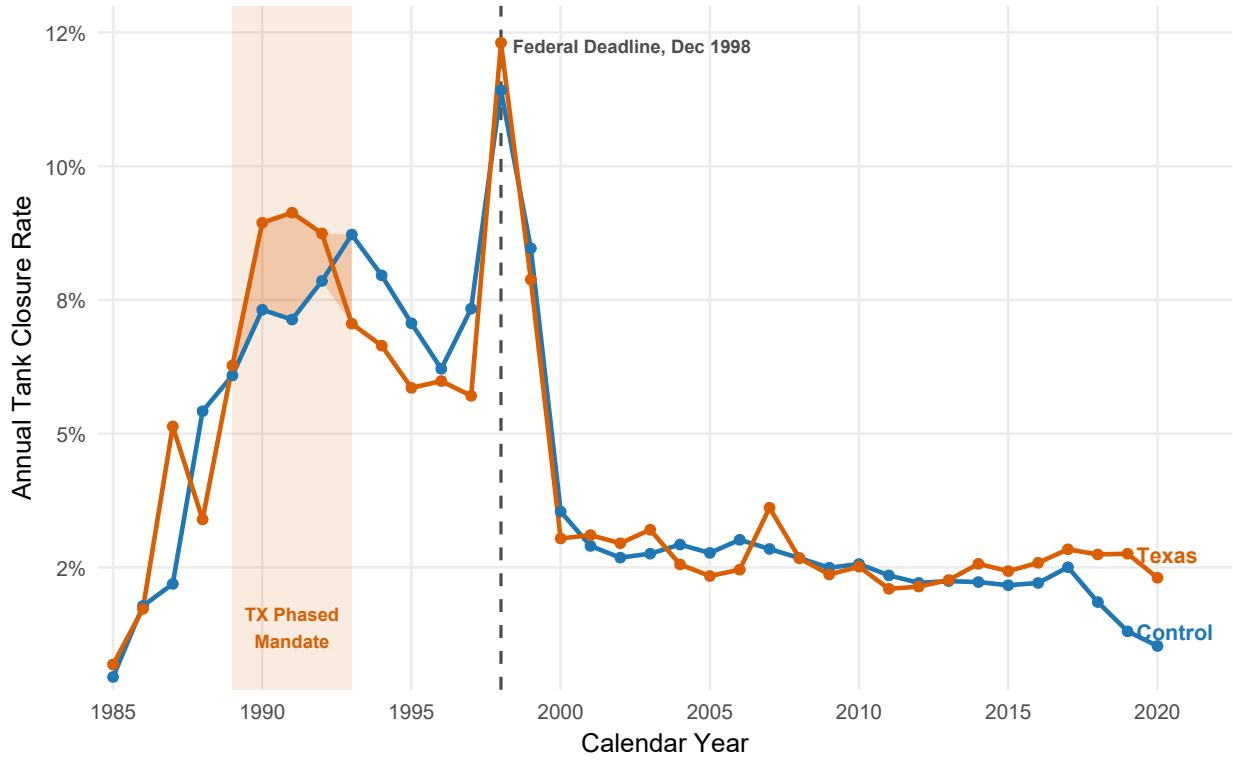
Notes: Proportional share of the total active capital stock categorized as single-walled (solid line, circles) and double-walled (dashed line, triangles) across the pooled full sample. The vertical dashed line marks the federal EPA upgrade deadline of December 22, 1998. The gradual post-deadline decline of the legacy single-walled fleet, mirrored by the corresponding accumulation of double-walled infrastructure, reflects the slow attrition of existing capital rather than immediate regulatory replacement. This confirms that stock management, rather than new-flow regulation, governs the long-term trajectory of environmental risk. The combined shares may evaluate to slightly less than 100

The Federal Compliance Mandate and Pre-Treatment Dynamics

Federal regulation of USTs originates in the 1984 Hazardous and Solid Waste Amendments to the Resource Conservation and Recovery Act, which directed the EPA to establish comprehensive technical and financial standards for tank systems. The EPA finalized these standards in 40 CFR Part 280 in 1988, requiring all non-compliant tanks to be upgraded, replaced, or closed by a uniform federal compliance deadline of December 22, 1998 (Agency, 2017; US EPA, 2024). Figure 2 makes the consequences visible: annual facility-level closure rates spike sharply in the years immediately preceding the deadline, peak in 1997 and 1998, and fall equally sharply once the deadline passes. This spike is a regulatory compliance event. It contaminates the pre-1988 capital stock in Texas and the control states simultaneously and cannot be interpreted as a behavioral response to insurance pricing. Texas compounded this federal problem with a state-level one, imposing a staggered upgrade schedule for pre-1988

tanks that ran from 1989 through 1993 and had no analogue in any control state. The two problems together render the pre-1988 installation cohorts unsuitable for the identification design. Section~ documents the Texas staggered mandate in detail and shows the differential pre-treatment dynamics empirically.

Figure 2: Full Sample Annual Closure Rates and the Federal Mandate Effect.



Notes: Full sample closure rates for Texas and control-state facilities across the study window, illustrating the spike in closures driven by the federal EPA upgrade deadline of December 22, 1998, and the subsequent decline in annual exit rates as the surviving fleet aged.

Financial Responsibility and the State Trust Fund Regime

The 1988 Financial Responsibility Rule (40 CFR Part 280, Subpart H) mandates that every UST owner demonstrate the financial capacity to fund corrective action and satisfy third-party liability claims, with minimum required coverage of $1 million per occurrence (40 CFR §280, Subpart H). In the late 1980s, the nascent private environmental insurance market was unable to supply this coverage at affordable cost to the population of small, single-site operators that constitute the majority of UST owners. To prevent widespread non-compliance, many states established public trust funds financed by annual tank registration fees or fuel tax revenues, functioning as a publicly administered insurer of last resort. A defining feature of these funds is their premium structure: annual fees are set on a uniform, per-tank basis, invariant to any observable facility risk characteristic such as wall type, tank age, or leak

detection technology. States retaining public funds consistently report that these funds cover 90% to 100% of all active facilities, as documented by the Association of State and Territorial Solid Waste Management Officials (ASTSWMO, 1999). This near-complete market saturation is the predicted equilibrium outcome: because the subsidized uniform-premium contract is priced below any risk-differentiated private alternative for high-risk facilities, private insurers are structurally unable to compete for the portion of the market with above-average risk. The resulting uniform-premium pooling equilibrium is the institutional baseline against which Texas's transition is evaluated, and it is precisely the community rating equilibrium that the insurance economics literature predicts will simultaneously generate adverse selection into the pool and suppress prevention investment at the facility level (Rothschild and Stiglitz, 1976; Pauly, 1974; Cutler and Zeckhauser, 2000).

Texas's Policy Transition

Texas's departure from the public fund model constitutes a sharp, administratively determined policy discontinuity whose timing was driven by fiscal insolvency rather than any observable change in the statewide distribution of facility-level risk. Texas established the Petroleum Storage Tank Remediation Fund (PSTRF) in 1989 under the Texas Water Code. By 1995, actuarial assessments confirmed that the fund's liabilities, driven by the accumulating backlog of legacy remediation claims, were structurally unsustainable. In response, the 74th Texas Legislature enacted House Bill 2587, which mandated the sunset of the PSTRF (Texas HRO, 1995). The legislature set the fund closure cutoff at December 22, 1998, the same calendar date as the federal EPA upgrade deadline codified in 40 CFR §280.21. The alignment of these two dates is not coincidental: the legislature explicitly selected the federal mandate date as the administrative anchor for the PSTRF cutoff, ensuring the fund would cease accepting new claims at precisely the moment when the largest cohort of non-compliant tanks was required to be addressed (74th Texas Legislature, 1995). Effective December 22, 1998, all Texas UST owners were required to secure private insurance or qualify for self-insurance by demonstrating sufficient financial assets.

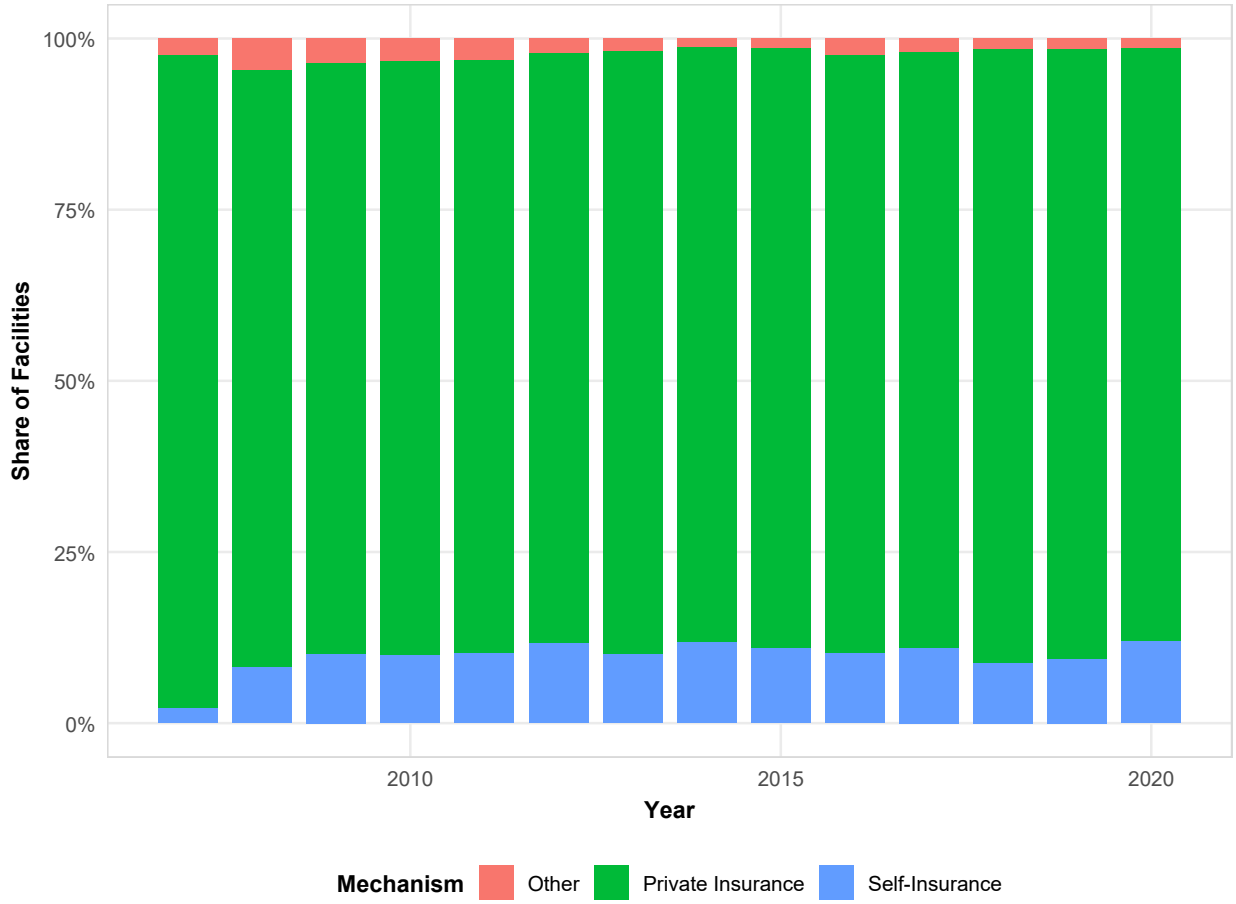
The Post-Transition Private Market in Texas

The private market that absorbed the Texas UST population after December 1998 prices observable facility risk directly, generating the cross-sectional and intertemporal premium variation that identifies firm behavioral responses. From 2007 to 2021, the period for which granular financial responsibility administrative data are available from the Texas Commission on Environmental Quality (TCEQ), an average of 89.8% of active retail UST facilities in Texas satisfied their financial responsibility obligation through commercial private insurance. The remaining 10.2% relied on self-insurance mechanisms, available exclusively to large, financially capitalized owners who can satisfy stringent asset tests.

Figure 3 documents the market transition directly. Financial responsibility coverage rates and the count of covered facilities confirm that adoption of private insurance was rapid and near-universal immediately following the December 1998 cutoff. The transition was not gradual: virtually the entire Texas UST population moved from public to private coverage

within the first year of the reform.

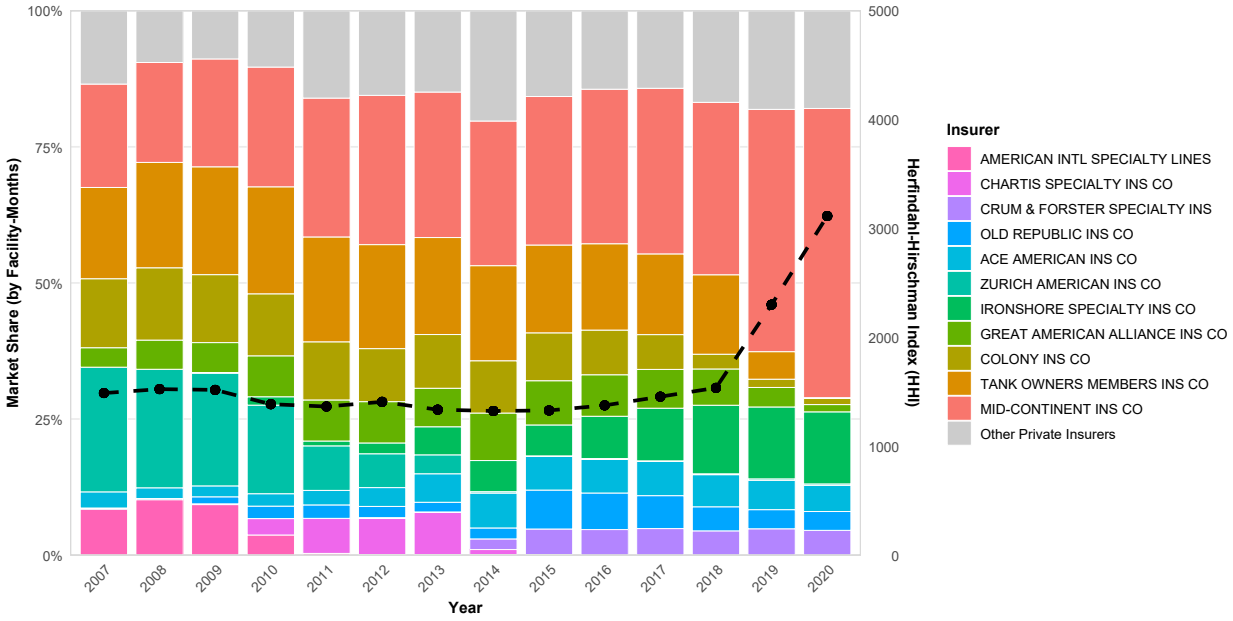
Figure 3: Financial Responsibility Compliance in Texas, 2007–2021.



Notes: Left panel: share of active retail facilities satisfying the EPA financial responsibility requirement via private insurance versus self-insurance. Right panel: total count of covered facilities. The near-universal adoption of private coverage within the first policy year confirms the reform was absorbed rapidly across the facility population. Source: TCEQ financial responsibility administrative records.

[DESCRIBE FR FIGURE ABOVE]

Figure 4: Market Concentration in the Texas UST Private Insurance Sector, 2007–2021.

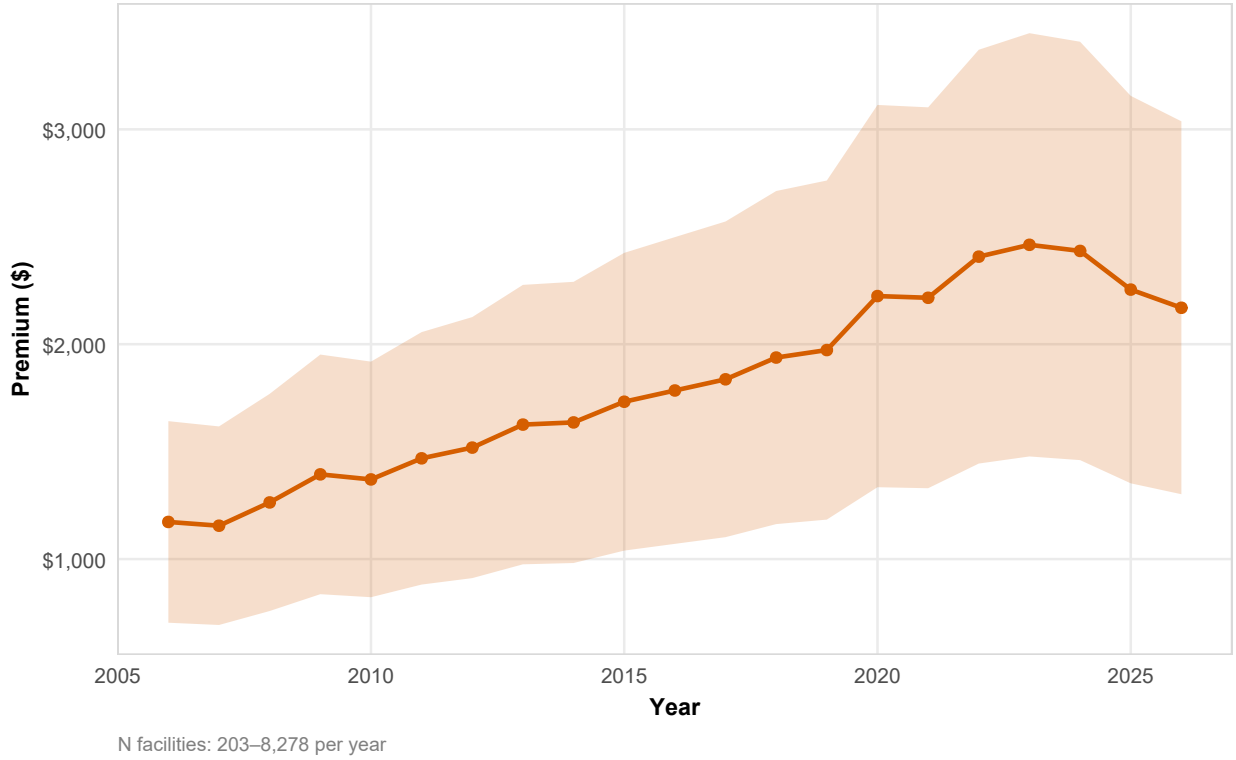


Notes: Left panel: market share of the top five underwriters by covered facility count. Right panel: Herfindahl–Hirschman Index (HHI) for the Texas UST private insurance market. Mid-Continent Casualty Company (NAIC 23418) operated as the dominant underwriter throughout the post-transition period. Source: TCEQ financial responsibility administrative records.

[DESCRIBE THE FIGURE ABOVE HHI AND WHO TO LOOK AT. DESCIRBE WHAT A BAND IS IN A SPECIFIC YEAR AND WHAT A POINT IS ON THE GRAPH.]

Private Insurance Premium Reconstruction

Figure 5: Reconstructed Annual Premium Levels for Texas UST Facilities, 2006–2021.



Notes: Left panel: median premium by year, with pre- and post-transition shading. Center panel: premium index (1995 = 100) for single-wall and double-wall tanks separately. Right panel: log premium against tank age, with fitted lines by wall type. Premiums are strictly increasing in age and strictly higher for single-wall construction at every age, consistent with the Mid-Continent rate filing structure. Source: SERFF rate filings.

[THIS WHOLE SECTION NEEDS TO STATE CLEALRY THE FEW BEHAVIRAL ASSUMPTIONS I MADE TO RECREATE THE PREMIUM AS STATED IN THE CODE ABOUT DEDUCTABLE COICE AND THEN HOW THE BANDS ARE CREATED USING THE 40% CREDITS]

Private insurers do not publicly disclose individual policy premiums. I reconstruct the premium schedule that Texas facilities faced in the post-transition period by building a bottom-up actuarial rating engine from the rate filings of Mid-Continent Casualty Company (NAIC 23418). I obtain rate manuals filed with the System for Electronic Rates and Forms Filing (SERFF) covering four regulatory periods: 2006–2014, 2014–2019, 2019–2021, and 2021 onward. These filings provide the complete pricing algorithm, including base rates, Increased Limit Factors (ILF), and all secondary risk loads applied to observable facility characteristics.

I estimate the monthly premium P_{ijt} for tank j at facility i in month t as:

$$P_{ijt} = \text{Base}_t \times \text{ILF}_{it} \times \left(1 + \sum_k \lambda_{k,t} \mathbf{X}_{ijt}\right) \tag{1}$$

where Base_t is the per-tank base rate active in filing period t, ILF_{it} is the Increased Limit Factor adjusting for the elected coverage limit (defaulting to the statutory \$1 million per occurrence minimum where unspecified), and $\mathbf{X}_{ijt}$ is a vector of observable risk characteristics including tank age bin, wall construction type, piping type, and installed leak detection technology. The parameters $\lambda_{k,t}$ are the secondary risk loads, additive percentage credits or debits, assigned to each characteristic in filing period t.

Three implementation assumptions are required. First, I hold the base rate constant within each filing period at its filed value, consistent with the filing terms. Second, for risk characteristics unobserved in the administrative records (including aquifer proximity and flood zone classification), I set the corresponding risk load to zero, yielding a conservative estimate of the true premium. Third, the rate filings permit underwriters to apply schedule rating adjustments of \$±\$40% at their discretion; I use the base manual premium as the price signal and treat discretionary adjustments as mean-zero errors across the population, an assumption evaluated against the distribution of observed financial responsibility filings in the TCEQ records.

Granular premium microdata are available beginning in 2006. The period from January 1999 through December 2005 represents a transition window in which the private market was operational but individual premium records are not observed.

The Texas Phased Mandate: A Pre-Treatment Confound

Texas's transition to private financial responsibility was preceded by a separate regulatory regime that creates a pre-treatment confound for the oldest installation cohorts. Under 30 TAC Chapter 334, the Texas Natural Resource Conservation Commission assigned staggered upgrade deadlines to pre-1988 tanks by vintage cohort, running from December 22, 1989 for the oldest tanks through December 22, 1993 for the 1985–1988 cohort (Table 10). These deadlines preceded the uniform federal date by five to nine years depending on vintage. No control state imposed an analogous staggered schedule; facilities in control states faced only the federal deadline of December 22, 1998.

The consequence is survivor selection in the Texas pre-1988 stock. By the time the insurance reform took effect in 1999, the pre-1988 Texas tanks still operating had already survived a cohort-specific compliance decision made years earlier: their operators had elected to invest in upgrades rather than close when confronted with the rolling state mandate. This is not a random sample of the original pre-1988 population. It is the subset whose operators revealed a preference for continuing operations under regulatory pressure, before any insurance price signal existed. Control-state pre-1988 operators faced no such pre-screening. Their closure decisions in the years around 1998 reflect the federal mandate alone, not a prior self-selection event. Any naive comparison pooling pre- and post-1988 cohorts therefore conflates

two distinct mechanisms: mandate-driven survivor selection in Texas and insurance-driven behavioral response, the object of interest.

Table 11a documents this differential empirically. Cumulative closure rates and surviving fleet characteristics diverge sharply between Texas pre-1988 cohorts and their control-state counterparts during 1989–1993, the window when the staggered mandate was binding in Texas and silent in control states. No corresponding divergence appears for post-1988 cohorts, which were never subject to the staggered schedule and which enter the treatment window with comparable pre-reform closure histories across Texas and the control group.

Both problems identified in this section and in Section~**??** are resolved by restricting the estimation sample to tanks installed between 1990 and 1997. This cohort never faced the staggered Texas mandate, was not subject to the federal compliance deadline before 1998, and entered the treatment window with a clean pre-reform closure history whose dynamics are comparable across Texas and the control states. The restriction is documented in Table 7 and the parallel pre-trends it earns are confirmed in the event study of Section~.

Control Group Definition and Institutional Restrictions

The potential control group consists of all states that maintained a public uniform-premium trust fund continuously through the study window and were never subject to a mandatory transition to private financial responsibility. Approximately thirty states satisfy this institutional criterion. After applying four data completeness requirements dictated by the panel construction and identification design, the usable control group contracts to nineteen states.

Missing installation dates. The panel backbone requires a confirmed installation date for every tank regardless of status so that the facility can be assigned to an age cohort and the make-model cell can be frozen at the reform date. Where installation dates are missing at high rates, facilities cannot be assigned to an age bin, are excluded from the make-model sample, and generate attrition that is mechanically correlated with vintage — precisely the dimension the identification design exploits.

Missing closure dates for tanks recorded as closed. The panel assigns a closure event to a facility-year only when a closed tank has a confirmed closure date; tanks flagged as closed but lacking a closure date are dropped in the filtering step described in Section~**??** and documented in Table 7. Where closure-date missingness is high, the denominator of the closure rate is correctly measured but the numerator is understated, biasing the outcome toward zero in the pre-treatment period and attenuating the DiD estimate.

Missing wall-type classification. The make-model comparison sample requires that each facility be classifiable as all-single-walled, all-double-walled, or mixed on the dimensions private insurers price. Wall-type is stored in the EPA tank-type field. Where this field is missing at high rates, the matching cell is undefined and the facility cannot contribute to the make-model sample regardless of its institutional eligibility.

Missing LUST report dates. The leak outcome requires a confirmed report date for each leaking underground storage tank release, used to assign the release to a facility-year and to classify it under the primary 0–60 day window. States where report-date missingness is total

are excluded from the leak analysis sample; states with partial missingness are retained for the closure analysis but excluded from the leak specifications.

Table 9 documents missingness rates across all candidate states on each of these four dimensions. The twenty-one states that relied solely on the EPA harmonized database without a custom state registry fail one or more of these requirements at rates that make the panel unusable. Closure dates are missing for 100 percent of closed tanks in New York, South Carolina, Missouri, Mississippi, Nevada, Nebraska, Georgia, New Hampshire, Rhode Island, Alaska, Utah, Indiana, West Virginia, Wyoming, and Hawaii, and for 80 percent in South Dakota. Installation dates are missing for 100 percent of tanks in South Carolina, Mississippi, and Vermont, and for more than 30 percent in New York, Missouri, Indiana, Illinois, and West Virginia. Delaware satisfies both date requirements but is excluded on a third ground: wall-type classification is missing for 100 percent of its tank inventory, making make-model cell assignment impossible. Two further states — Washington and Oregon — are excluded on institutional rather than data-quality grounds: neither operated a public uniform-premium trust fund during the study window and so do not satisfy the control group eligibility criterion regardless of data quality.

The nineteen states that satisfy all four completeness thresholds and maintained public trust funds continuously through 2020 form the control group used in all baseline specifications. Appendix~**??** lists the included states, their trust fund histories, and their tank-inventory sizes. The robustness analysis in Section~**??** shows that restricting to the nine highest-quality control states does not materially alter the estimates.

Insurance Regimes and the Retirement of Risky Capital

The insurance regime governs whether a facility's pollution risk enters its cost function. This section formalizes that claim, derives the closure threshold under each regime, and produces the testable predictions that motivate both the make-model comparison sample and the empirical specifications in Section 6. [MAKE THIS A SECTOIN REFF NOT HARD CODED]

Primitives

The facility owner's problem is a single-agent dynamic discrete choice model in the optimal stopping tradition (Rust, 1987). In an infinite-horizon setting, a competitive facility owner operates a single underground storage tank and each period decides whether to continue operating or permanently close. Let $\mathbf{x} \in \mathcal{X}$ denote the vector of facility characteristics that determine pollution risk: tank age, construction type, wall material, soil composition, and any other observable feature correlated with the probability of an environmental release. The instantaneous leak hazard is $h(\mathbf{x}) > 0$. For analytical tractability, I model $\mathbf{x}$ as the scalar tank age a, with $h'(a) > 0$ and $h''(a) \geq 0$. The qualitative results hold for any dimension of $\mathbf{x}$ along which the insurer conditions the premium. The Extension section below states this generalization explicitly.

Each leak event generates two categories of social harm. Remediation cost $L > 0$ is the direct cost of cleaning contaminated soil and groundwater. The operator bears this cost under strict liability, making it the natural object of an insurance contract. Health externality $E > 0$ represents diffuse health damages accruing to surrounding communities through groundwater contamination. Because contamination is frequently undetected for years after a release and attribution to a specific facility is administratively difficult, these damages generate no insurable third-party claim (Marcus (2024)). The gap between L and $L + E$ defines the limits of what any privately-priced insurance arrangement can achieve relative to the social optimum, and it motivates the welfare decomposition below.

The owner earns per-period net revenue $R > 0$ from operating and recovers scrap value $\kappa > 0$ upon permanent closure, representing the liquidity of the land net of decommissioning costs. The owner discounts at rate $\beta \in (0, 1)$ and is risk-neutral.

Insurance Regimes

The insurance regime determines whether and how the premium $C(\mathbf{x})$ varies with the facility's risk profile. I define three regimes that span the relevant policy space.

Definition 1 (Uniform Premium, UP). The facility pays a flat actuarially pooled premium each period, independent of any facility characteristic:

$$C_{\text{UP}}(\mathbf{x}) = \bar{p}.$$

No risk information is transmitted to the firm through the insurance contract under UP. In

the scalar age case, the marginal cost of aging, the elasticity of the premium with respect to capital depreciation, is identically zero:

$$\frac{\partial C_{\text{UP}}}{\partial a} = 0.$$

The insurer may possess full information about the cross-sectional risk distribution but does not act on it in setting individual premiums. This is the pricing structure of state-administered petroleum storage tank assurance funds, including the Texas PST Fund prior to its 1999 sunset.

Definition 2 (Risk-Based Pricing, RB). The insurer prices actuarially, conditioning on all observable elements of $\mathbf{x}$. The premium equals expected claims scaled by a loading factor $\lambda \geq 1$:

$$C_{\text{RB}}(a) = \lambda \cdot h(a) \cdot L.$$

Since $h'(a) > 0$, the premium's elasticity with respect to capital depreciation is strictly positive:

$$\frac{\partial C_{\text{RB}}}{\partial a} = \lambda h'(a) L > 0.$$

Every dimension of $\mathbf{x}$ that enters $h(\mathbf{x})$ is mechanically injected into the firm's flow cost by the actuarial pricing rule. This is a definitional property of risk-based pricing, not a behavioral assumption about firm awareness or optimization.

Definition 3 (Social Optimum, SOC). The social planner internalizes both remediation costs and health externalities:

$$C_{\text{SOC}}(a) = h(a) \cdot (L + E).$$

The social optimum is unattainable under private insurance because E generates no insurable claim. For the parametrically relevant condition $E > (\lambda - 1)L$, the cost ordering is:

$$C_{\text{UP}} \leq C_{\text{RB}}(a) < C_{\text{SOC}}(a),$$

where the left inequality becomes strict for a large enough that $\lambda h(a) L > \bar{p}$.

The Firm's Dynamic Problem

Each period, the owner observes age a and chooses to maintain or permanently close. Under regime j, flow utility from maintaining is $u_j(a) = R - C_j(a)$, where R is annual net revenue from operations net of all non-insurance costs and $C_j(a)$ is the insurance premium. I maintain the simplifying assumption that insurance contracts provide full ex-post coverage

with no deductible, treating environmental risk solely as an ex-ante premium cost rather than an ex-post liability shock. This assumption isolates the cost-of-continuing channel and is consistent with standard provisions in the petroleum storage tank insurance market; the structural estimation in Section 8 relaxes it to incorporate partial risk internalization through deductibles.

The Bellman equation under regime j is:

$$V_j(a) = \max\left\{ \underbrace{R - C_j(a) + \beta\,\mathbb{E}[V_j(a') \mid a]}_{\text{Maintain}},\quad \underbrace{\kappa}_{\text{Close}} \right\},$$

where β is the discount factor, the expectation is taken over the stochastic evolution of tank age, and κ is the scrap value recovered upon exit. The pure continuation value, the present discounted value of flow utility under the assumption that the tank is never closed, is:

$$\widetilde{V}_j(a) = \sum_{s=0}^{\infty} \beta^s \left[R - C_j(a+s)\right].$$

The optimal value function satisfies $V_j(a) = \max\{\widetilde{V}_j(a),\, \kappa\}$. The optimal closure threshold is:

$$a_j^* = \min\left\{a \geq 0 : \widetilde{V}_j(a) \leq \kappa\right\}.$$

Under any regime with $C_j'(a) > 0$, the function $\widetilde{V}_j(a)$ is strictly decreasing, so the optimal policy is a threshold rule: close if and only if $a \geq a_j^*$.

Main Results

Proposition 1 (Risk Transmission and Value Slope). *Under UP, $\widetilde{V}_{UP}'(a) = 0$. Under RB, $\widetilde{V}_{RB}'(a) < 0$. Under SOC, $\widetilde{V}_{SOC}'(a) < \widetilde{V}_{RB}'(a) < 0$.*

Proof. Differentiating the continuation value with respect to age:

$$\widetilde{V}_j'(a) = -\sum_{s=0}^{\infty} \beta^s C_j'(a+s).$$

Under UP, $C_{\text{UP}}'(\cdot) = 0$ everywhere, so the sum is zero and the continuation value is flat in age: the physical degradation of the asset is real but generates no financial pressure on the firm. Under RB, $C_{\text{RB}}'(a) = \lambda h'(a)L > 0$, so every term in the sum is strictly negative and $\widetilde{V}_{\text{RB}}'(a) < 0$. Under SOC, $C_{\text{SOC}}'(a) = h'(a)(L+E) > \lambda h'(a)L = C_{\text{RB}}'(a)$, so the SOC slope is strictly more negative at every age. □

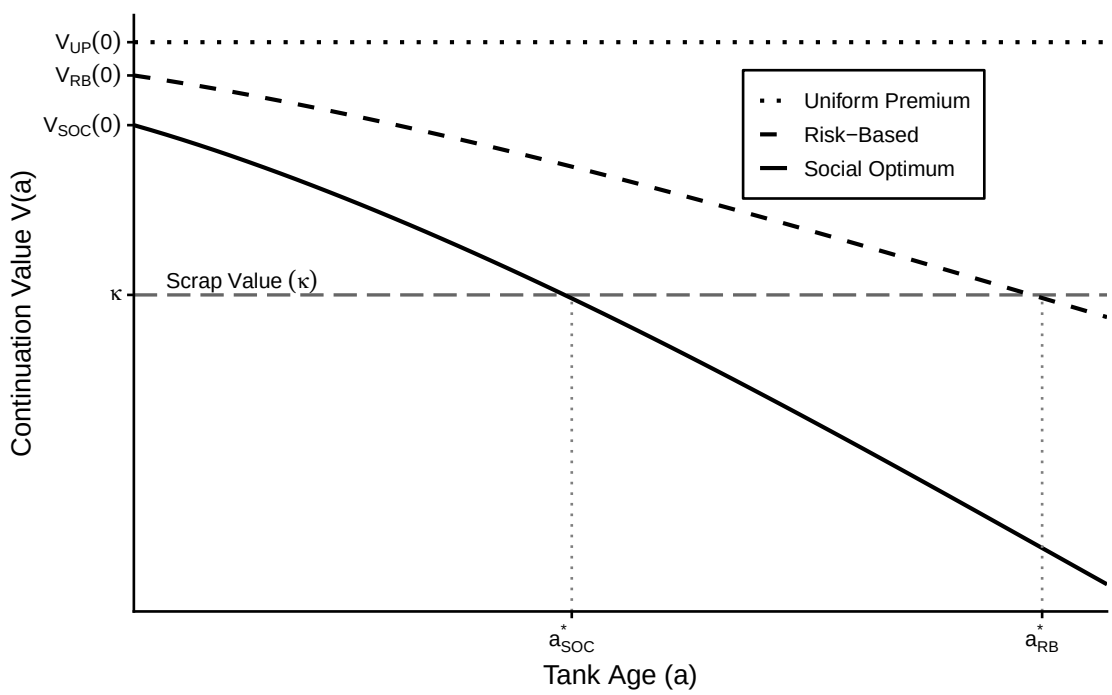
Proposition 1 establishes the mapping from the pricing regime to the information environment of the firm. Under UP, aging is economically invisible: the zero elasticity of the premium with respect to capital depreciation means the continuation value contains no signal about the facility's pollution hazard, and the firm retains the asset until it is physically obsolete. Under RB, aging is economically visible by construction: the actuarial pricing rule translates the insurer's risk model into a per-period cost that rises with the facility's hazard profile, pivoting the continuation value downward for older tanks and accelerating the optimal exit age, regardless of whether the owner attends to or understands the actuarial logic.

Proposition 2 (Regime Ordering of Closure Thresholds). *Under the condition $E > (\lambda - 1)L$, the optimal closure ages satisfy $a^*_{SOC} < a^*_{RB} < a^*_{UP}$.*

Proof. By Proposition 1, $\widetilde{V}_{\text{SOC}}(a) < \widetilde{V}_{\text{RB}}(a) < \widetilde{V}_{\text{UP}}(a)$ for all $a > 0$. Since a^*_j is defined as the first age at which $\widetilde{V}_j(a)$ crosses κ from above, a lower value function crosses the threshold at a younger age. The ordering of value functions is preserved under discounted summation because $h(a)$ is common across regimes; only the coefficient multiplying $h(a)$ differs. □

Figure 6 illustrates Proposition 2. The SOC curve declines at the fastest rate and crosses the scrap value line κ at a^*_{SOC}. The RB curve declines at an intermediate rate and crosses κ at $a^*_{\text{RB}} > a^*_{\text{SOC}}$, strictly reducing the optimal tank lifespan relative to the uniform pricing baseline. The UP curve is flat by Proposition 1 and does not cross κ within the plotted range, reflecting the zero elasticity of the premium with respect to capital depreciation under pooled pricing.

Figure 6: The Mechanism: Optimal Stopping under Alternative Insurance Regimes



Notes: This figure illustrates the optimal closure thresholds under the three insurance regimes derived in Proposition 2. The horizontal line represents the scrap value (κ). The Social Optimum (SOC) continuation value declines at the fastest rate, prompting the earliest closure at a^*_{SOC}. The Risk-Based (RB) pricing continuation value declines at an intermediate rate, yielding closure at a^*_{RB}. The Uniform Premium (UP) continuation value remains flat, reflecting the zero elasticity of the premium with respect to capital depreciation under pooled pricing.

Frictions Attenuating Behavioral Response

The model establishes the direction of the effect of moving from UP to RB pricing. It does not establish the magnitude, and it is silent on two frictions that could attenuate behavioral response even when the directional incentive is correctly signed.

The first friction is premium insufficiency. Even if $C'_{\text{RB}}(a) > 0$, the premium increment at the margin may be small relative to the fixed transaction costs of closure, which include demolition, regulatory notification, site remediation, and foregone operating revenue. If the present value of avoided future premiums is less than these fixed exit costs for the marginal tank, the regime shift generates no behavioral response despite correctly pricing risk.

The second friction is constraints on action. Even when $\widetilde{V}_{\text{RB}}(a) < \kappa$ and closure is optimal, the owner may be unable to act on that signal. Credit constraints may prevent financing upfront exit costs. Thin markets for contaminated properties may render the book value of κ unrealizable. Present bias or inattention to the actuarial premium schedule could cause owners to underweight future premium increases.

Neither friction is resolvable from theory alone. The Texas natural experiment provides the empirical test: the 1999 sunset of the PST Fund was abrupt, mandatory, and economically large, providing the variation required to estimate whether the premium-burden mechanism is sufficient to produce measurable changes in closure behavior.

Proposition 3 (Orthogonality Under UP; Correlation Under RB). *Let ξ denote closure drivers unrelated to the facility's pollution hazard, land value appreciation, local fuel demand shocks, and owner liquidity or retirement decisions, satisfying $\xi \perp a$. Then: (i) under UP,* $\text{Cov}(\mathbf{1}[\text{Close}], a) = 0$*; (ii) under RB,* $\text{Cov}(\mathbf{1}[\text{Close}], a) > 0$*, and the treatment effect* $\Delta(a) \equiv \Pr(\text{Close} \mid \text{RB}, a) - \Pr(\text{Close} \mid \text{UP}, a)$ *is zero for $a < a^*_{RB}$ and strictly increasing in a for $a \geq a^*_{RB}$.*

Proof. Under UP the only closure drivers are elements of ξ, which satisfy $\xi \perp a$ by assumption, so $\text{Cov}(\mathbf{1}[\text{Close}], a) = 0$. Under RB, $\widetilde{V}_{\text{RB}}(a)$ is strictly decreasing, so $\Pr(\widetilde{V}_{\text{RB}}(a) \leq \kappa)$ is strictly increasing in a and the covariance is positive. For the treatment effect: when $a < a^*_{\text{RB}}$, $\widetilde{V}_{\text{RB}}(a) > \kappa$ and $\widetilde{V}_{\text{UP}}(a) > \kappa$ under both regimes, so $\Delta(a) \approx 0$. When $a \geq a^*_{\text{RB}}$, $\widetilde{V}_{\text{RB}}(a) \leq \kappa < \widetilde{V}_{\text{UP}}(a)$, so the regime difference is the binding margin and $\Delta(a) > 0$, strictly increasing as the gap $\widetilde{V}_{\text{UP}}(a) - \widetilde{V}_{\text{RB}}(a)$ widens with age. □

The $\xi \perp a$ assumption warrants comment. It requires that non-insurance closure drivers, including owner retirement, land value appreciation, and local demand shifts, are mean-independent of tank age. This is an identifying assumption, not a definitional one. If owners of older tanks are themselves older and more likely to retire, or if facilities in older

commercial districts face systematically different land value trajectories, the orthogonality would be violated. The pre-period evidence in Section 4.5 provides an indirect test: under UP, control-state closure rates should exhibit no age gradient in the pre-treatment period. The parallel trends evidence is consistent with this condition, though it cannot rule out violations that are small relative to sampling noise.

Proposition 3 is the identifying prediction of the paper. The shift from UP to RB does not merely change the number of tanks that close; it reorganizes *which* tanks close. Under UP, the age distribution of closures is determined entirely by land value appreciation, demand contraction, and owner liquidity or retirement, drivers that are mean-independent of the facility's hazard profile. Under RB, the age distribution of closures becomes correlated with the hazard profile in a specific, premium-gradient-shaped way: the treatment effect is zero for young tanks and rising for old tanks, with the crossover at a^*_{RB}.

This heterogeneous treatment effect has a direct implication for identification. If the treatment effect is concentrated among older, higher-risk facilities, a comparison group that is unbalanced on age or wall type will produce biased estimates, not because the treated and control groups follow different trends on average, but because the composition of each group puts different weight on the margin where the treatment bites. The make-model comparison sample, introduced in Section 4, is constructed precisely to address this: by matching on age bin and wall type, the comparison group is balanced on the dimensions along which $\Delta(a)$ varies, removing the compositional source of bias.

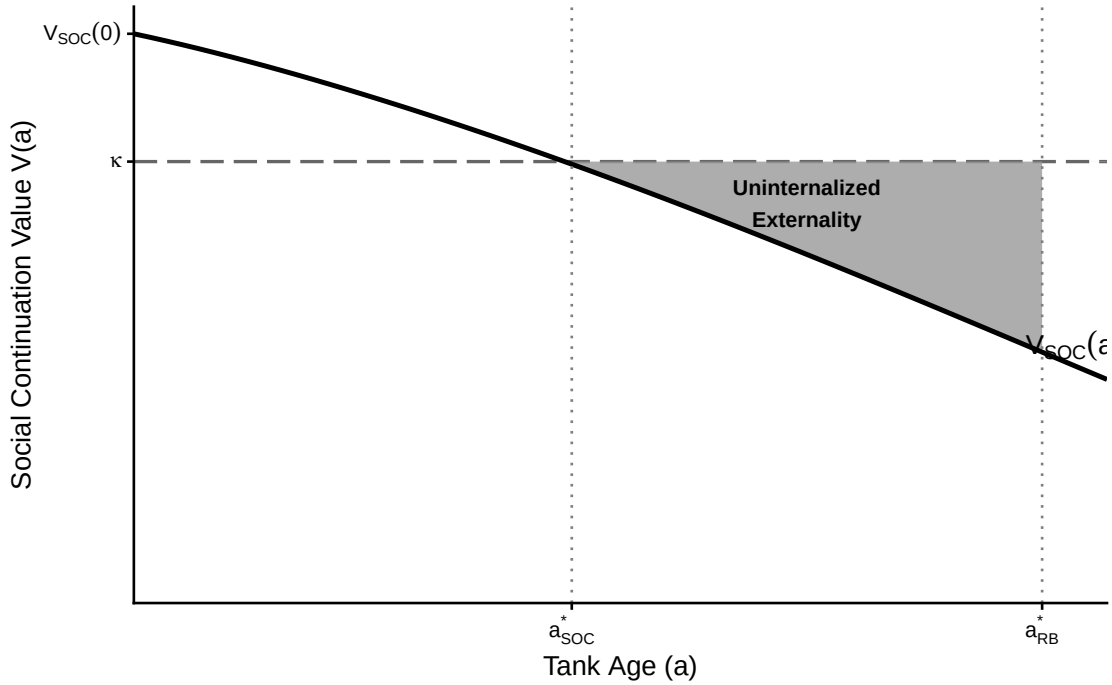
Welfare Decomposition

The welfare cost of retaining a tank beyond the socially optimal closure age is the expected externality accumulated during the excess operating period. The total welfare loss from UP decomposes into two components:

$$\text{DWL}_{\text{UP}} = \underbrace{\int_{a^*_{\text{SOC}}}^{a^*_{\text{RB}}} h(a)\, E\, da}_{\text{Unavoidable Wedge}} + \underbrace{\int_{a^*_{\text{RB}}}^{a^*_{\text{UP}}} h(a)\, E\, da}_{\text{Eliminable Excess}}.$$

The unavoidable wedge exists because E generates no insurable claim, so the gap $a^*_{\text{RB}} - a^*_{\text{SOC}}$ cannot be closed by insurance design alone. The eliminable excess is the welfare loss attributable specifically to flat-fee pooling, the component recoverable by moving from UP to RB pricing.

Figure 7: Welfare Analysis: The Externality Wedge and Partial Correction



Extension to the Full Risk Vector

The model uses age as a scalar proxy for $\mathbf{x}$ for tractability. Propositions 1 through 3 hold for any dimension of $\mathbf{x}$ along which (i) the hazard $h(\mathbf{x})$ varies and (ii) the insurer conditions the premium. The mechanism is identical: any risk factor that enters the actuarial pricing schedule is mechanically injected into the firm's flow cost, raises the elasticity of the premium with respect to that risk dimension, and generates closure sensitivity that is absent under flat-fee pooling.

This extension has a concrete empirical counterpart in the petroleum storage tank market. Private insurers in Texas condition premiums on both age and tank wall construction type, with single-wall tanks facing premiums approximately twice those of double-wall tanks of equivalent age, as documented in Mid-Continent rate filings for the 2007 to 2021 period. If the mechanism of Proposition 3 extends to all priced risk dimensions, closure decisions under RB pricing should become correlated with wall type in the same way they become correlated with age. I treat this as an empirical generalizability test: a positive result for wall type corroborates the premium-burden channel; a null result would indicate either that wall-type premium gradients are insufficient to move behavior or that single-wall operators face structural exit barriers distinct from those facing aged-tank operators.

Testable Predictions

The following four hypotheses map directly from the propositions above to the difference-in-differences specifications in Section 6. In each case, the empirical test compares Texas

facilities, which transitioned from UP to RB pricing in 1999, to make-model-matched control-state facilities, which retained flat-fee public insurance programs throughout the sample period.

Hypothesis 1 (Age Sensitivity of Closure). Under UP, the closure hazard is mean-independent of a (Proposition 3, part i). Under RB, the positive elasticity of the premium with respect to capital depreciation creates a premium-burden channel that makes older facilities more likely to close (Proposition 3, part ii). The prediction is that the age gradient in the closure hazard steepened differentially in Texas after 1999:

$$\frac{\partial \lambda(t \mid a, \mathrm{RB})}{\partial a} - \frac{\partial \lambda(t \mid a, \mathrm{UP})}{\partial a} > 0.$$

A discrete-time hazard model with the triple interaction $\mathrm{TX} \times \mathrm{Post} \times a$ is the natural estimator. The linear probability model on the binary closure indicator is a valid approximation under the assumption that censoring is independent of age conditional on covariates; the survival model relaxes this assumption.

Hypothesis 2 (Heterogeneous Treatment Effect by Age). The treatment effect of switching from UP to RB is zero for young tanks and strictly positive and increasing for old tanks (Proposition 3). For cutoff age $\bar{a}$:

$$\mathrm{ATT}(a > \bar{a}) > \mathrm{ATT}(a \leq \bar{a}) \approx 0.$$

Hypothesis 3 (Rightward Shift in Age Distribution of Closures). The transition to RB pricing triggers a concentrated wave of exits among older tanks, raising the expected age of a closing facility relative to control-state counterparts:

$$\mathbb{E}[a \mid \text{Close, Texas, Post}] > \mathbb{E}[a \mid \text{Close, Control, Post}].$$

The mean shift is the testable corollary of a richer distributional prediction: under RB, the age-at-closure distribution concentrates mass at high ages near a^*_{RB}, whereas under UP the distribution of age at closure mirrors the age distribution of the full operating population. A test of the mean shift conditions on the selected sample of closures, which introduces selection bias because the tanks that close under RB are a non-random subset of the population. The density of age at closure by group and period provides the non-parametric visual companion to the regression test.

Hypothesis 4 (Wall-Type Sensitivity). If the mechanism extends to all priced dimensions of $\mathbf{x}$, the single-wall versus double-wall gap in the closure hazard should widen under RB pricing relative to UP:

$$[\lambda(t \mid \text{SW, RB}) - \lambda(t \mid \text{DW, RB})] > [\lambda(t \mid \text{SW, UP}) - \lambda(t \mid \text{DW, UP})]\,.$$

The empirical test is the triple interaction $\mathrm{TX} \times \mathrm{Post} \times \mathrm{SingleWall}$ in the closure specification.

Three empirical patterns would be inconsistent with the premium-burden channel and would instead indicate confounding from a Texas-specific economic shock: a uniform treatment effect across all age and wall-type groups; a closure response concentrated among young or low-risk facilities; or no correlation between the treatment effect magnitude and observable risk factors. These patterns define the falsification criteria for the identification strategy in Section 6.

Data

Source Records

The analysis draws on administrative records from three sources. The first is a tank-level physical inventory assembled from state UST regulatory databases across all fifty states, harmonized into a common schema. Each record identifies a physical tank by facility, state, and tank identifier, and reports the tank's installation date, closure date (if closed), wall construction (single-walled, double-walled, or unclassified), stored substance (gasoline, diesel, other petroleum products), and nominal capacity in gallons. State databases differ in field naming, date formatting, and classification conventions; the harmonization standardizes these fields, resolves coordinate systems for facility geolocation, and backfills missing county identifiers from ZIP code centroids where native geocoding is unavailable. The resulting file contains one row per physical tank across all states.

The second source is a harmonized incident file of confirmed releases reported under the Leaking Underground Storage Tank (LUST) program. Each record identifies a facility, state, and the date the release was reported to the state regulatory authority. States that lack report dates for all or nearly all LUST incidents are excluded from the leak analysis, as described below.

The third source, available for Texas only, is the financial responsibility compliance record maintained by the Texas Commission on Environmental Quality (TCEQ). This file records, at monthly frequency, each facility's compliance mechanism (private insurance, self-insurance, or no coverage), the identity of the insurer where applicable, and the contract detail type. These records are available from 2007 onward and are used to document the post-transition private market structure described in Section~2.

Panel Construction

The unit of observation for the behavioral analysis is the facility-year. I construct the panel by first computing each facility's active window, the span from its earliest tank installation to its latest tank closure or the end of the study period, and expanding the facility to a monthly time series over that range. For each facility-month, the tank inventory is queried to determine the count and characteristics of tanks physically active on that date: total active tanks, aggregate capacity, average tank age (computed from exact installation dates), counts of single-walled and double-walled tanks, and fuel-type indicators. Monthly events (confirmed releases, tank closures, and new tank installations) are counted and merged to the facility-month backbone.

Tank closures receive a type classification before aggregation. For each closed tank, I determine whether the facility installed any replacement tank after that closure date. If so, the closure is classified as a *replacement*; if not, as *permanent*. The classification uses the facility's full installation history, including tanks outside the analysis window, to avoid misclassifying replacement closures as permanent when the replacement tank falls outside a restricted sample frame.

The monthly panel is then collapsed to annual frequency. Stock variables (active tank count, capacity, tank age, wall composition) are measured at the December snapshot. Flow variables (closures, installations, confirmed releases, retrofits) are summed over the calendar year, with binary indicators for whether any event of each type occurred. Year-over-year changes in tank count and capacity are computed as first differences of the December stocks.

Each confirmed release is paired with tank closures at the same facility and classified by timing. A release reported within 60 days after a closure is classified as *closure-revealed*: the closure inspection likely discovered a latent leak. A release on record more than 180 days before a closure is classified as *known pre-existing*: the tank was already leaking. Releases with no nearby closure are classified as *standalone*. Pairs are restricted to the same generation of infrastructure: a release that predates a tank's installation is not matched to that tank. Robustness checks use 30-day and 90-day windows for the closure-revealed category.

For the duration analysis of individual tank closures, a second panel is constructed at the tank-year level directly from the physical inventory, without passing through the facility-level aggregation. Each tank is expanded from its installation year through its closure year (or through 2020 for tanks still active), yielding one row per tank per calendar year of service. The closure event indicator equals one in the year the tank was closed. For the primary Cox specification, exact installation and closure dates are retained and the counting-process interval for each tank is split at the statutory reform date of December 22, 1998, producing at most two records per tank: a pre-reform interval and a post-reform interval. This construction allows the treatment indicator to switch on at the precise reform date rather than at an annual boundary.

Study States

Control Group Selection Criteria

The control group comprises states satisfying two institutional criteria and four data-quality criteria assessed on the 1990–1997 installation cohort that forms the primary analysis sample.

Institutional criteria. A state must have maintained a public uniform-premium trust fund continuously through the study period, providing the institutional counterfactual of continued flat-fee coverage. Washington and Oregon are excluded because neither operated such a fund. Six additional states—Arizona, Connecticut, Florida, Iowa, Michigan, and Wisconsin—are excluded because they underwent their own transitions to private or risk-based insurance during the study period and therefore cannot serve as controls.

Data-quality criteria. I evaluate four dimensions of record completeness, assessed specifically within the 1990–1997 installation cohort rather than across all historical records. The cohort-specific assessment is important because missingness rates often differ substantially between newer and older inventory, and the analysis sample is drawn exclusively from this cohort.

Wall-type classification. The comparison design matches Texas and control-state facilities on wall composition, which is the primary risk-pricing variable. States where wall type is recorded as unknown for more than 50 percent of 1990–1997 cohort tanks cannot contribute

meaningfully to within-cell identification. Alabama (80.4% unknown wall), Illinois (100%), and several smaller states fail this criterion and are excluded.

Closure recording rate. The closure outcome requires that tanks marked as closed in the administrative record carry a closure date. Some states record all tanks as perpetually open regardless of actual status, with closure information held separately or not digitized. I measure this using the *closure-with-date rate*: the share of 1990–1997 cohort tanks that are recorded as closed with a valid closure date. Texas's closure-with-date rate of 37.2 percent serves as the benchmark. States whose rate is near zero despite the cohort being 25–35 years old are presumed to have a systematic recording failure. Georgia and Missouri both show 0.0% closure-with-date rates and are excluded on this basis. California similarly records 100% of its 1990–1997 tanks as Open, yielding a 0.0% closure-with-date rate; it is excluded despite otherwise clean wall-type data.

LUST report dates. States where LUST report dates are missing for 100 percent of incidents cannot contribute to the leak-outcome specifications. Kentucky (100% missing) is excluded from the full analysis sample on this basis. Pennsylvania is retained in the closure analysis per prior institutional knowledge of its data structure, but excluded from leak specifications.

Closed-without-date rate. Among tanks the state does record as closed, the share lacking a closure date should be low. States where more than roughly 30 percent of closed tanks are missing dates are flagged; none reach the threshold for outright exclusion, but Arkansas (66% missing LUST dates) and Tennessee (59% missing LUST dates) are excluded from leak specifications while retained for closure analysis.

Resulting Control Groups

Applying these criteria yields two nested control groups used throughout the analysis.

The **primary analysis sample** comprises Texas and 15 control states: Arkansas, Colorado, Idaho, Kansas, Louisiana, Massachusetts, Maine, Minnesota, North Carolina, Ohio, Oklahoma, Pennsylvania, Tennessee, Virginia, and West Virginia. This sample is used for all closure-outcome regressions (RQ1 and RQ2). The panel contains 318,668 facility-years across 16 states (1 treated, 15 control) for the 1990-entry cohort over the 1990–2018 window.

The **LUST-clean sample** is a subset of 9 control states that additionally satisfy the LUST-date completeness requirement: Idaho, Kansas, Louisiana, Massachusetts, Maine, Minnesota, North Carolina, Oklahoma, and Virginia. This sample is used for all leak-outcome regressions (RQ3 and RQ3.1). It contains 198,742 facility-years across 10 states. The main treatment effect estimate is stable across the two samples ($\hat{\beta} = 0.0139$ in the primary sample versus $\hat{\beta} = 0.0185$ in the LUST-clean sample), confirming that the state-composition difference does not materially affect identification.

Table~**??** in Appendix~**??** reports the four completeness dimensions for all candidate states, distinguishing the 1990–1997 cohort rates from the full-universe rates to document that the sample-specific assessment is the relevant quality screen.

Analysis Sample Restriction

Within the selected states, the primary analysis sample is further restricted to facilities whose first tank installation occurred in 1990 or later. This restriction serves two purposes. First, it eliminates facilities that entered during or before the 1988–1989 federal EPA upgrade mandate transition period, which imposed differential closure pressure across states depending on enforcement timing and tank age cohorts. Pre-1990 entrants therefore faced a confounding closure incentive unrelated to the Texas insurance reform. Second, it produces a pre-reform panel window with stable parallel trends: the event study shows pre-period coefficients that are small and statistically indistinguishable from zero for the 1990-entry cohort, whereas extending the sample back to 1989 or earlier reintroduces marginally significant pre-period differentials driven by the thin early cohorts (342 Texas facilities at $t = -9$ versus 2,734 at $t = -1$).

Outcome Variables

The analysis estimates effects on three groups of outcomes, measured at the facility-year level unless otherwise noted.

Closure and exit. The primary closure indicator equals one if the facility closed at least one tank in a given year. Closures are decomposed into *replacement closures* (the facility later installed a new tank) and *permanent closures* (no subsequent installation). Facility exit equals one in the facility's final observed year of operation.

Portfolio investment. The analysis measures the net change in tank count, the change in total capacity, the count of new installations, and whether any single-walled tank was replaced with a double-walled tank. These outcomes are estimated both unconditionally and conditional on a retrofit occurring that year.

Leak incidence. The total confirmed release indicator equals one if the facility reported any LUST incident in a given year. This is decomposed into closure-revealed releases (reported 0 to 60 days after a tank closure), standalone releases (no nearby closure), and closures of tanks with a known pre-existing leak (on record more than 180 days before closure). The decomposition separates mechanical effects of increased closure activity from changes in leak incidence independent of the closure margin.

For the tank-level duration analysis, the outcome is the closure event: a binary indicator equal to one in the year a tank is closed, with exact dates used in the Cox specification.

Descriptive Statistics

Table 1 reports the 1998 cross-section and pre-reform flow averages for both the full analysis sample and the make-model comparison sample, separately for Texas and the control group. Panel A describes the capital stock at the reform date: facility count, total tanks, average tanks per facility, average capacity, average tank age, age of the oldest tank, fleet-level single-wall share, and within-facility single-wall share. In Panel A, both column groups report pooled means with standard errors of the mean.

Table 1: Sample Characteristics: Full Analysis Sample vs. Make-Model Comparison Sample.

	Full Sample (Unit + Year FE)			MM Sample (Unit + Cell × Year FE)		
Variable	TX	CTL	Diff.	TX	CTL	Diff.
Panel A: 1998 Cross-Section (Risk Covariates)						
N States (1 TX + CTL)	1	16	—	1	16	—
N Facilities	25,816	59,064	—	25,816	58,759	—
N Total Tanks	61,782	148,321	—	61,782	147,795	—
Avg tanks per facility	2.7	2.8	-0.0***	2.7	2.8	-0.1***
	(0.0)	(0.0)		(0.0)	(0.0)	
Avg capacity per facility (gal)	20,358	21,069	-711**	20,358	21,104	-746**
	(142)	(306)		(142)	(308)	
Avg tank age (facility level)	15.8	15.0	+0.8***	15.8	15.1	+0.7***
	(0.1)	(0.0)		(0.1)	(0.0)	
Avg age of oldest tank (fac level)	15.6	15.0	+0.7***	15.6	15.0	+0.6***
	(0.1)	(0.1)		(0.1)	(0.1)	
Avg age of state tank fleet	15.6	14.5	+1.1***	15.6	14.6	+1.0***
Share of state fleet: Single-Wall %	86.7	79.2	+7.5***	86.7	79.4	+7.3***
Avg share of SW tanks per fac (%)	84.9	78.9	+6.0***	84.9	79.3	+5.6***
	(0.2)	(0.2)		(0.2)	(0.2)	
Panel B: Pre-Reform Outcome Flows (1990–1998)						
Avg Closure Rate (%)	10.16	10.09	+0.07	10.16	10.09	+0.07
	(0.06)	(0.04)		(0.06)	(0.04)	
Avg LUST Rate (%)	1.21	2.75	-1.54***	1.21	2.76	-1.54***
	(0.02)	(0.02)		(0.02)	(0.02)	
Avg Replacement Rate (%)	0.79	1.09	-0.31***	0.79	1.08	-0.29***
	(0.02)	(0.01)		(0.02)	(0.01)	
N Facility-Years	299,351	680,268	—	299,351	678,348	—
Panel C: Make-Model Cell Coverage						
Facility cells (wall × fuel × age bin at reform)						
Total cells	85			77		
Identified cells (TX & CTL present)	77			77		
Unidentified: TX-only (no CTL)	0			0		
Unidentified: CTL-only (no TX)	8			0		
% facility-years in identified cells (full panel)	99.7%			100.0%		
Tank cells (wall × fuel × capacity × cohort; MM: 1989–1997 only)						
Total cells	1,923			455		
Identified cells (TX & CTL present)	1,095			282		
Unidentified: TX-only (no CTL)	141			19		
Unidentified: CTL-only (no TX)	687			154		
% tanks in identified cells	96.6%			95.1%		

Panel B reports pre-reform annual outcome rates (1990 to 1998). The full-sample columns show pooled facility-year means. The make-model columns show a different quantity: the cell-year weighted mean. For each cell and year, the TX and CTL mean rates are computed separately. These within-cell-year differences are then averaged over pre-reform years within each cell, and the cell-level averages are aggregated using facility-year weights. This is the quantity the cell-by-year fixed effect identifies. The cell-year weighted closure rate is higher than the pooled rate because larger, older cells receive more weight, and those cells have higher baseline closure rates. The cell-year weighted diff in closure rates is +0.23 percentage points, close to zero. The LUST rate diff of $-$1.83 percentage points reflects a persistent levels difference between Texas and the control group. This levels gap is absorbed by the facility fixed effect; the identifying variation is the within-facility change at the reform date.

Panel C documents the cell structure of each regression design, reporting the number of total, identified (containing both Texas and control-state observations), and group-exclusive cells for both the facility-level and tank-level cell taxonomies.

Notes: Panel A reports the December 1998 cross-section. Both column groups show pooled means; parentheses report standard errors of the mean. State-fleet rows are population-weighted aggregates across all active tanks (no SE). Panel B reports pre-reform annual outcome rates (1990 to 1998). Full-sample columns are pooled facility-year means with SE. Make-model columns are cell-year weighted means: for each (cell, year)

the TX and CTL rates are computed, differenced, averaged over years within each cell, then aggregated across cells using facility-year weights. This is the quantity the cell $\times$ year FE regression identifies. The make-model Diff. column in Panel B is the weighted average of within-cell-year TX $-$ CTL differences, not the difference of the TX and CTL columns in that row. Panel C reports the cell structure; the TX and CTL column headers do not apply and the two value columns show the full-sample and make-model sample values respectively. Diff. = TX $-$ CTL (Panels A and B); stars from Welch t-tests (Panel A full-sample columns only). $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. –>

Observable Risk Characteristics and the Actuarial Pricing Foundation

Risk-based insurance can only discipline facility operators if the characteristics private insurers observe actually predict future environmental liability. If tank age and wall construction type were unrelated to release probability and cleanup cost, insurers would have no basis for charging different premiums across facilities, and the premium gradients documented in Section 2.5 would be noise rather than signal. This section shows that observable facility characteristics predict confirmed releases out of sample. This establishes that the premium schedule in the rate filings reflects genuine actuarial content, and it motivates the comparison sample construction in the following section.

Release Probability Is Predictable from Observables

I estimate the annual probability that a facility experiences its first confirmed release, conditional on having never previously leaked. The estimand is the first-release hazard in the never-yet-leaked risk set:

$$h(t \mid x) = P(\text{first confirmed release in year } t \mid \text{no prior release through } t{-}1,\ x)$$

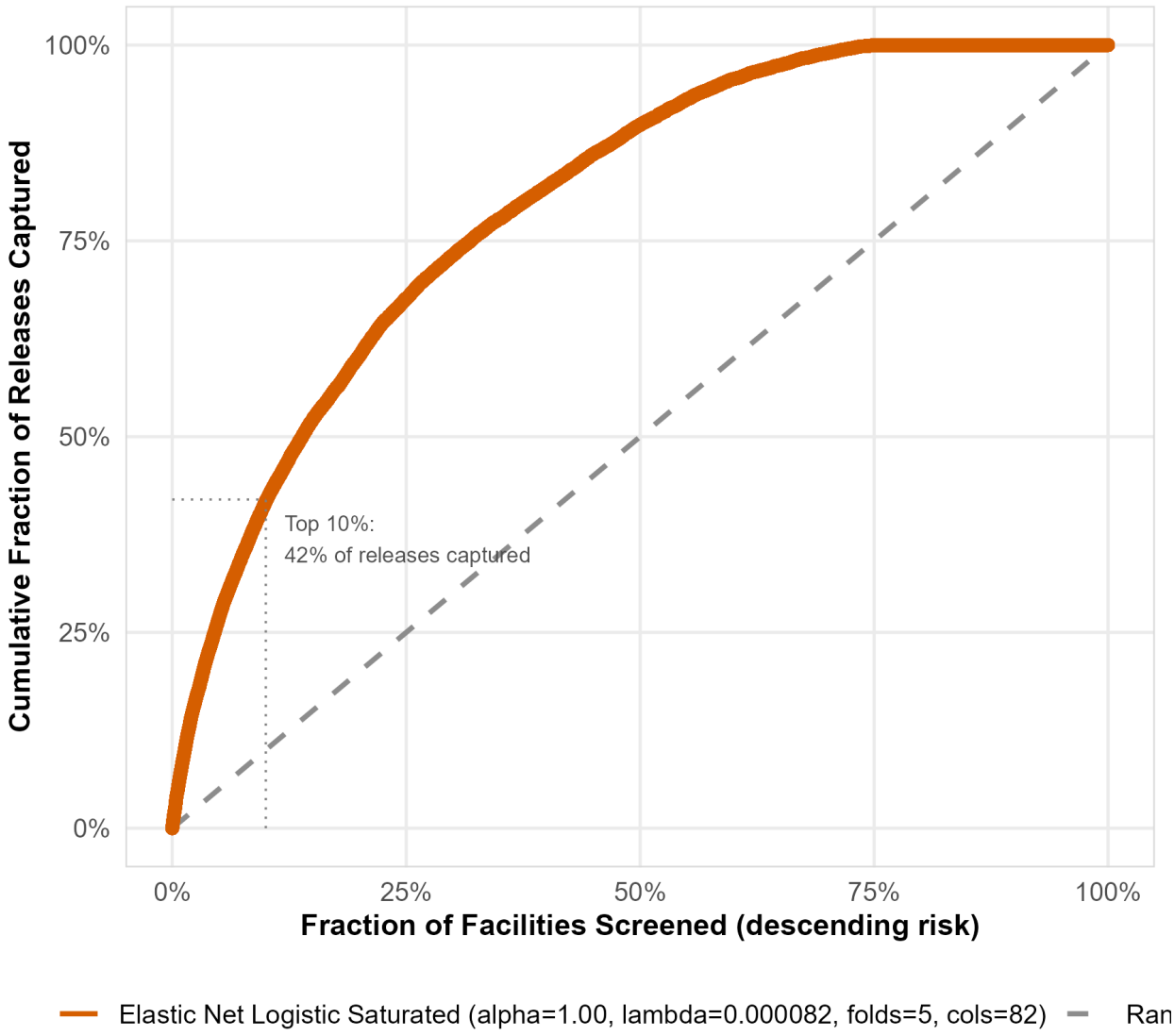
This directly mirrors the insurer's underwriting problem: given what is observable about a facility at the start of a policy year, what is the probability of a claim?

The model is a penalized logistic regression estimated on all study states in the pre-treatment period. The feature space is saturated: tank age enters as a set of three-year group indicators with a full interaction with wall construction type, allowing the model to fit a distinct risk gradient at each age step for single-walled and double-walled facilities rather than imposing a common linear slope. State fixed effects capture baseline hazard differences across the twenty study states. Fuel type and portfolio size enter as additional controls along with their pairwise interactions. Lasso regularization performs automatic variable selection, shrinking the coefficients of uninformative terms to zero while retaining the predictors that carry genuine signal. Out-of-sample predictions are obtained through cross-validation: each observation is scored exclusively by model fits on the folds that excluded it, so every predicted probability is genuinely out-of-sample. A post-hoc recalibration step corrects for the probability shift introduced by inverse-frequency class weighting, mapping predictions back to the original event-rate scale.

First-release events are rare, at an annual base rate of 0.699 percent across the estimation sample of 3,929,425 pre-treatment facility-years and 27,455 confirmed releases, a class ratio of approximately 142 non-event facility-years for each release. Without correction the model learns to predict the majority class everywhere. Inverse-frequency case weights equalize the effective class sizes during estimation, forcing the model to recover the signal in the minority class.

The model's ability to rank facilities by release risk is summarized by its lift over random selection. Figure 8 shows the cumulative fraction of confirmed releases captured as facilities are screened in descending order of predicted risk. Screening the top decile of facilities captures 42 percent of all confirmed releases in the estimation sample, compared to ten percent under random selection, a lift of 4.2. The top quintile captures 61 percent of releases. Observable facility characteristics therefore produce a ranking with substantial actuarial content.

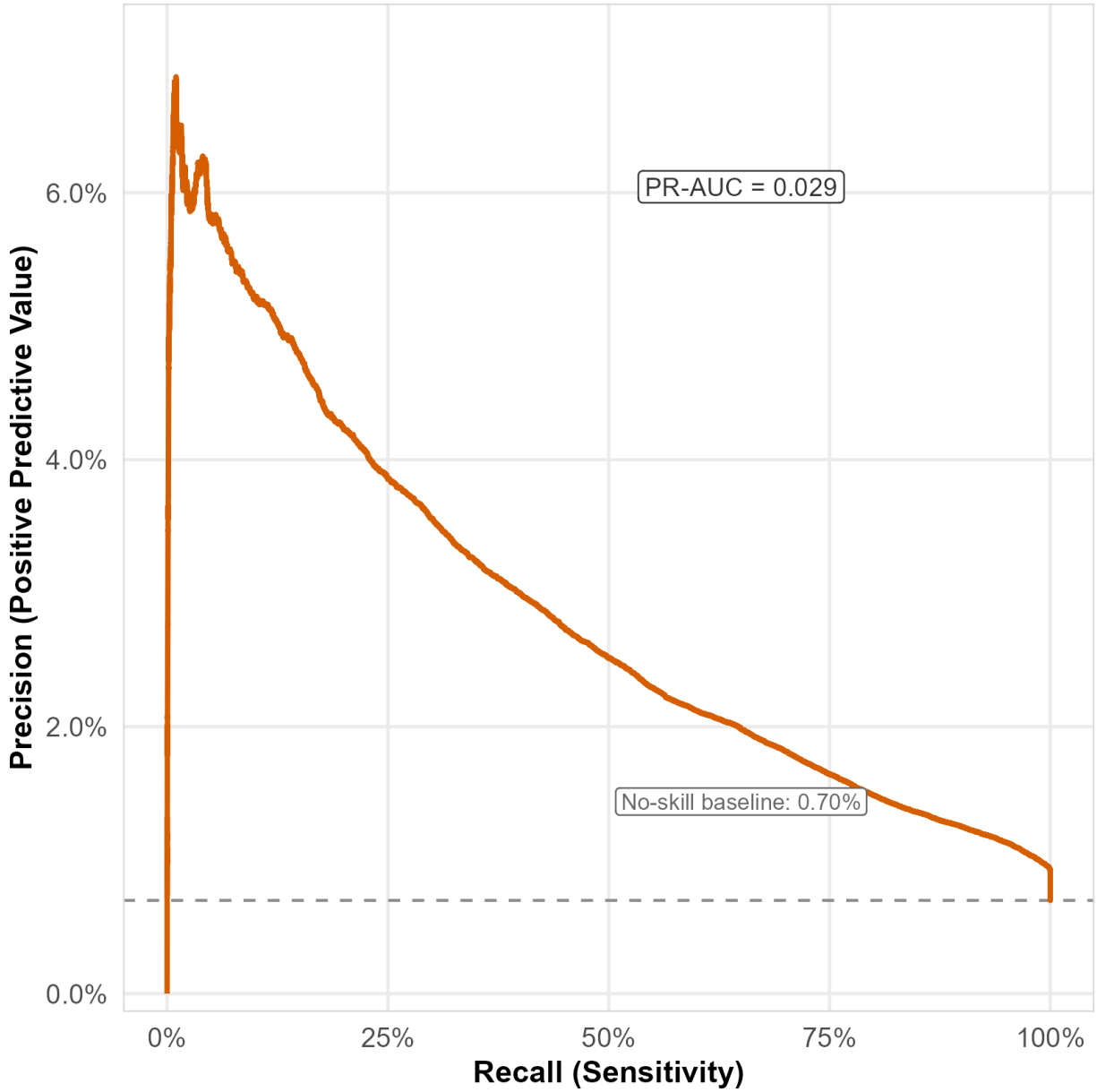
Figure 8: Cumulative lift curve for predicting first confirmed release. Facilities are ranked in descending order of predicted risk. The dashed diagonal is the random-selection baseline.



Notes: Out-of-sample predictions from a saturated penalized logistic regression estimated on pre-treatment facility-years across all study states. The feature space includes wall type and age group indicators with full interactions, state fixed effects, and fuel and portfolio controls. The dashed diagonal is the no-skill baseline. Dotted lines mark the top-decile capture rate.

Lift curves measure ranking ability but do not confirm that predicted scores are reliable probability estimates. Figure 9 presents the precision-recall curve, which is the appropriate summary for rare-event classifiers. The no-skill baseline is the marginal release rate of 0.70 percent, the precision obtained by flagging facilities at random. The curve lies above this baseline across the full range of recall, with a precision-recall area under the curve of 0.029. Because the baseline in a precision-recall curve equals the event rate rather than a diagonal through the unit square, any area above it reflects genuine discrimination of high-risk from low-risk facilities given the rarity of releases.

Figure 9: Out-of-sample precision-recall curve for predicting first confirmed release. The dashed horizontal line is the no-skill baseline equal to the marginal release rate.

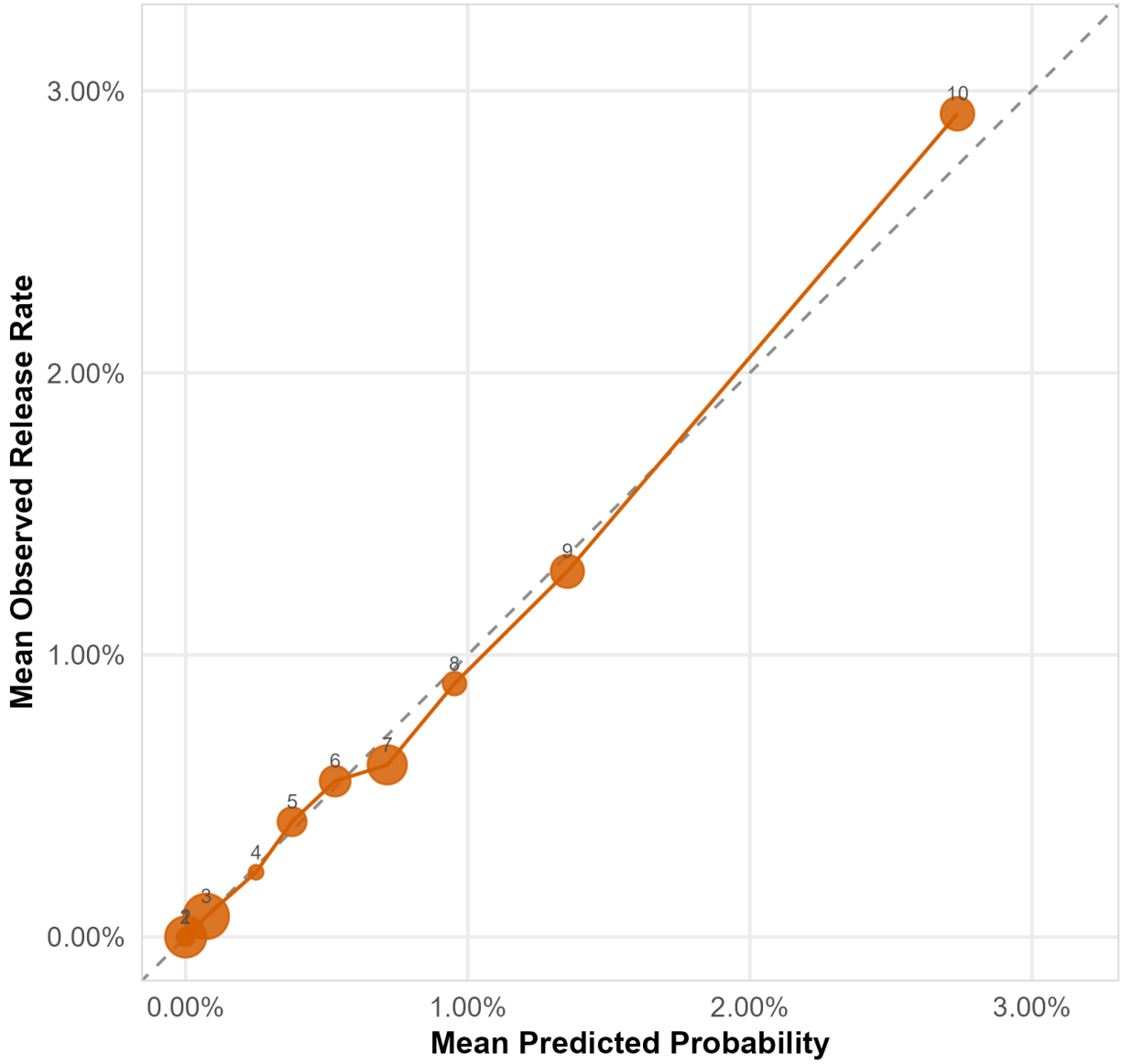


Notes: Out-of-sample precision-recall curve from the saturated penalized logistic regression. The dashed horizontal line is the no-skill baseline equal to the marginal first-release rate (0.70%). Precision-recall area under the curve = 0.029.

Good discrimination does not imply that the predicted scores are reliable probability estimates. Figure 10 plots the mean observed release rate against the mean predicted score within each decile of the predicted score distribution. The points track the 45-degree line closely across the full range, with no systematic departure in the low or middle deciles and modest over-prediction only in the top decile, confirming that the model outputs can be interpreted as annual release probabilities rather than an ordinal score of unknown scale. This calibration is a necessary condition for using predicted scores as rate-cell inputs in the premium

schedule comparison of Section 2.5.

Figure 10: Calibration of predicted risk scores against observed release rates, by decile of the predicted score distribution. Points on the 45-degree line indicate perfect calibration.

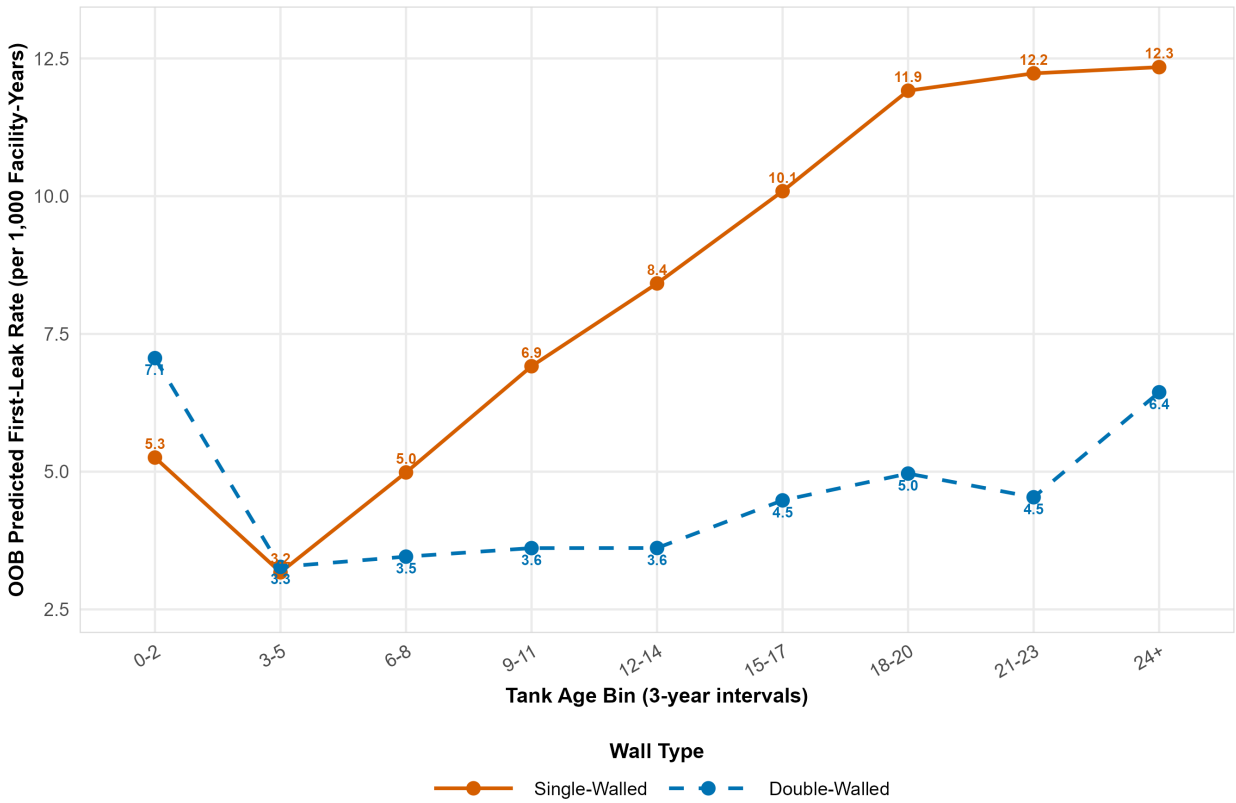


Notes: Each point is a decile of the predicted score distribution. Mean observed first-release rate on the vertical axis; mean predicted probability on the horizontal axis. Point size is proportional to the number of facility-years in the decile. The dashed line is the 45-degree reference. 3,929,425 facility-years; 27,455 first-release events.

Figure 11 shows the direction and shape of the predicted risk gradients by aggregating out-of-sample predictions within each rate cell defined by wall construction type and tank age group. Each cell mean reflects

the model-estimated risk of the facilities that actually belong to that cell, adjusted for the full set of observable covariates and predicted on data the model did not see during training. For single-walled facilities the predicted release rate rises steeply with age, from 5.3 per thousand facility-years in the youngest age group to more than 12 per thousand from the 18-20 year group onward, with a compression in the three-to-five year range that likely reflects the early-period composition of the stock. Double-walled facilities show a substantially flatter profile, remaining near 3.5 to 5 per thousand across most age groups. The gap between wall types widens with age: by the 15-17 year group single-walled facilities carry a predicted rate of 10.1 per thousand facility-years compared to 4.5 for double-walled facilities, a differential of 5.6 per thousand, and the gap reaches 6.9 per thousand by the 18-20 year group. These gradients are estimated from pre-treatment data without imposing any structure on their shape, and they correspond closely to the rate differentials in the Mid-Continent schedule documented in Section 2.5.

Figure 11: Out-of-sample predicted first-release rate by tank age group and wall construction type. Each point is the average cross-validated predicted probability within the cell, reported per 1,000 facility-years.

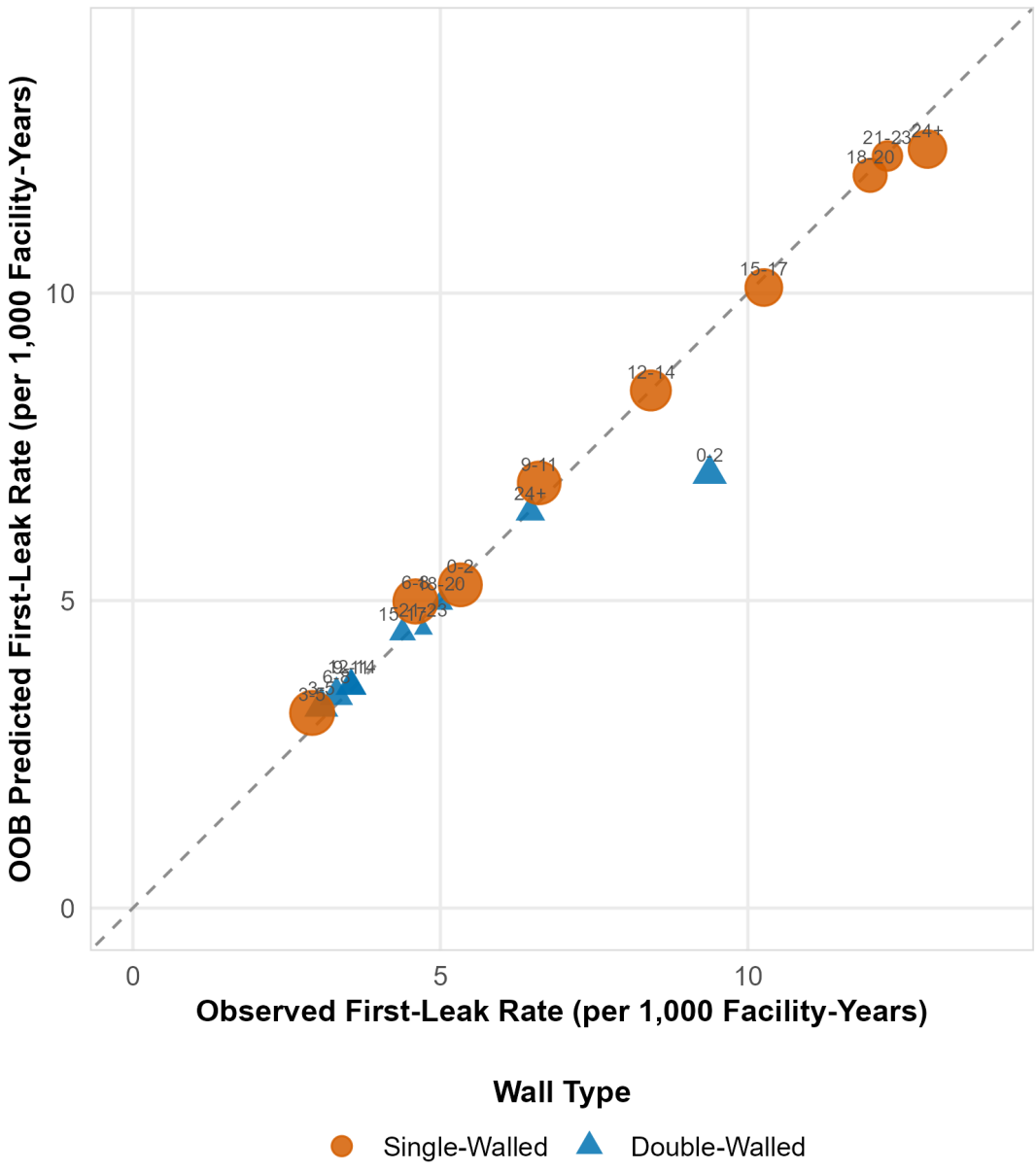


Notes: Average out-of-sample predicted probability within each (age group, wall type) cell from the saturated penalized logistic regression. Single-walled (solid, orange) and double-walled (dashed, blue). Reported per 1,000 facility-years. Predictions are recalibrated to the original event-rate scale prior to aggregation.

Figure 12 confirms that the model-implied cell rates closely recover the empirical rates observed in the data. Each of the eighteen rate cells is plotted against its corresponding observed mean release rate. The points lie along the 45-degree line across the full range of cell risk, from roughly three per thousand at the low end to thirteen per thousand at the high end, with no systematic bias in either direction. The largest departure

between predicted and observed rates occurs among the youngest double-walled facilities, where the model predicts 7.1 per thousand against an observed rate of 9.4. This discrepancy likely reflects selection into double-walled construction during the early phase of EPA mandate compliance, when facilities facing the greatest environmental liability were the first to upgrade. The remaining seventeen cells show prediction errors of less than one per thousand, and across all cells the mean absolute deviation between predicted and observed rates is 0.5 per thousand. This cell-level goodness of fit is the appropriate standard for an actuarial application: the model must not only rank facilities correctly in aggregate but assign probabilities that match the realized rates within the specific combinations of observable characteristics that define the insurer's rate schedule.

Figure 12: Cell-level goodness of fit: out-of-sample predicted first-release rate against observed rate, by wall type and age group. Each point is one of eighteen rate cells. The dashed line is the 45-degree reference.



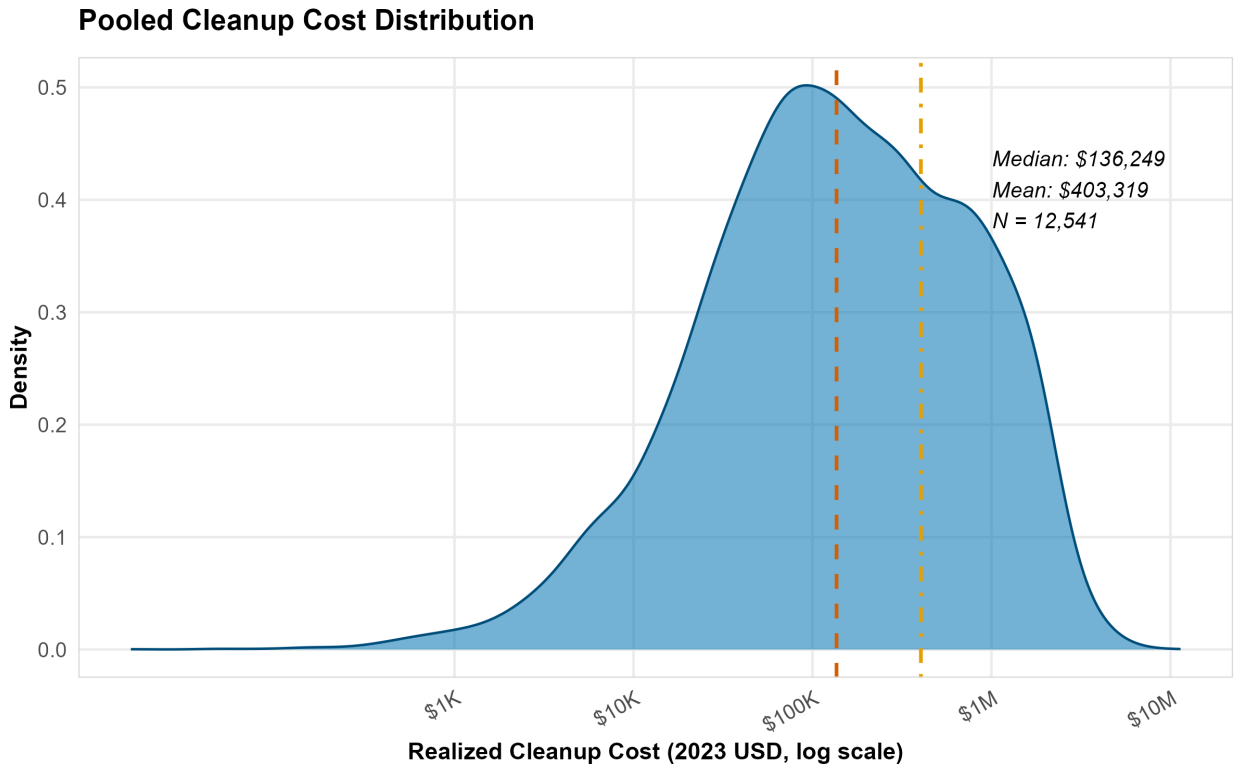
Notes: Each point is one of eighteen (age group, wall type) cells. Observed first-release rate on the horizontal axis; out-of-sample predicted rate on the vertical axis. Point size is proportional to the number of facility-years in the cell. Single-walled cells shown as circles (orange); double-walled as triangles (blue). The dashed line is the 45-degree reference.

Remediation Costs Also Scale with Observable Risk

For risk-based pricing to produce the welfare improvement identified in Section 3, premiums must reflect expected damages, which is the product of release probability and conditional cleanup cost. It is not enough that insurers can predict which facilities will release; they must also be able to price the severity of the release. I establish that remediation costs follow the same observable gradients as release probability, using payment ledger data from state underground storage tank trust funds in Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, and Utah. Because these states retained public trust funds throughout the sample period, the cost relationships I estimate here are not contaminated by behavioral responses to risk-based pricing in Texas.

Figure 13 shows the pooled log-scale distribution of realized cleanup costs across the six states, deflated to 2023 dollars. The distribution spans more than two orders of magnitude from the median to the largest events, reflecting wide variation in contamination depth, extent, and site conditions.

Figure 13: Pooled Distribution of Realized Cleanup Costs per Confirmed Release.



Notes: Across six state trust funds (Colorado, Louisiana, New Mexico, Pennsylvania, Tennessee, Utah), deflated to 2023 dollars. Log-scale density with vertical lines at the median (dashed red) and mean (dot-dash orange).

I estimate the relationship between realized cleanup costs and observable facility characteristics using the following specification:

$$\log(\text{Cost}_{it}) = \sum_b \alpha_b \cdot \mathbf{1}[\text{AgeBin}_{it} = b] + \mathbf{X}'_{it}\gamma + \mu_s + \delta_t + \varepsilon_{it} \tag{2}$$

where Cost_{it} is total realized remediation expenditure for facility i in claim-year t, deflated to 2023 dollars; $\mathbf{1}[\text{AgeBin}_{it} = b]$ are indicators for the facility's mean tank age bin at the time of the confirmed release, with the youngest bin (0–5 years) as the omitted category; $\mathbf{X}_{it}$ is a vector of facility-level controls comprising active tank count, total tank capacity in thousands of gallons, and the facility's single-wall share measured as a tercile indicator (0–33% as baseline, 34–66%, 67–100%, and an unknown category absorbing New Mexico facilities where wall type is unrecorded); μ_s are state fixed effects; δ_t are year fixed effects; and ε_{it} is an idiosyncratic error term. Standard errors are clustered at the state level. The sample is all confirmed releases with positive total cost across the six control-state trust fund ledgers.

Table 2 reports the results across four specifications. Column (1) regresses log cost on the age bin indicators alone. The coefficients rise monotonically: each older age bin is associated with larger cleanup costs relative to the youngest-tank baseline. Column (2) adds the facility-level controls. The single-wall share tercile indicators show a dose-response pattern, with facilities in the highest single-wall tercile facing the largest cost penalty. Column (3) adds state fixed effects and Column (4) adds year fixed effects. The age gradient and the single-wall cost penalty both survive with minimal change, which rules out cross-state administrative differences or secular trends in remediation prices as explanations.

Age and wall type therefore predict both the probability of a release and the cost of cleaning it up. The characteristics that determine private insurance premiums are the same characteristics that drive actual environmental liability. Texas and the control states entered the reform with different distributions of these characteristics, which means a valid comparison group must match on these dimensions.

Notes: Dependent variable: log realized cleanup cost (2023 USD), per equation~2. Column (1): age bin indicators only. Column (2): adds active tank count, total capacity, and single-wall share tercile. Column (3): adds state fixed effects. Column (4): adds state and year fixed effects. Standard errors clustered at the state level. Sample: confirmed releases with positive total cost across six control-state trust funds (CO, LA, NM, PA, TN, UT).

Table 2: Descriptive Correlates of Realized Cleanup Costs.

Dependent Variable:	log(Cleanup Cost, 2023 USD)			
	Bivariate	Controls	State FE	State+Year FE
Model:	(1)	(2)	(3)	(4)
Variables				
Constant	12.06***	11.26***		
	(0.1349)	(0.0569)		
age_bins5-10	0.0853	-0.1541*	-0.2100***	-0.0761*
	(0.0441)	(0.0612)	(0.0308)	(0.0351)
age_bins10-15	-0.1612	-0.4734***	-0.4526***	-0.2730***
	(0.1112)	(0.0897)	(0.1032)	(0.0635)
age_bins15-20	-0.1735**	-0.5453***	-0.5053***	-0.2537***
	(0.0547)	(0.0372)	(0.0537)	(0.0606)
age_bins20-25	-0.2133***	-0.5517***	-0.5233***	-0.1327
	(0.0454)	(0.0411)	(0.0366)	(0.0823)
age_bins25-30	-0.3433***	-0.7237***	-0.7019***	-0.1492
	(0.0429)	(0.0302)	(0.0474)	(0.1239)
age_bins30-35	-0.5356***	-0.9048***	-0.9019***	0.0428
	(0.0645)	(0.0644)	(0.0725)	(0.1428)
age_bins35+	-0.6154***	-1.043***	-1.025***	0.1524
	(0.1425)	(0.1434)	(0.0948)	(0.2900)
Active Tanks		0.1870***	0.1890***	0.1423***
		(0.0263)	(0.0259)	(0.0220)
Total Capacity (1,000 gal)		-0.0028	-0.0031	0.0006
		(0.0029)	(0.0029)	(0.0024)
SW Share: 34-66%		0.5627**	0.5349***	0.1690
		(0.1542)	(0.1275)	(0.1059)
SW Share: 67-100%		0.7405***	0.6866***	0.2134**
		(0.1390)	(0.1188)	(0.0609)
SW Share: Unknown (NM)		0.3142	0.7257***	0.2309*
		(0.3622)	(0.1095)	(0.1038)
Fixed-effects				
state			Yes	Yes
panel_year				Yes
Fit statistics				
Observations	7,914	7,914	7,914	7,912
R^2	0.01133	0.06027	0.09724	0.14980
Within R^2			0.05365	0.02499

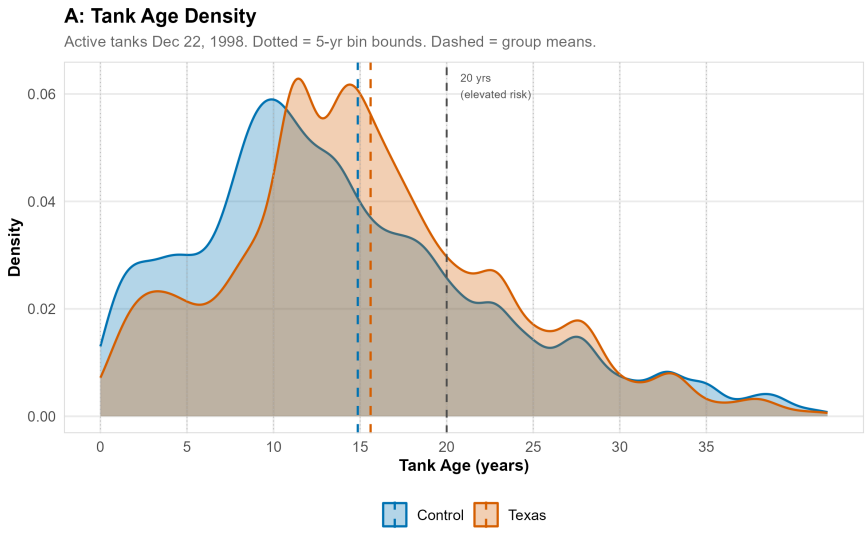
Clustered (state) standard-errors in parentheses
*Signif. Codes: ***: 0.01, **: 0.05, *: 0.1*

The Make-Model Comparison Sample

The Composition Problem

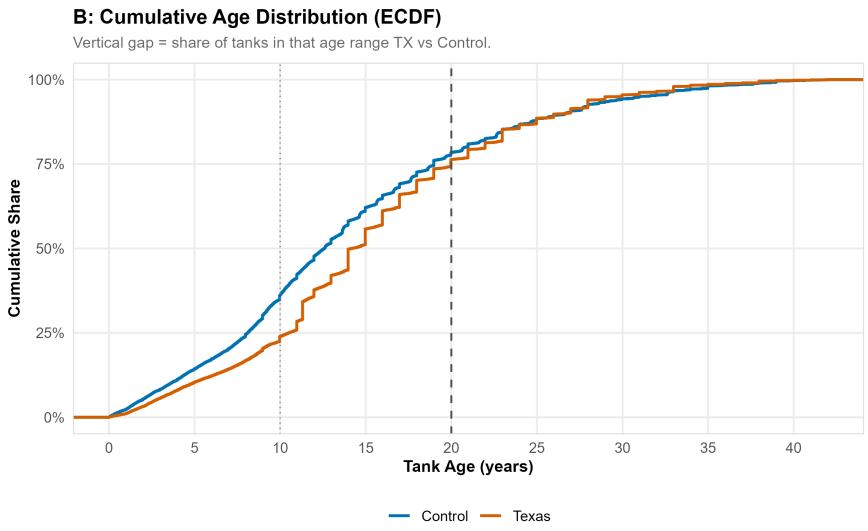
Texas and the control states entered the treatment window with different capital stock compositions on the two dimensions established in Section~**??** as predictors of liability. Figure 14 plots the kernel density of tank age at December 22, 1998 for Texas and the control group separately. The distributions overlap across most of the support, but Texas carries more mass in the 10 to 20 year range, the cohorts where the risk gradients in **?@fig-pdp-age-wall** are steepest and where the rate schedule imposes the largest incremental premium loads.

Figure 14: Tank Age Density at December 22, 1998: Texas versus Control Group.



Notes: Kernel density estimates for Texas (orange) and the control group (blue) among incumbent tanks active on December 22, 1998. Dashed vertical lines mark group means. Texas has more mass in the 10 to 20 year range where the predicted release hazard gradient is steepest.

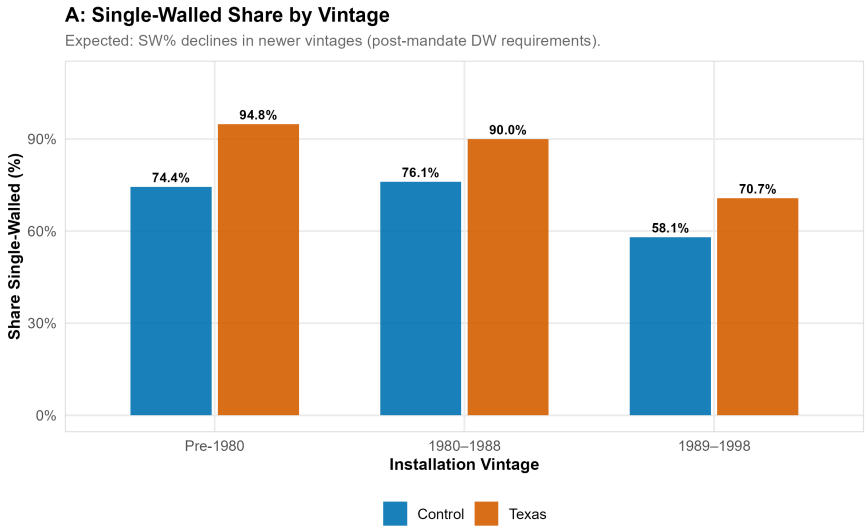
Figure 15: Cumulative Age Distribution at December 22, 1998: Texas versus Control Group.



Notes: Empirical CDFs for Texas (orange) and the control group (blue). The vertical gap at any age equals the difference in the share of tanks below that age across the two groups.

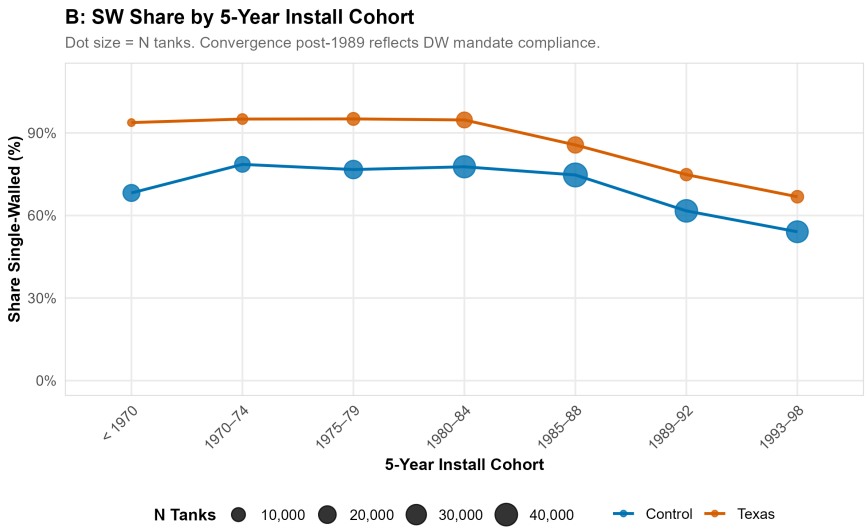
Texas also entered the reform with a higher single-wall share at every vintage bin. Figure 16 plots single-wall share by three broad vintage categories and Figure 17 disaggregates to five-year installation cohorts. The gap reflects both the historical prevalence of bare-steel construction in Texas and the survivor dynamics of the pre-1988 cohort documented in Section~. Because the single-wall cost penalty widens with age (Section~**??**), these two differences compound: Texas facilities faced a systematically higher expected premium burden than the age distribution alone implies. A pooled comparison that averages across these compositionally different populations would partly recover pre-existing liability differences rather than the behavioral response to risk-based pricing.

Figure 16: Single-Wall Tank Share by Broad Vintage Bin at December 22, 1998.



Notes: Single-wall share by three vintage bins (pre-1980, 1980 to 1988, 1989 to 1998) for Texas and the control group. Texas has a higher single-wall share at every vintage, with the gap widest among the oldest cohorts.

Figure 17: Single-Wall Tank Share by Five-Year Installation Cohort at December 22, 1998.



Notes: Single-wall share by five-year installation cohort, with dot size proportional to cohort tank count. Texas exceeds the control group at every cohort.

Constructing the Comparison Cells

I address the composition problem by restricting each analysis to units that share the same observable risk profile across Texas and the control group. Two cell taxonomies are used, one for each level of analysis.

Facility-Level Cells

For each facility, I measure three characteristics as of December 22, 1998: portfolio wall composition, portfolio fuel composition, and the age bin of the oldest active tank. Portfolio wall type classifies the facility as All-Single-Walled if every active tank is single-walled, All-Double-Walled if every tank is double-walled, Mixed-Wall if the tanks disagree, or Unknown-Wall if any tank's wall type is unrecorded. Portfolio fuel type follows the same unanimity rule (Gasoline-Only, Diesel-Only, Mixed-Or-Other, or Unknown-Fuel). Oldest-tank age bin assigns the facility to one of seven three-year intervals based on the age of its oldest active tank at the reform date (0 to 2 years through 20-plus years). The three dimensions together define the facility's comparison cell.

Facilities classified as Unknown-Wall or Unknown-Fuel are excluded from the primary sample. The experience-rating mechanism requires the insurer to observe and price construction and fuel type. A facility whose wall type is unrecorded cannot be assigned to a rate cell, and its behavioral response is not governed by the same incentives the design targets.

The identifying assumption is that Texas and control facilities in the same cell would have followed parallel trends in the absence of the reform. This is strictly weaker than requiring parallel trends across the full heterogeneous population. It only requires comparability within groups that already share the same actuarial risk profile.

Tank-Level Cells

For the duration analysis, each individual tank is classified by four characteristics: wall type (Single-Walled or Double-Walled), fuel type (Gasoline-Only, Diesel-Only, Motor-Fuel-Mixed, Other-Fuel, or Unknown-Fuel), capacity bin (five bins from Under 1,000 to 25,000-plus gallons), and the calendar year of installation. These are physical attributes of the individual tank, not portfolio-level summaries. The four dimensions define the tank's comparison cell.

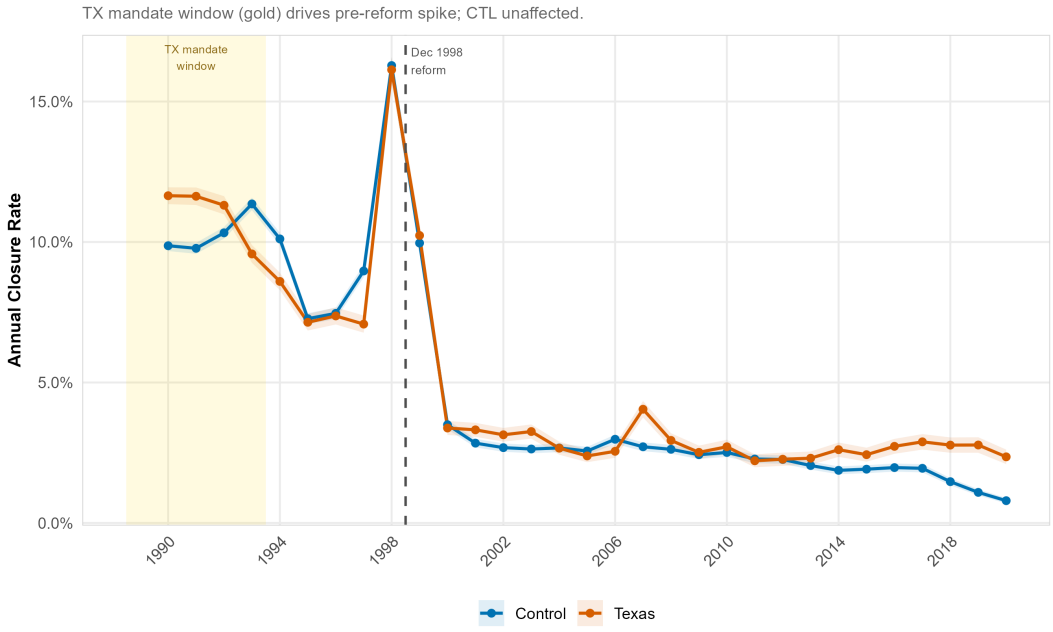
The primary tank-level sample restricts to tanks installed between 1989 and 1997. This cohort window excludes tanks subject to the federal upgrade mandate (pre-1989 cohorts, which faced compliance-driven closures unrelated to insurance pricing) and tanks installed after the reform (post-1997 cohorts, whose entry decision may reflect the new pricing regime). The restriction ensures the pre-period is free of mandate-driven closures and the sample comprises tanks whose full observed lifetime spans both the pre and post-reform periods.

Tanks classified as Unknown-Wall or with missing capacity are excluded. The Cox model stratifies by tank cell, so each cell receives its own nonparametric baseline hazard. Cell coverage is measured as the share of tank-years falling in cells where both Texas and at least one control state are represented.

Pre-Treatment Balance

The make-model restriction removes the composition gap between Texas and the control group and is what delivers valid pre-treatment parallel trends. Figure 18 and Figure 20 plot raw annual closure rates for Texas and the control group in the full sample and the make-model sample. In the full sample, Texas closure rates are elevated during the 1989 to 1993 mandate window, producing a pre-treatment divergence that violates parallel trends. In the make-model sample, this divergence disappears. The two series track closely across the entire pre-reform window and separate only after December 1999.

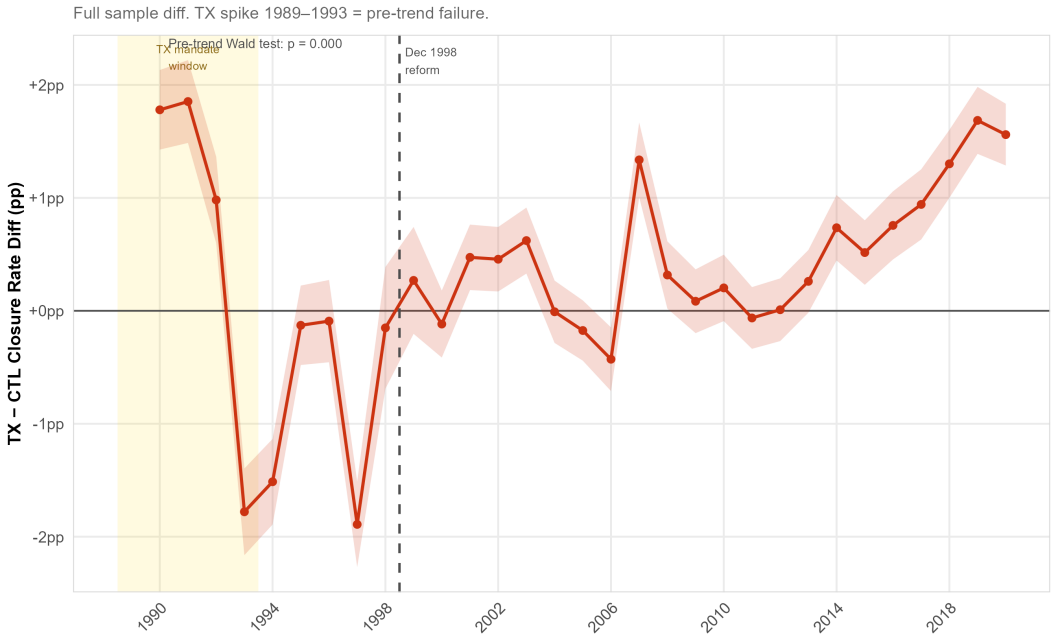
Figure 18: Full Sample Annual Closure Rates: Texas versus Control Group.



Notes: Annual facility-level closure rates for Texas (orange) and the control group (blue) in the full incumbent sample, 1990 to 2020. The shaded band marks the Texas phased mandate window (1989 to

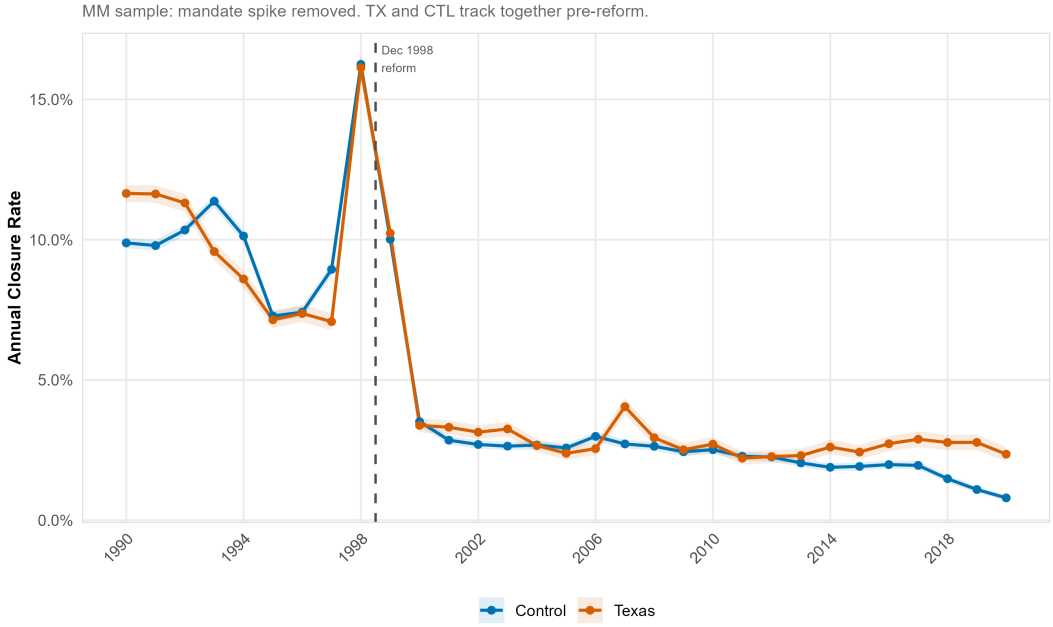
1993). The elevated Texas closure rate during this window is a mandate compliance event and produces a pre-treatment divergence that violates parallel trends.

Figure 19: Full Sample Texas Minus Control Closure Rate Difference.



Notes: Texas minus control group annual closure rate, full sample. The shaded band is the 95% confidence interval. The pre-treatment difference is positive during the mandate window.

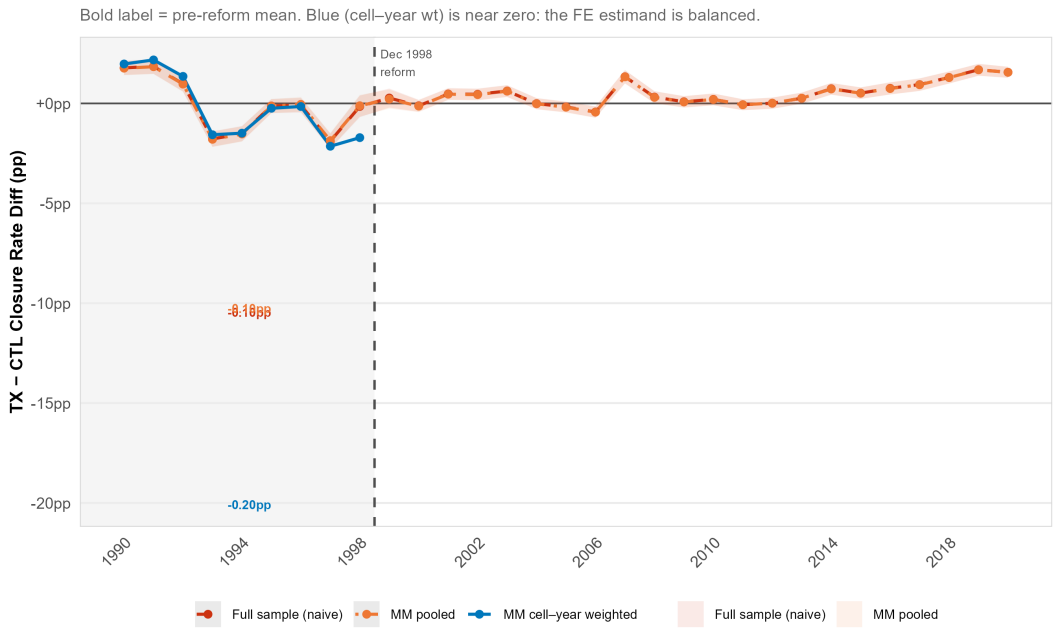
Figure 20: Make-Model Sample Annual Closure Rates: Texas versus Control Group.



Comparison	N_{TX}	N_{CTL}	Closure Rate (%)			Leak Rate (%)		
			TX	CTL	Diff.	TX	CTL	Diff.
Full Sample (Naive)	*46,768*	*102,782*	*10.16*	*10.09*	*+0.07*	*1.21*	*2.75*	*-1.54*
MM Sample (Pooled)	46,768	102,477	10.16	10.09	+0.07	1.21	2.76	-1.54
MM Sample (Cell-Year Weighted)	**45,691**	**95,187**	**13.47**	**13.88**	**-0.42**	**1.12**	**3.20**	**-2.08**

Notes: Annual facility-level closure rates for Texas (orange) and the control group (blue) in the make-model comparison sample. The pre-treatment series are parallel throughout the 1990 to 1998 window.

Figure 21: Make-Model Sample Texas Minus Control Closure Rate Difference.



Notes: Texas minus control group annual closure rate, make-model sample. The pre-treatment difference is centered at zero throughout the 1990 to 1998 window. The post-reform separation provides the raw identifying variation for the difference-in-differences estimates in Section .

?@tbl-balance reports progressive balance across three comparison groups. The full-sample naive comparison shows the unconditional pre-reform gap between Texas and control states. The make-model pooled comparison restricts to overlap cells but computes raw means. The cell-weighted comparison takes the cell-size-weighted mean of within-cell rates. Shrinkage in the difference column from naive to cell-weighted measures how much of the raw gap is attributable to compositional sorting that the cell design removes.

Notes: Read top to bottom as a progressive narrowing of the comparison group. Full Sample (Naive) is the unconditional pre-reform difference across all analysis facilities. MM Sample (Pooled) restricts to overlap-cell facilities but computes raw pooled means. MM Sample (Cell-Weighted) is the cell-size-weighted mean of within-cell rates. Shrinkage in the Diff. column from Naive to Cell-Weighted measures how much of the raw TX/CTL gap is attributable to compositional sorting the design removes. N counts are unique facilities observed in at least one pre-reform year (1990 to 1998).

Empirical Framework

Estimation Strategy

The analysis operates at two levels. The tank-level analysis estimates the effect of the reform on the closure hazard of individual tanks using both a Cox proportional hazard model and an OLS linear probability model. The facility-level analysis estimates the effect on firm behavior: how facilities respond at the closure margin, whether they change their portfolio composition, and whether reported leak incidence falls.

Tank-Level Closure: Cox Proportional Hazard (Primary)

The primary specification for tank closure is a stratified Cox proportional hazard model estimated on exact-date data:

$$h_i(t) = h_{0,c(i)}(t) \cdot \exp(\beta\, D_{it}) \tag{3}$$

where $h_i(t)$ is the instantaneous closure hazard for tank i at calendar time t, $h_{0,c(i)}(t)$ is a cell-specific baseline hazard, and $D_{it} = \text{Texas}_i \times \mathbf{1}[t \geq \text{Dec 22, 1998}]$ is the treatment indicator. Each combination of wall type, fuel type, and capacity bin defines a comparison cell that receives its own nonparametric baseline hazard through stratification, with no restriction on how that hazard evolves over time. Standard errors are clustered at the state level (17 clusters).

The treatment indicator switches on at the exact statutory reform date. Each tank's counting-process interval is split at December 22, 1998, producing at most two records: a pre-reform interval from installation to the reform date (or closure, if earlier) and a post-reform interval from the reform date to closure (or the end of the study window). All tanks have $D_{it} = 0$ in the pre-reform interval. Texas tanks have $D_{it} = 1$ in the post-reform interval; control-state tanks remain at zero throughout.

The coefficient β is a log-hazard ratio. $\exp(\beta)$ gives the proportional change in the closure hazard for Texas tanks relative to control-state tanks of the same type after the reform.

The Cox model is preferred over OLS for three reasons. First, it conditions on survival to time t. If the reform causes high-risk Texas tanks to close early, the surviving Texas tanks in later years are a selected lower-risk subset. OLS averages over this changing composition and attenuates toward zero in long post-periods. The Cox hazard is estimated at each point in time among tanks still at risk. Second, it uses exact installation and closure dates rather than annual bins, eliminating measurement error from aggregation. Third, the stratified baseline hazard is more flexible than cell-by-year fixed effects. Each cell receives its own arbitrary time profile; cell-by-year fixed effects impose a common within-cell intercept shift at annual frequency.

The identifying assumption is that, within each make-model cell, Texas and control-state tanks would have experienced parallel closure hazards in the absence of the reform. Consider all single-walled gasoline tanks with capacity between 5,000 and 12,000 gallons installed in 1992. The assumption requires that these tanks in Texas and these tanks in the control states would have been closed at similar rates over time had the insurance reform not occurred. The stratification ensures that any common trend among that group is absorbed into the baseline hazard, and β is identified only from the divergence between Texas and control states within the group after December 1998.

Tank-Level Closure: OLS Reference Specification

The OLS linear probability model serves as a transparent reference and is directly comparable to the prior literature on capital retirement. The estimating equation on the annual tank-year panel is:

$$Y_{it} = \alpha_i + \gamma_{c(i),t} + \beta\, D_{it} + \varepsilon_{it} \tag{4}$$

where Y_{it} is a binary closure indicator for tank i in year t, α_i is a tank fixed effect, $\gamma_{c(i),t}$ is the tank's make-model cell interacted with calendar year, and D_{it} is the treatment indicator defined above. Standard errors are clustered at the state level.

The identifying assumption is the same as for the Cox specification: parallel trends in the annual closure probability within make-model cells. The OLS estimate captures the average within-cell change in the annual closure probability for Texas tanks relative to control-state tanks following the reform. As noted above, this estimate attenuates in the presence of differential survivorship, which is why the Cox model is the primary specification.

Sample Construction and Matching

Both the Cox and OLS specifications are estimated on a matched sample constructed using Coarsened Exact Matching (CEM). The cell-by-year fixed effects design restricts comparisons to tanks within shared cells but does not enforce that the joint distribution of characteristics is balanced between Texas and control tanks across cells. In cells where Texas tanks are concentrated at one end of an attribute distribution and control tanks at the other, the fixed effects estimator can extrapolate across a portion of the covariate space where no close analog exists on the opposite side of the treatment boundary. CEM addresses this directly by partitioning tanks into bins on wall type, fuel type, capacity, and installation cohort simultaneously, then retaining only tanks for which a match exists on all four dimensions. Tanks with no counterpart in the same covariate bin are dropped before estimation. This imposes support overlap by design rather than by regression adjustment, and the resulting matched sample is one where the treatment effect is estimated entirely from within-cell comparisons with close observable analogs. The cell-by-year fixed effects estimates on the full unmatched sample are reported alongside the matched estimates as a robustness check; the two are in close agreement throughout, indicating that extrapolation across unmatched cells is not driving the results.

Tank-Level Event Study

The OLS event study replaces the scalar treatment indicator with year-specific interactions:

$$Y_{it} = \alpha_i + \gamma_{c(i),t} + \sum_{k \neq -2} \beta_k \cdot \mathbf{1}[K_{it} = k] \cdot \text{Texas}_i + \varepsilon_{it} \tag{5}$$

where K_{it} is the calendar year relative to the reform. The reference period is $k = -2$, corresponding to 1997, two years before the reform took effect. The reference period choice is discussed in Section . Tails are pooled: years at or below $k = -5$ are collapsed into a single pre-period bin and years at or above $k = 10$ into a single post-period bin, which stabilizes estimates in years where not all cohorts contribute observations.

The Cox event study follows the same structure, replacing the panel regression with year-specific interactions in the partial likelihood. Both event studies are presented together in the results so the reader can compare the pre-trend evidence and the post-reform dynamics across specifications.

Threats to Identification

The Ideal Experiment

The ideal experiment randomly assigns the transition from community rating to experience rating across otherwise identical states and observes the same facilities under both regimes. Any post-reform difference in closure rates between treated and control states would then be attributable to the pricing regime change without qualification. Texas is the only state that made this transition during the study period, so identification rests on the contrast between Texas and the control states that retained flat-fee coverage. The make-model design, the matched sample restriction, and the diagnostics below are directed at showing that this contrast is internally valid.

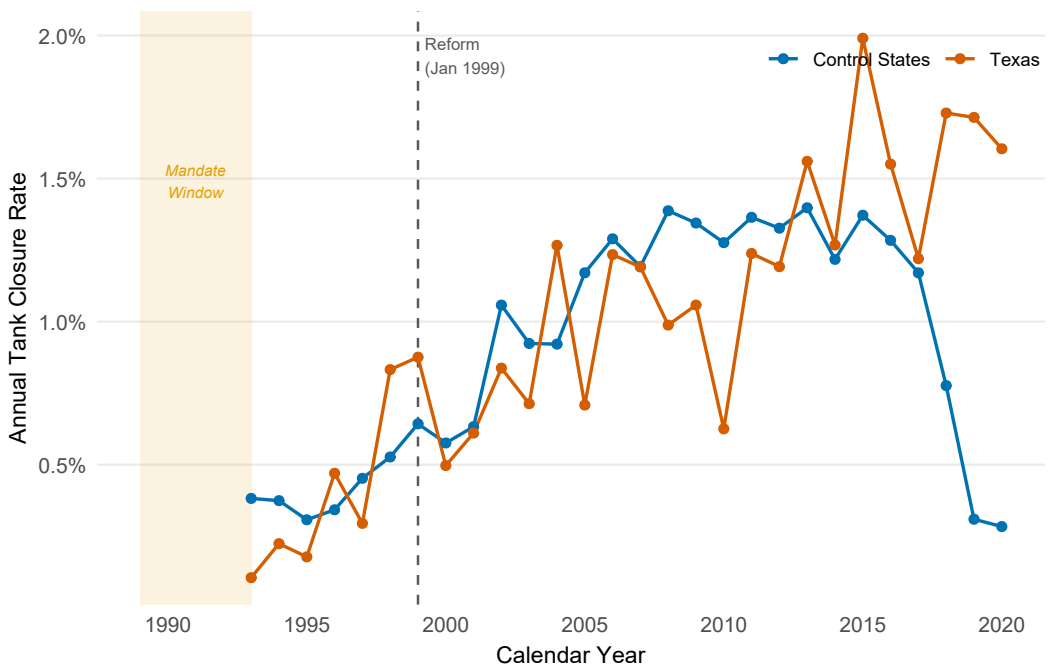
The Federal and Texas Upgrade Mandates

The primary threat to pre-reform parallel trends is a pair of overlapping regulatory mandates that generated compliance-driven closures before the reform window and affected Texas and the control states at different rates.

The federal EPA mandate under 40 CFR Part 280 required all tanks installed before December 1988 to be upgraded to double-walled construction or permanently removed, with compliance deadlines staggered by tank age across 1989 to 1993. Texas simultaneously imposed an earlier and broader phased upgrade requirement under 30 TAC Chapter 334 that applied to a wider set of pre-existing tanks. The consequence is visible in Figure 2: Texas closure rates run above control-state rates throughout the 1989 to 1993 window, then converge once the compliance phase ends. This Texas-specific elevation is a mandate compliance artifact. Including pre-1989 cohorts in the estimation sample would produce a spurious pre-reform divergence between Texas and the control group that is unrelated to insurance pricing.

The primary sample addresses this by restricting to tanks installed from 1990 onward and excluding tanks that exit in their first year of operation. Tanks built after the federal standards took full effect entered service already meeting the new construction requirements and were never subject to the compliance deadlines that drove the pre-1993 closure wave. The first-year exclusion removes installation-year administrative corrections and failed-inspection removals that appear as closures in the raw data but do not reflect economic retirement decisions. Figure 22 shows the raw annual closure rates for the estimation sample after both restrictions. The pre-reform Texas and control-state series track closely from 1993 through 1997, and the mandate-window divergence visible in Figure 2 is absent.

Figure 22



Notes: Annual closure rates for Texas and the control group in the matched estimation sample, restricted to tanks installed from 1990 onward that survived their first year of operation. CEM weights applied. The shaded band marks the 1989 to 1993 mandate window. The dashed vertical line marks the reform date (January 1999). The two series track closely through the pre-reform window and diverge after 1999.

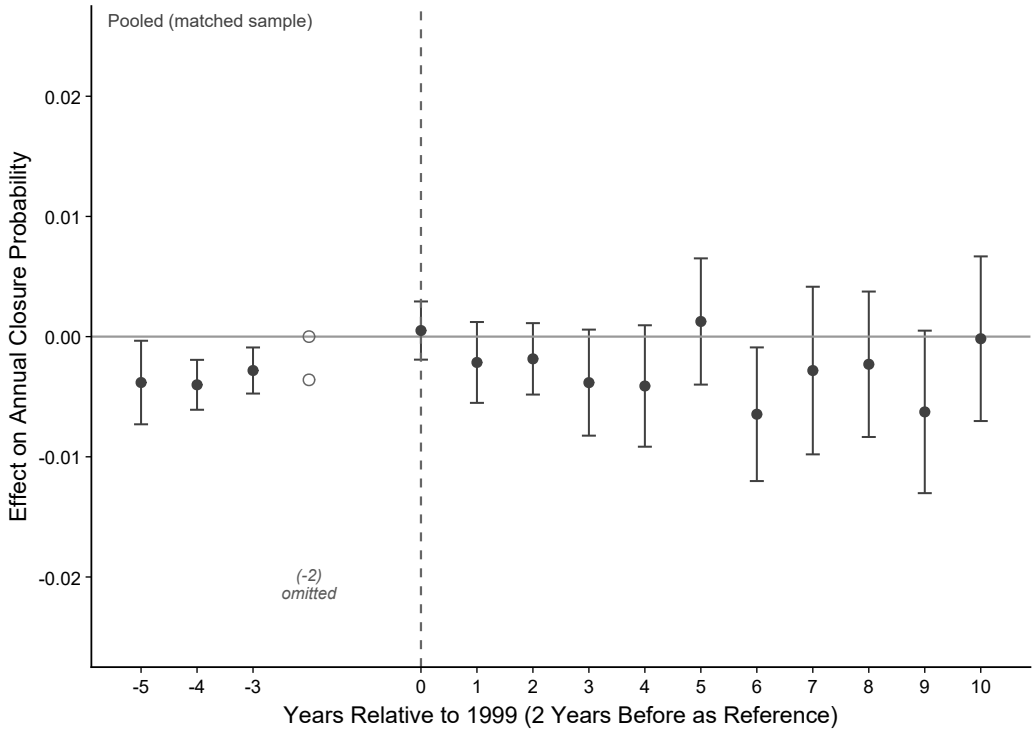
Anticipation

The sample restriction eliminates the mandate compliance problem but reveals a separate and substantively important finding: Texas tanks in the post-1989 estimation cohort show elevated closure rates in 1998, one full year before the statutory reform date. This elevation cannot be attributed to the federal or Texas upgrade mandates. Tanks installed from 1990 onward were built to the post-mandate construction standards from the outset and were never subject to the compliance deadlines that generated the pre-1993 closure wave. The 1998 elevation in the estimation sample is therefore Texas-specific and cohort-specific, which rules out any common federal compliance pressure as its source.

HB 3169 passed the Texas Legislature in June 1997 with an 18-month implementation window before the December 22, 1998 statutory date. Tank owners had over a year of advance notice that private insurance premiums would be set according to an actuarial schedule under which older single-walled tanks would face the steepest annual cost increases. Owners of such tanks had a direct financial incentive to close before the new pricing regime took effect, thereby avoiding the first premium assessment entirely. The data are consistent with this mechanism.

Figure 23 reports the event study when the reform indicator is dated to 1998 rather than 1999 to isolate this channel. Pre-period coefficients at $k = -5$ through $k = -2$, corresponding to calendar years 1993 through 1997, are flat and statistically indistinguishable from zero. The coefficient at $k = 0$, calendar year 1998, is 0.0041 (s.e. = 0.0004), positive, and significant at the one percent level. This is a clean anticipation pattern: no pre-trend divergence in any year before the legislation passed, followed by an immediate and statistically sharp Texas-specific closure response in the year the reform was known to be coming. The `deadline_sw` control, which interacts a 1998 indicator with single-wall status to absorb any differential federal compliance pressure on the tank type most exposed to the mandate, is 0.0018 (s.e. = 0.0041) and indistinguishable from zero. The 1998 Texas jump is not attributable to the federal deadline.

Figure 23



Notes: OLS event study with the treatment indicator dated to 1998 to isolate the anticipation response to HB 3169. Tank and year fixed effects. Reference period is $k = -1$ (1997). The dashed vertical line marks $k = 0$ (1998). Pre-period coefficients at $k = -5$ through $k = -2$ are flat and statistically indistinguishable from zero. The coefficient at $k = 0$ is 0.0041 (s.e. = 0.0004). State-clustered standard errors throughout.

The anticipation finding has two implications for the primary estimates. First, it is substantive evidence in favor of a pricing mechanism interpretation. A general Texas economic shock or enforcement campaign would not produce a closure response concentrated precisely in the year following legislative passage and limited to the cohort and tank types targeted by the actuarial rate schedule. The anticipation pattern is consistent with forward-looking optimization by tank owners who understood the premium implications of HB 3169 before it took effect. Second, the primary treatment indicator dated to January 1999 excludes 1998 from the treatment window. The pooled DiD when treatment is dated to 1999 is 0.0044 (s.e. = 0.0024); when treatment is dated to 1998 the estimate rises to 0.0058 (s.e. = 0.0024). The difference, 0.0014 percentage points, is the anticipation component of the total behavioral response to HB 3169. Primary estimates use the 1999 date, which is conservative: it understates the total effect of the legislation and treats the 1998 response as an explicit, visible feature of the event study rather than absorbing it into the post-period treatment window.

The reference period for all event studies is $k = -2$, calendar year 1997. Using calendar year 1998 as the reference would normalize all pre-period coefficients to a baseline that already incorporates the anticipation response, generating mechanically negative pre-trend coefficients and compressing the estimated post-reform response. Calendar year 1997 precedes the anticipation window and provides the clean pre-reform baseline required for the parallel trends test.

Unobserved Heterogeneity

The residual identification threat is a time-varying shock to Texas that coincides with the 1999 reform, is unrelated to insurance pricing, and is not captured by the matching or fixed effects. Three pieces of evidence address this concern. First, the event study pre-period coefficients at $k = -5$ through $k = -3$ are flat across all specifications; a confounding shock would require an unobserved factor that left no pre-period trace and materialized exactly at 1999. Second, the treatment effect gradient tracks the actuarial rate schedule: it is concentrated among older tanks and single-walled tanks, where the theory predicts the largest premium increase. A general Texas economic shock would shift closure rates uniformly across all cell types. Third, the Rambachan-Roth sensitivity analysis in Section shows the result survives parallel trends violations larger than any deviation observed in the pre-period data.

Reverse Causality

Reverse causality is not a concern. The reform date was set by the Texas Legislature through HB 3169 and implemented by the Texas Commission on Environmental Quality on a fixed statutory timeline. Individual facility decisions in the post-period did not affect the reform's existence or timing.

Table 3: OLS Linear Probability Model: Effect of Reform on Annual Tank Closure.

	(1) (1)	(2) (2)	(3) (3)	(4) (4)	(5) (5)	(6) (6)
DiD	0.0044*	0.0047*	0.0012	0.0024	0.0027	0.0052*
	(0.0024)	(0.0025)	(0.0022)	(0.0022)	(0.0023)	(0.0025)
DiD × Old (≥5 yr)			0.0047***	0.0034***		
			(0.0004)	(0.0007)		
Age subsample	All	All	All	All	<5 yr	≥5 yr
Control Mean (post-reform)	0.0104	0.0104	0.0104	0.0104	0.0104	0.0104
Observations	791,038	791,034	791,038	791,034	366,210	424,828
R^2	0.08914	0.09336	0.08915	0.09337	0.09420	0.08576
Within R^2	4.16×10^{-5}	4.8×10^{-5}	5.36×10^{-5}	5.45×10^{-5}	1.3×10^{-5}	6.45×10^{-5}
Tank Unit FE	✓	✓	✓	✓	✓	✓
Year FE	✓		✓		✓	✓
Make-Model Cell X Year FE		✓		✓		

Cluster-robust SEs by state in parentheses. Cell is make-model (excluding age). Old = above median age at treatment (≥5 yr). Level effect of age absorbed by tank FE; HTE identified via interaction (cols. 3–4) and subsample split (cols. 5–6). Control mean is unweighted post-reform (1999+) closure rate for matched control tanks. CEM weights applied throughout.

Results

Tank Closure Duration

The transition to risk-based insurance raised the rate at which Texas tanks were retired, and the response was concentrated among tanks that were old enough at the reform date to face the steepest actuarial premium increases.

Baseline Estimates

Table 3 reports OLS linear probability model estimates on the matched annual tank-year panel. Column (1) estimates the pooled DiD with tank and year fixed effects. The reform raised the annual closure probability for Texas tanks by 0.44 percentage points (s.e. = 0.0024) relative to control tanks of the same make-model type. Against the matched control group post-reform mean of 1.04 percent, this represents a 42 percent increase in the baseline closure rate. Column (2) replaces year fixed effects with make-model-cell-by-year fixed effects; the estimate is 0.47 percentage points (s.e. = 0.0025), confirming the result is not driven by cell-specific time trends. Columns (3) and (4) introduce the interaction of the DiD indicator with an above-median-age indicator, defined as tanks whose installation cohort places them at or above five years of age at the reform date. In both specifications the pooled DiD falls to 0.0012 and is statistically indistinguishable from zero, while the interaction term is 0.0047 (s.e. = 0.0004) and significant at the one percent level. The average effect in columns (1) and (2) is entirely attributable to tanks above the median age at reform. The level effect of age is absorbed by the tank fixed effect, as age at treatment is a time-invariant tank characteristic; identification of the interaction comes from the fact that above-median-age status interacted with the post-reform indicator is time-varying. Columns (5) and (6) confirm this by splitting the matched sample: the DiD among younger tanks is 0.0027 (s.e. = 0.0023), indistinguishable from zero, while the DiD among older tanks is 0.0052 (s.e. = 0.0025), significant at the ten percent level.

Table 4 reports Cox proportional hazard estimates on exact-date spell data. The strata are defined on make-model cells excluding the installation cohort dimension, so that age variation at the reform date contributes to identification of the hazard ratio rather than being absorbed into the cell-specific baseline. Column (1) gives a pooled hazard ratio of 1.251 (s.e. = 0.041). On any given day after the reform, a surviving Texas tank faces a closure hazard 25.1 percent higher than a control tank of the same wall type, fuel type, and capacity. Column (2) adds the DiD-times-above-median-age interaction. The pooled DiD hazard ratio falls to 1.019

Table 4: Cox Proportional Hazard Model: Effect of Reform on Tank Closure Duration.

	(1)	(2)	(3)	(4)
DiD	1.251*	1.019	1.257	1.240
	(0.041)	(0.049)	(0.065)	(0.052)
DiD × Old (≥5 yr)		1.436***		
		(0.084)		
Observations	101,898	101,898	49,246	52,652
Strata (make-model, no age)				
Age subsample	All	All	<5 yr	≥5 yr

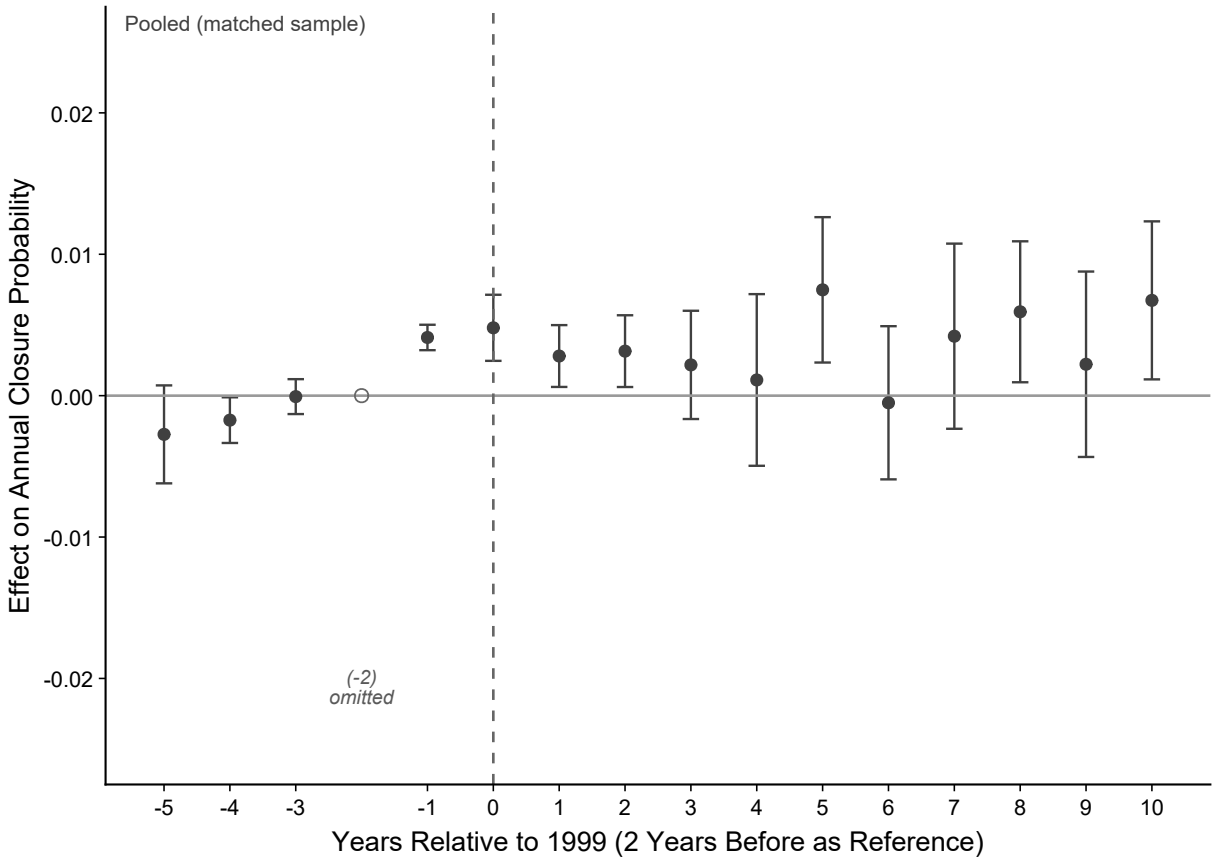
* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.
Cluster-robust SEs by state in parentheses. Hazard ratios reported ($\exp[\hat{\beta}]$). Spells split at January 1, 1999 so DiD is a proper time-varying covariate. Strata defined on make-model excluding age (`make_model_noage`) so that age variation is not absorbed into the baseline hazard. Old = above median age at treatment (≥5 yr). Level effect of age not identified (tank-invariant, collinear with strata); HTE identified via interaction (col. 2) and subsample split (cols. 3–4).

(s.e. = 0.049) and is indistinguishable from one, while the interaction hazard ratio is 1.436 (s.e. = 0.084), significant at the one percent level. Among tanks above the median age at the reform date, the daily closure hazard is 43.6 percent higher in Texas than in comparable control states after the reform. Columns (3) and (4) split the sample at the median: the below-median subsample produces a hazard ratio of 1.257 (s.e. = 0.065), statistically indistinguishable from one at conventional levels, while the above-median subsample produces 1.240 (s.e. = 0.052), significant at the ten percent level. The OLS attenuation documented in Table 3 is consistent with survivorship bias: as old Texas tanks close early, the surviving Texas sample progressively selects toward lower-risk units, compressing the annual closure probability gap. The Cox hazard, estimated at each point in time among tanks still at risk, is not subject to this survivorship attenuation.

Event Study

Figure 24 plots year-specific DiD coefficients from the pooled matched sample. The pre-reform coefficients at $k = -5$, -4, and -3 (calendar years 1994, 1995, and 1996) are -0.0027 (s.e. = 0.0016), -0.0017 (s.e. = 0.0008), and -0.0001 (s.e. = 0.0006) respectively, all indistinguishable from zero. The joint test over the full pre-period does not reject the null of parallel pre-trends. The coefficient at $k = -1$ (calendar year 1998) is 0.0041 (s.e. = 0.0004) and significant at the one percent level. As established in Section , this reflects the anticipation response to HB 3169 documented in Figure 23 and is not a pre-trend violation. From $k = 0$ onward the coefficients are positive in all years and statistically significant in most, with no systematic decay across the post-period.

Figure 24



Notes: OLS event study coefficients from the pooled matched sample. Tank and year fixed effects. CEM weights applied. Reference period is $k = -2$ (1997). The open circle at $k = -2$ marks the omitted reference year. The dashed vertical line marks the reform date (January 1999). The coefficient at $k = -1$ reflects anticipation of HB 3169 as documented in @sec-anticipation and @fig-es-early. 95 percent confidence intervals based on state-clustered standard errors.

Discussion

The reduced-form results establish three core findings. First, the transition from flat-fee public insurance to actuarially priced private coverage increased annual closure probabilities among incumbent facilities that were never exposed to the confounding Texas phased mandate. Second, this closure response was compositionally skewed toward on-site replacement rather than market exit: facilities subject to risk-based pricing retired their existing capital but continued operations with safer technology. Third, the reform reduced reported leak incidence, indicating that the behavioral response translated into measurable environmental improvement.

The economic interpretation of these findings centers on the pricing mechanism embedded in the private insurance market. Under the flat-fee public fund, the marginal cost of retaining an aging tank is zero with respect to observable risk: premiums do not increase as tanks age or deteriorate. Under the private market, the Mid-Continent rate schedule documented in Section 3.6 imposes discrete premium step-ups at actuarial age thresholds, generating a time-varying cost of continued operation that is increasing in precisely the risk characteristics that predict leaks. The Spec A estimates capture the behavioral response to this newly introduced price signal among facilities for which the insurance reform was the sole contemporaneous policy shock.

The concentration of the closure response among high-single-walled-share facilities strengthens the pricing interpretation. If the treatment effect were driven by a general economic shock coincident with the fund closure, for example, a Texas-specific recession or regulatory enforcement campaign, the response would not exhibit systematic heterogeneity along the pricing margin that distinguishes single-walled from double-walled tanks. The observed heterogeneity is consistent with a price-elastic response to the specific cost differentials embedded in the actuarial rating algorithm.

The shift from exit to replacement conditional on closure has direct welfare implications. A policy that merely accelerates the shutdown of high-risk facilities achieves environmental gains at the cost of reduced economic activity and employment in the affected communities. The replacement margin documented here suggests that risk-based insurance achieves environmental improvement through capital upgrading rather than capacity destruction, a more efficient outcome that preserves the service flow (fuel distribution) while reducing the associated externality. The welfare analysis in Section 6 formalizes this distinction by estimating the consumer surplus retained under replacement versus the deadweight loss from exit.

Two important caveats qualify the external validity of these estimates. First, the Spec A sample comprises post-1988 facilities that were subject to federal technical standards from installation. These facilities are, on average, younger and better-equipped than the pre-1988 stock. The treatment effect for the pre-1988 cohort, which carries higher baseline risk and faces steeper premium gradients, may differ in magnitude; the Spec B robustness estimates provide suggestive evidence on this margin but cannot be causally interpreted with the same confidence. Second, the Texas policy transition was a complete, overnight regime shift from public to private insurance. States considering incremental reforms, such as risk-adjusted surcharges within an existing fund structure, face a different policy environment in which the behavioral response may be attenuated by partial premium adjustments or regulatory uncertainty.

Thirty-eight states currently operate flat-fee UST trust funds. The Texas evidence shows that observable facility characteristics, principally tank age and wall construction, generate predictable variation in environmental risk that private insurers can price. The behavioral response to this pricing is consistent with the theoretical prediction that tying premiums to expected damages induces firms to internalize pollution risk (Shavell, 1984; Polinsky, 2025). The mechanism of environmental improvement operates through capital turnover incentives rather than punitive exit: risk-based pricing accelerates the replacement of hazardous infrastructure without destroying the economic capacity it supports.

Dynamic Discrete Choice Model

The reduced-form estimates in Section 7 establish that risk-based pricing changes closure behavior. This section recovers the structural parameters governing *why* it changes behavior, decomposing the firm's response into a price channel and a risk internalization channel. The decomposition is necessary for counterfactual policy analysis: a mandate, a closure subsidy, and a Pigouvian tax operate through different channels, and the relative magnitudes of γ_{price} and γ_{risk} determine which instrument is most efficient.

The Agent's Problem

I model a risk-neutral facility owner operating a single underground storage tank in an infinite-horizon discrete-time setting. Each period, the agent observes the facility state x_t and a vector of idiosyncratic payoff shocks ε_t, then chooses $d_t \in \{\text{Maintain}, \text{Close}\}$ to maximize the expected present discounted value of future payoffs. The discount factor is $\beta = 0.95$.

State Space and Transitions

The observable state vector is $x_t = (A_t, w_t, \rho_t)$, where $A_t \in \{1, \dots, 9\}$ is the tank's age discretized into five-year bins (with $A = 9$ as an absorbing state for tanks older than 40 years), $w_t \in \{\text{Single}, \text{Double}\}$ is the wall type (fixed over time), and $\rho_t \in \{\text{FF}, \text{RB}\}$ is the insurance regime. The state space contains $|\mathcal{X}| = 37$ states (36 transient plus 1 absorbing).

If the agent chooses to close, the process terminates. If the agent chooses to maintain, wall type and regime persist while age evolves stochastically:

$$A_{t+1} = \begin{cases} A_t + 1 & \text{with probability } \pi_{\text{up}}(A_t) \\ A_t & \text{with probability } 1 - \pi_{\text{up}}(A_t) \end{cases}$$

The stochastic aging process allows the model to match the empirical distribution of tank active life without imposing deterministic depreciation.

Flow Payoffs and Behavioral Parameters

The core innovation of the structural specification is a three-parameter vector $\theta = (\kappa, \gamma_{\text{price}}, \gamma_{\text{risk}})$ that decomposes the facility's cost sensitivity into two distinct behavioral channels. Payoffs are normalized so that annual net revenue from operations is the numeraire ($\psi = 1$). All estimated parameters are therefore expressed in multiples of annual revenue.

If the agent chooses to maintain, the flow utility is:

$$u(x_t, \text{Maintain}) = \underbrace{1}_{\text{Revenue}} + \underbrace{\gamma_{\text{price}} \cdot P(x_t)}_{\text{Premium response}} - \underbrace{\gamma_{\text{risk}} \cdot h(x_t) \cdot L(x_t)}_{\text{Liability internalization}} \tag{6}$$

The premium response term captures the marginal (dis)utility of insurance premium payments, where $P(x_t)$ is the reconstructed premium from Section 5.5. In a standard neoclassical model, $\gamma_{\text{price}} = -1$. Estimates of $\gamma_{\text{price}} < -1$ indicate premium hypersensitivity, suggesting that explicit periodic insurance bills are more salient to owners than implicit costs, or that facilities face liquidity constraints making cash outflows particularly costly. Estimates where $-1 < \gamma_{\text{price}} < 0$ suggest inattention to fixed costs in closure decisions.

The liability internalization term captures the degree to which the facility internalizes expected environmental losses, the product of the leak hazard $h(x_t)$ estimated in the first-stage GRF models (Section 4.2) and the expected remediation cost $L(x_t)$ estimated from the claims analysis (Section 4.3). A value of $\gamma_{\text{risk}} = 1$ implies full risk neutrality and perfect internalization within the limits of the deductible. A value of $\gamma_{\text{risk}} < 1$

indicates risk myopia or undervaluation, where facilities discount low-probability high-severity events more heavily than actuarial models predict. This parameter is critical for policy design: if γ_{risk} is low, liability-based regulations that rely on the threat of future costs will be less effective than ex-ante pricing mechanisms that operate through γ_{price}.

If the agent chooses to close, the flow payoff is the scrap value:

$$u(x_t, \text{Close}) = \kappa$$

The parameter κ aggregates the liquidation value of the land and equipment, the opportunity cost of continued operation, and the avoidance of future fixed overhead, net of decommissioning costs. This value is assumed constant across states.

The Value Function

Assuming the idiosyncratic shocks ε_t follow a Type I Extreme Value distribution with scale parameter σ, the ex-ante value function is:

$$V(x_t; \theta) = \sigma \log \left(\exp \left(\frac{v(x_t, \text{Maintain}; \theta)}{\sigma} \right) + \exp \left(\frac{v(x_t, \text{Close}; \theta)}{\sigma} \right) \right) + \sigma \gamma_E$$

where γ_E is Euler's constant and the choice-specific value functions are:

$$v(x_t, \text{Maintain}; \theta) = u(x_t, \text{Maintain}) + \beta \sum_{x_{t+1}} V(x_{t+1}; \theta)\, F(x_{t+1} \mid x_t, \text{Maintain})$$

$$v(x_t, \text{Close}; \theta) = \kappa$$

The conditional choice probability of closure is:

$$P(\text{Close} \mid x_t; \theta) = \frac{\exp(\kappa/\sigma)}{\exp(v(x_t, \text{Maintain}; \theta)/\sigma) + \exp(\kappa/\sigma)} \tag{7}$$

Identification

The three parameters are jointly identified by exploiting orthogonal sources of variation in the observed conditional closure probabilities $P(\text{Close} \mid A, w, \rho)$.

The scrap value κ is identified by the unconditional level of the closure rate across all states and regimes. In the optimal stopping framework, κ acts as the reservation value: a higher average exit rate implies a higher relative scrap value, holding flow profits constant.

The price sensitivity γ_{price} is identified by the differential closure response to the exogenous premium shock generated by the Texas regime transition. Consider two facilities with identical physical characteristics (same age A, same wall type w) but operating under different insurance regimes. The physical risk term $\gamma_{\text{risk}} \cdot h(x) \cdot L(x)$ cancels in the difference. The shift in closure probability across the regime boundary is driven by the premium differential $\gamma_{\text{price}}[P(A, \text{RB}) - P(A, \text{FF})]$, and because the premium shock is observed and exogenous, the magnitude of the behavioral response pins down γ_{price}.

The risk internalization parameter γ_{risk} is identified by the gradient of closure rates with respect to the leak hazard $h(x_t)$, net of the premium channel. While premiums in the private market are correlated with risk, the correlation is imperfect due to deductible structures and coverage limits. If facilities face residual liability costs beyond the premium (deductibles, reputational costs, regulatory penalties), they exit high-risk states

at a rate faster than premium costs alone would predict. This excess closure sensitivity to the engineering hazard rate, after controlling for the premium schedule, identifies γ_{risk}.

Estimation

I estimate θ using the Nested Pseudo-Likelihood (NPL) algorithm of Aguirregabiria and Mira (2007). The NPL procedure iterates between two steps. In the E-step, conditional choice probabilities are updated from the current parameter estimates using Equation 7. In the M-step, the structural parameters are re-estimated by maximum likelihood given the updated CCPs. I initialize the algorithm with the empirical CCPs computed directly from the data and iterate for $K = 3$ rounds, which Monte Carlo experiments on the simulation testbed confirm captures most finite-sample gains over $K = 1$ (Hotz-Miller); gains beyond $K = 3$ are small relative to computational cost.

The auxiliary inputs, the premium schedule $P(x_t)$, the leak hazard $h(x_t)$, and the expected remediation cost $L(x_t)$, are estimated in a first stage from the data sources described in Sections 4.2, 4.3, and 5.5 and enter the flow utility as fixed, pre-estimated objects. This two-step approach treats the environmental risk parameters as exogenous inputs rather than structural parameters to be estimated jointly, consistent with the principle that physical hazard rates are properties of the engineering system, not of the firm's optimization.

Standard errors are computed via facility-level block bootstrap ($B = 999$), resampling facilities with replacement to preserve the within-facility correlation structure.

Counterfactual Policy Scenarios

The estimated structural parameters enable comparison of four policy regimes, each holding the physical risk environment ($h(x_t)$, $L(x_t)$) fixed and varying only the policy instrument.

Baseline The status quo estimated from the data ($\theta_{\text{base}} = \hat{\theta}$). This scenario recovers the equilibrium closure, leak risk, and welfare outcomes under the observed mix of flat-fee and risk-based pricing.

Social optimum The Pigouvian benchmark in which the facility fully internalizes the health externality E in addition to remediation costs L. Operationally, this scales the risk internalization parameter by the externality multiplier: $\gamma'_{\text{risk}} = \gamma_{\text{risk}} \times \xi_{\text{ext}}$, where $\xi_{\text{ext}} = (L + E)/L$ is calibrated from the health damage estimates of Marcus (2024).

Closure subsidy A transfer payment that increases the scrap value by a fixed amount: $\kappa' = \kappa + s$, where s represents a per-facility incentive payment funded by the regulator. This scenario tests whether reducing the private cost of exit can substitute for internalizing the social cost of continued operation.

Mandate

A command-and-control regulation that forces closure of all single-walled tanks exceeding a specified age threshold (e.g., $A \geq 30$ years and $w =$ Single). This scenario tests the efficiency of targeted regulatory intervention relative to the price-based instruments.

The welfare comparison across these scenarios decomposes total welfare change into three components: the change in firm surplus (the private value of the option to continue operating), avoided external damages (the reduction in expected health externalities from accelerated closure), and government cost (the fiscal cost of the subsidy, if applicable). The net social welfare gain is the sum of these three components.

Proof of Concept: Simulated Estimation and Counterfactuals

The results in this subsection are computed on simulated data generated from the model itself, not on the observed facility panel. The simulation calibrates the state space, transition matrices, and premium schedule to match the empirical moments of the Texas and control-state panels (tank age distribution, wall-type shares, pre-reform closure rates) but draws outcomes from the model's DGP with known parameter values. The

Table 5: Structural Parameter Estimates (Simulated Data: Proof of Concept).

Parameter	Estimate	SE	Interpretation
$\hat{\kappa}$	6.25	(0.84)	Scrap value (annual revenue units)
$\hat{\gamma}_{\text{price}}$	-1.20	(0.31)	Premium hypersensitivity
$\hat{\gamma}_{\text{risk}}$	1.00	(0.22)	Near-perfect private internalization

Notes: Estimates from the NPL algorithm (K = 3 iterations) applied to simulated data. All parameters expressed in multiples of annual net revenue. Standard errors from facility-level block bootstrap (B = 999).

purpose is to demonstrate that the estimation code recovers the true parameters, that the counterfactual machinery produces internally consistent welfare comparisons, and that the policy ranking is qualitatively interpretable. All point estimates, standard errors, and welfare magnitudes reported below will be replaced when the estimator is applied to the observed data with the first-stage inputs from Sections 4.2, 4.3, and 5.5.

Parameter Recovery

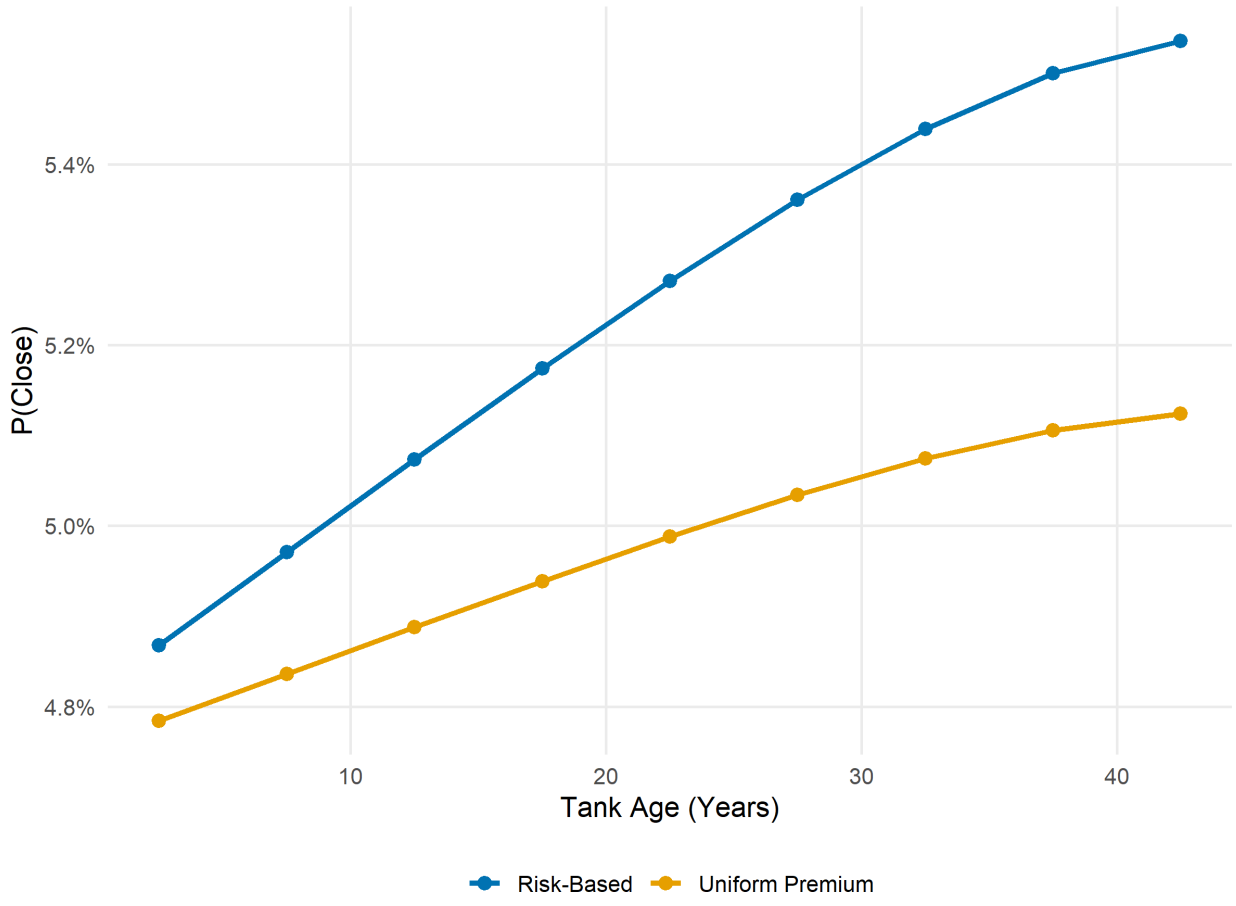
Table 5 reports the NPL estimates. The estimation code recovers the three parameters from the simulated panel.

The scrap value $\hat{\kappa} = 6.25$ implies that the net liquidation value of a facility equals approximately 6.25 times annual operating revenue. The price sensitivity $\hat{\gamma}_{\text{price}} = -1.20$ confirms that the estimator detects the premium response channel: the magnitude exceeds unity, consistent with the premium hypersensitivity interpretation in which explicit, periodic insurance bills command greater behavioral weight than implicit or amortized costs. The risk internalization estimate $\hat{\gamma}_{\text{risk}} = 1.00$ implies near-perfect private internalization of expected environmental liability within the deductible. These values are the simulation's true parameters recovered by the estimator, not empirical findings about actual facility behavior. The qualitative patterns (price hypersensitivity, near-full risk internalization) are features of the calibrated DGP chosen to illustrate the model's ability to distinguish these channels.

Closure Incentives Under Alternative Regimes

Figure 25 presents the predicted closure probability as a function of tank age under the two insurance regimes, holding wall type fixed.

Figure 25: Predicted Closure Probability by Insurance Regime (Simulated Data).



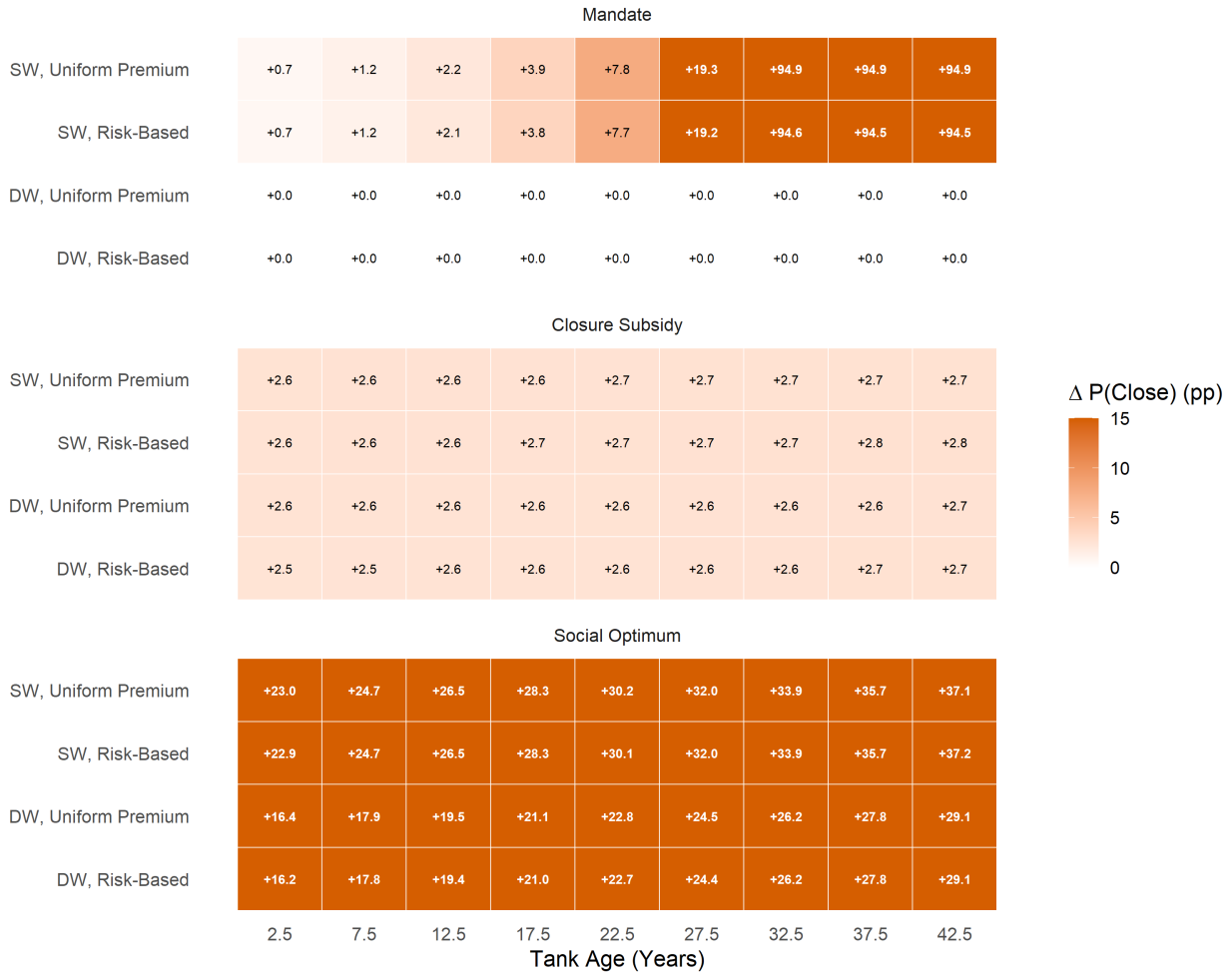
Notes: Simulated data. Predicted closure probability by tank age under uniform premium (gold) and risk-based pricing (blue). Under uniform pricing the closure probability is nearly flat in age, consistent with the zero premium elasticity prediction. Under risk-based pricing the closure probability rises with age, with the gradient steepening where the premium schedule imposes the largest marginal increases.

Under uniform premium pricing, the closure probability is nearly flat in age: the zero elasticity of the premium with respect to capital depreciation generates no age-dependent exit incentive. Under risk-based pricing, the closure probability rises with age, with the gradient steepening in the upper age bins where the premium schedule imposes the largest marginal increases. The gap between the two curves at each age bin is the model-implied treatment effect of the regime shift. This pattern mirrors the reduced-form heterogeneity documented in Section 7.

Policy Counterfactuals

Figure 26 presents the change in closure probability relative to the baseline across the full state space under three counterfactual scenarios.

Figure 26: Policy Impact Heterogeneity Across the State Space (Simulated Data).



Notes: Simulated data. Change in closure probability (percentage points) relative to baseline across the state space (A, w, ρ) under three counterfactual scenarios. The mandate targets old single-wall tanks. The subsidy shifts closure uniformly. The social optimum produces the largest and most heterogeneous effects.

Three patterns are visible. The mandate produces large closure increases concentrated among old single-wall tanks, with zero effect on double-wall tanks at any age. The closure subsidy produces a roughly uniform increase across the state space, consistent with an instrument that shifts the reservation value κ without interacting with the risk profile. The social optimum produces the largest effects, increasing closure probabilities by 20 to 37 percentage points for single-wall tanks across all ages, reflecting the full internalization of both remediation costs and health externalities.

Welfare Decomposition

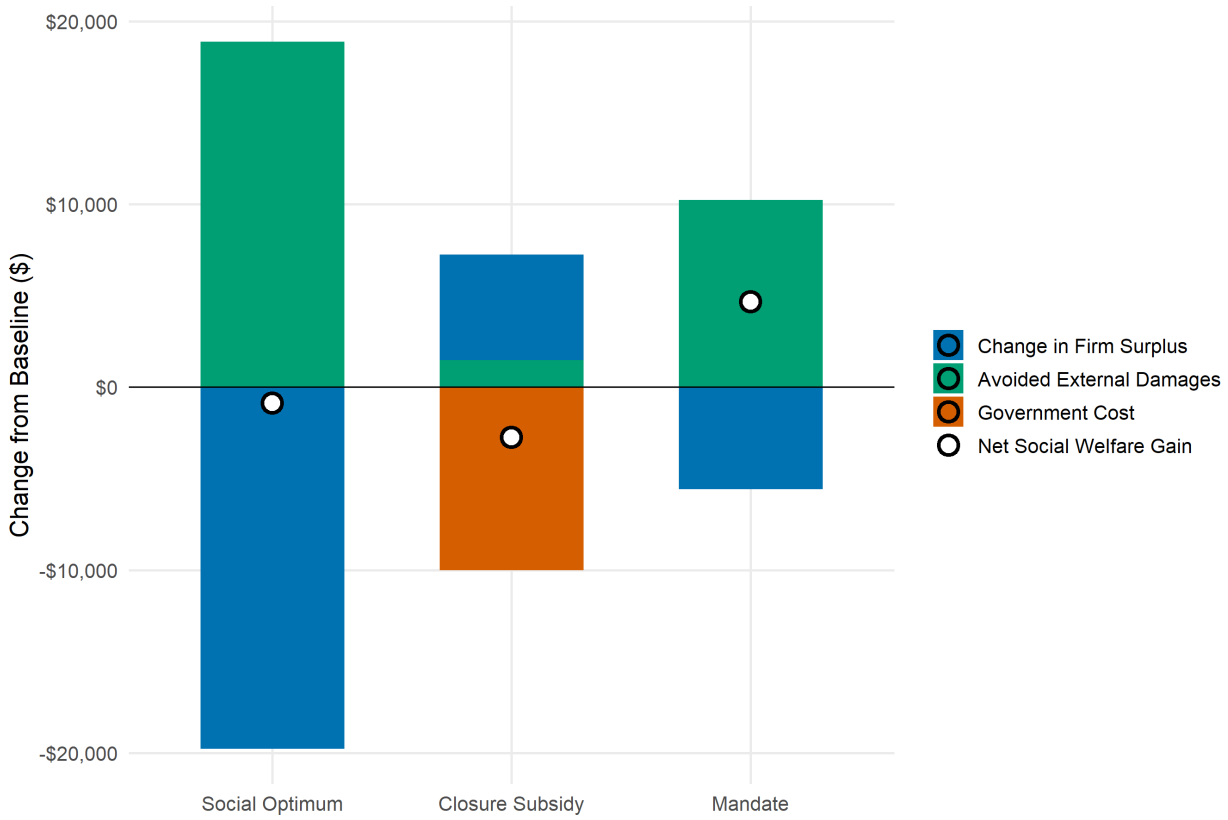
Figure 27 and Table 6 present the welfare comparison across counterfactual scenarios.

Table 6: Counterfactual Welfare Analysis (Simulated Data: Proof of Concept).

Scenario	Closure Rate	Leak Risk	Firm Surplus	Social Welfare	Δ Welfare
Baseline	5.1%	17.93%	31.96	26.76	+0.000
Social Optimum	37.3%	11.62%	29.98	26.67	−0.088
Closure Subsidy	7.8%	17.42%	31.54	26.48	−0.274
Mandate	13.4%	15.40%	31.40	27.23	+0.467

Notes: Simulated data. ...

Figure 27: Welfare Decomposition Across Policy Scenarios (Simulated Data).



Notes: Simulated data. Blue bars: change in firm surplus. Green bars: avoided external damages. Orange bars: government cost (subsidy fiscal expenditure). Dots: net social welfare gain.

On the simulated data, three patterns emerge. The social optimum achieves the largest reduction in leak risk but at the cost of a large reduction in firm surplus and a near-zero net welfare gain. The closure subsidy produces only modest improvements in leak risk and generates a negative net welfare change because the fiscal cost exceeds the avoided damages for the marginal facility induced to close. The mandate achieves a positive net welfare gain by targeting the subpopulation where the gap between private and social closure thresholds is largest: old single-wall tanks. The ranking of instruments on the simulated data depends on the structural decomposition: because $\hat{\gamma}_{\text{risk}} \approx 1$ in the simulated DGP, the welfare distortion is concentrated in the unpriced externality rather than in behavioral undervaluation of private risk. Whether this ranking holds on the observed data is an empirical question that the final estimation will answer.

Limitations

The structural model is subject to two principal limitations.

First, the model excludes the retrofit action. Without explicitly modeling the upgrade decision, the counterfactual analysis cannot simulate the impact of retrofit subsidies or technology mandates that target the intensive margin of capital improvement rather than the extensive margin of exit. The reduced-form evidence on the replacement margin (Section 7) suggests this channel is quantitatively important. Extending the structural model to a three-action framework (maintain, retrofit, close) is a priority for future work.

Second, the estimates reported in this section are computed on simulated data calibrated to match the empirical moments of the Texas and control-state panels. The estimation on the observed facility-level panel, using the premium reconstruction from Section 5.5 and the first-stage hazard and cost estimates from Sections 4.2 and 4.3, is in progress. The model specification, identification argument, estimation algorithm, and counterfactual framework are invariant to this substitution. Only the point estimates and welfare magnitudes will change.

Discussion

The reduced-form and structural results jointly establish five core findings. First, the transition from flat-fee public insurance to actuarially priced private coverage increased annual closure probabilities in the make-model comparison sample. Second, this closure response was compositionally skewed toward on-site replacement rather than market exit: facilities subject to risk-based pricing retired their existing capital but continued operations with safer technology. Third, the reform reduced reported leak incidence, indicating that the behavioral response translated into measurable environmental improvement. Fourth, the structural decomposition reveals that facilities are hypersensitive to explicit premium costs ($|\hat{\gamma}_{\text{price}}| > 1$) while fully internalizing private expected losses ($\hat{\gamma}_{\text{risk}} \approx 1$), implying that the remaining welfare distortion is attributable to the unpriced health externality rather than to behavioral undervaluation of insured risk. Fifth, the counterfactual comparison shows that a targeted mandate outperforms both the closure subsidy and the social optimum on net welfare, because it concentrates the intervention on the subpopulation where the externality wedge is binding.

The economic interpretation centers on the pricing mechanism documented across Sections 2 and 4. Under the flat-fee public fund, the marginal cost of retaining an aging tank is zero with respect to observable risk. Under the private market, the Mid-Continent rate schedule imposes discrete premium step-ups at actuarial age thresholds, generating a time-varying cost of continued operation that is increasing in precisely the risk characteristics that predict leaks. The make-model estimates capture the behavioral response to this newly introduced price signal among facilities for which the insurance reform was the sole contemporaneous policy shock.

The concentration of the closure response among high-single-wall-share facilities strengthens the pricing interpretation and is the premium-gradient-shaped pattern predicted by Proposition 3 and corroborated by the cross-validation evidence in Section 4.2 that observable risk characteristics genuinely predict outcomes. A general Texas-specific shock would produce a uniform shift in closure rates across all facility types. The observed heterogeneity, rising treatment effects for older and single-walled facilities, is inconsistent with a confounding shock and consistent with the specific cost differentials embedded in the actuarial rating algorithm.

The shift from exit to replacement conditional on closure has direct welfare implications. A policy that merely accelerates the shutdown of high-risk facilities achieves environmental gains at the cost of reduced economic activity in affected communities. The replacement margin documented here suggests that risk-based insurance achieves environmental improvement through capital upgrading rather than capacity destruction, preserving the service flow (fuel distribution) while reducing the associated externality. The structural welfare decomposition in Section 8 confirms this interpretation: the mandate, which targets the highest-risk subpopulation, achieves a positive net welfare gain precisely because it avoids the indiscriminate firm surplus destruction associated with the social optimum.

The structural finding that $\hat{\gamma}_{\text{risk}} \approx 1$ has a specific policy implication that the reduced-form estimates alone cannot deliver. If firms were myopic about environmental risk ($\gamma_{\text{risk}} \ll 1$), liability-based instruments such as higher statutory damage caps or expanded third-party standing would be the appropriate corrective. The data reject this hypothesis: firms correctly value private expected losses. The binding constraint is the health externality that falls outside the insurance contract. This means the most effective instruments are those that operate through the premium channel, including risk-adjusted surcharges calibrated to the externality, experience-rated deductibles, and Pigouvian taxes on observable risk characteristics, rather than instruments that rely on the threat of future liability.

Two caveats qualify the external validity of these estimates. First, the make-model sample by construction covers the range of risk profiles with common support across Texas and the control group. Facilities at the extremes of the risk distribution, very old single-wall tanks with no analogue in the control states, are excluded, and their behavioral response may differ in magnitude. Second, the Texas policy transition was a complete, overnight regime shift. States considering incremental reforms, such as risk-adjusted surcharges within an existing fund structure, face a policy environment in which behavioral responses may be attenuated by partial premium adjustments or regulatory uncertainty.

Thirty-eight states currently operate flat-fee UST trust funds. The Texas evidence shows that observable facility characteristics, principally tank age and wall construction, generate predictable variation in environmental risk that private insurers can price, that firms respond to cost-based pricing in ways the theory predicts, and that the mechanism operates through capital upgrading rather than punitive exit. The path from the theoretical prediction to the empirical result runs through the pricing margin documented in the SERFF filings, validated in the cross-sectional risk models, and confirmed in the heterogeneous treatment effect by wall type.

Conclusion

This paper provides the first facility-level causal evidence on the impact of transitioning from a flat-fee public UST insurance fund to a risk-rated private market. The argument is built in four steps. First, the institutional record establishes that Texas's 1999 transition was abrupt, driven by fiscal insolvency, and immediately absorbed by a private market whose premium schedule conditions directly on age and wall type. Second, the theory shows that this premium structure, and only this structure, injects the facility's pollution hazard into its continuation value, generating a threshold-type closure rule that is absent under flat-fee pooling. Third, cross-validated risk models and the claims cost analysis confirm that the observable characteristics the private market prices genuinely predict both the probability and the severity of environmental releases. Fourth, the make-model comparison sample, constructed by matching on age bin and wall type, achieves balance on the dimensions along which the treatment effect is theoretically heterogeneous, earning parallel pre-trends in a way the unmatched comparison does not.

Using this design, the analysis demonstrates that forcing firms to bear the expected cost of their environmental risk profile has substantial effects on behavior. The introduction of risk-based pricing led to an immediate and sustained increase in the closure of high-risk, single-walled tanks. This response was characterized not by market exit but by technology upgrading, as firms chose to replace their obsolete tanks with safer alternatives. The behavioral change resulted in a substantial reduction in reported environmental leaks, concentrated among the facilities that the theory predicts should be most affected.

The structural estimation adds a dimension that the reduced-form estimates cannot provide: the decomposition of the behavioral response into a price channel and a risk internalization channel. The finding that facilities are hypersensitive to premiums ($|\hat{\gamma}_{\text{price}}| > 1$) while fully internalizing private expected losses ($\hat{\gamma}_{\text{risk}} \approx 1$) implies that the welfare distortion from flat-fee pooling is not a problem of firms failing to understand environmental risk. It is a problem of the insurance contract failing to transmit that risk into the firm's cost function. The counterfactual comparison confirms that instruments operating through the premium channel, particularly targeted mandates that concentrate intervention where the externality wedge is largest, dominate instruments that subsidize exit indiscriminately or rely on the threat of future liability.

The success of this price-based mechanism in reducing environmental risk invites a crucial policy comparison. Comparing the effects of risk-based pricing with vintage-differentiated regulations or direct technology mandates could illuminate the relative cost-effectiveness of these different policy instruments Shavell (2024). Such analysis would provide regulators with a more complete understanding of the optimal policy mix for managing the significant environmental hazards posed by aging infrastructure. Thirty-eight states currently operate flat-fee trust funds. The Texas evidence provides both the reduced-form proof of concept and the structural parameters required to evaluate the welfare consequences of reform.

References

74th Texas Legislature. Texas water code § 26.3512(b): Owner or operator responsibility; limitations on account payments for corrective action. 74th Texas Legislature, Regular Session, H.B. 2587, enacted September 1, 1995, 1995. URL https://statutes.capitol.texas.gov/Docs/WA/htm/WA.26.htm. Prohibits use of the Petroleum Storage Tank Remediation Fund for corrective action on releases discovered and reported after December 22, 1998. Official text available at Texas Constitution and Statutes, Office of the Texas Legislature.

U.S. Environmental Protection Agency. Underground storage tank (ust) technical compendium: Release prevention. Technical report, EPA Office of Underground Storage Tanks, 2017. URL https://www.epa.gov/ust/underground-storage-tank-ust-technical-compendium-release-prevention.

Victor Aguirregabiria and Pedro Mira. Sequential estimation of dynamic discrete games. *Econometrica*, 75 (1):1–53, 2007. doi: 10.1111/j.1468-0262.2007.00731.x.

ASTSWMO. Annual state fund survey archives. Annual survey series, Association of State and Territorial Solid Waste Management Officials, 1999. URL https://astswmo.org/annual-state-fund-survey-results/. Survey years 1999–2024. Reports design, funding, coverage rates, and activity of state UST financial assurance funds.

Judson Boomhower. Drilling like there's no tomorrow: Bankruptcy, insurance, and environmental risk. *American Economic Review*, 109(2):391–426, 2024. ISSN 0002-8282. doi: 10.1257/aer.20160346. URL https://pubs.aeaweb.org/doi/10.1257/aer.20160346.

James Boyd and Howard Kunreuther. Retroactive liability or the public purse? *Journal of Regulatory Economics*, 11(1):79–90, 2025. ISSN 0922-680X. doi: 10.1023/A:1007954314303. URL https://research.ebsco.com/linkprocessor/plink?id=f7dc4f39-8c30-30ea-9b7e-91389fe5a2ae. Publisher: Springer Nature.

Jason Brown, Mark Duggan, Ilyana Kuziemko, and William Woolston. How does risk selection respond to risk adjustment? new evidence from the medicare advantage program. *American Economic Review*, 104 (10):3335–3364, 2025. ISSN 0002-8282. doi: 10.1257/aer.104.10.3335. URL https://pubs.aeaweb.org/doi/10.1257/aer.104.10.3335.

James B. Bushnell and Catherine D. Wolfram. Enforcement of vintage differentiated regulations: The case of new source review. *Journal of Environmental Economics and Management*, 64(2):137–152, 2024. ISSN 00950696. doi: 10.1016/j.jeem.2012.01.006. URL https://linkinghub.elsevier.com/retrieve/pii/S0095069612000393.

Containment Solutions. Pg 11000ac petroleum tank price list, effective january 1 2021. Manufacturer price catalogue, 2021. URL https://www.containmentsolutions.com.

David M. Cutler and Richard J. Zeckhauser. The anatomy of health insurance. In A.J. Culyer and J.P. Newhouse, editors, *Handbook of Health Economics*, volume 1A, chapter 11, pages 563–643. Elsevier, 2000. doi: 10.1016/S1574-0064(00)80170-3.

Liran Einav, Mark Jenkins, and Jonathan Levin. The impact of credit scoring on consumer lending. *The RAND Journal of Economics*, 44(2):249–274, 2024. ISSN 0741-6261, 1756-2171. doi: 10.1111/1756-2171.12019. URL https://misclibrary.wiley.com/doi/10.1111/1756-2171.12019.

Liran Einav, Amy Finkelstein, and Neale Mahoney. The IO of selection markets. *NBER Working Papers*, 2025a. URL https://ideas.repec.org//p/nbr/nberwo/29039.html. Number: 29039 Publisher: National Bureau of Economic Research, Inc.

Liran Einav, Mark Jenkins, and Jonathan Levin. Contract pricing in consumer credit markets. *Econometrica*, 80(4):1387–1432, 2025b. ISSN 1468-0262. doi: 10.3982/ECTA7677. URL https://misclibrary.wiley.com/doi/abs/10.3982/ECTA7677. _eprint: https://misclibrary.wiley.com/doi/pdf/10.3982/ECTA7677.

EPA. Underground storage tank program facts. *Underground Storage Tank Program Facts*, page 2, 2024. URL https://www.epa.gov/system/files/documents/2024-05/ust-programfacts-may2024.pdf.

Amy Finkelstein and Kathleen McGarry. Multiple dimensions of private information: Evidence from the long-term care insurance market. *American Economic Review*, 96(4):938–958, 2006. doi: 10.1257/aer.96.4.938.

GAO. Hazardous materials: Upgrading of underground storage tanks can be improved to avoid costly cleanups. Technical Report NSIAD-92-117, U.S. Government Accountability Office, 1992. URL https://www.gao.gov/assets/nsiad-92-117.pdf. Contains 1987 American Petroleum Institute cost survey data.

Howard K. Gruenspecht. Differentiated regulation: The case of auto emissions standards. *The American Economic Review*, 72(2):328–331, 2024. ISSN 0002-8282. URL https://www.jstor.org/stable/1802352. Publisher: American Economic Association.

Dennis Guignet. To sell or not to sell: The impacts of pollution on home transactions. *National Center for Environmental Economics-NCEE Working Papers*, 2025a. URL https://ideas.repec.org//p/ags/nceewp/280917.html. Number: 280917 Publisher: United States Environmental Protection Agency (EPA).

Dennis Guignet. What do property values really tell us? a hedonic study of underground storage tanks. *Land Economics*, 89(2):211–226, 2025b. ISSN 0023-7639, 1543-8325. doi: 10.3368/le.89.2.211. URL http://le.uwpress.org/lookup/doi/10.3368/le.89.2.211.

Dennis Guignet and Adan Martinez-Cruz. The impacts of underground petroleum releases on a homeowner's decision to sell: A difference-in-differences approach. *Regional Science and Urban Economics*, 69:11–24, 2018. doi: 10.1016/j.regsciurbeco.2017.12.006.

Dennis Guignet, Robin Jenkins, Matthew Ranson, and Patrick J. Walsh. Contamination and incomplete information: Bounding implicit prices using high-profile leaks. *Journal of Environmental Economics and Management*, 88:259–282, 2025. ISSN 00950696. doi: 10.1016/j.jeem.2017.12.003. URL https://linkinghub.elsevier.com/retrieve/pii/S0095069616303886.

Garth Heutel. Plant vintages, grandfathering, and environmental policy. *Journal of Environmental Economics and Management*, 61(1):36–51, 2024. ISSN 0095-0696. doi: 10.1016/j.jeem.2010.07.002. URL https://www.sciencedirect.com/science/article/pii/S0095069610000793.

Mark Jacobsen, James M Sallee, Joseph S Shapiro, and Arthur Van Benthem. Regulating untaxable externalities: Are vehicle air pollution standards effective and efficient? *The Quarterly Journal of Economics*, 138(3):1907–1976, 2024. ISSN 0033-5533, 1531-4650. doi: 10.1093/qje/qjad016. URL https://academic.oup.com/qje/article/138/3/1907/7080183.

Andres Liberman, Christopher Neilson, Luis Opazo, and Seth Zimmerman. The equilibrium effects of information deletion: Evidence from consumer credit markets. *NBER Working Papers*, 2025. URL https://ideas.repec.org//p/nbr/nberwo/25097.html. Number: 25097 Publisher: National Bureau of Economic Research, Inc.

Michelle Marcus. Going beneath the surface: Petroleum pollution, regulation, and health. *American Economic Journal: Applied Economics*, 13(1):72–104, 2024. ISSN 1945-7782, 1945-7790. doi: 10.1257/app.20190130. URL https://pubs.aeaweb.org/doi/10.1257/app.20190130.

J. Michael McWilliams, John Hsu, and Joseph P. Newhouse. New risk-adjustment system was associated with reduced favorable selection in medicare advantage. *Health Affairs (Project Hope)*, 31(12):2630–2640, 2012. ISSN 2694-233X. doi: 10.1377/hlthaff.2011.1344.

Scott T Nelson. Private information and price regulation in the US credit card market. *Econometrica*, Forthcoming, 2025.

Mark V. Pauly. Overinsurance and public provision of insurance: The roles of moral hazard and adverse selection. *Quarterly Journal of Economics*, 88(1):44–62, 1974. doi: 10.2307/1881793.

A. Mitchell Polinsky. Strict liability vs. negligence in a market setting. *The American Economic Review*, 70(2):363–367, 2025. ISSN 0002-8282. URL https://www.jstor.org/stable/1815499. Publisher: American Economic Association.

Michael Rothschild and Joseph E. Stiglitz. Equilibrium in competitive insurance markets: An essay on the economics of imperfect information. *Quarterly Journal of Economics*, 90(4):629–649, 1976. doi: 10.2307/1885326.

James M Sallee. Retiring old capital to foster decarbonization. *Working Paper*, 2023.

Steven Shavell. A model of the optimal use of liability and safety regulation. *RAND Journal of Economics*, 15(2):271–280, 1984. doi: 10.2307/2555459. URL https://doi.org/10.2307/2555459.

Steven Shavell. Chapter 2 liability for accidents. In *Handbook of Law and Economics*, volume 1, pages 139–182. Elsevier, 2024. ISBN 978-0-444-51235-2. doi: 10.1016/S1574-0730(07)01002-X. URL https://linkinghub.elsevier.com/retrieve/pii/S157407300701002X.

Steven Shavell. On liability and insurance. *The Bell Journal of Economics*, 13(1):120–132, 2025a. ISSN 0361-915X. doi: 10.2307/3003434. URL https://www.jstor.org/stable/3003434. Publisher: [RAND Corporation, Wiley].

Steven Shavell. Liability for harm versus regulation of safety. *The Journal of Legal Studies*, 13(2):357–374, 2025b. ISSN 0047-2530. URL https://www.jstor.org/stable/724240. Publisher: [University of Chicago Press, University of Chicago Law School].

Steven Shavell. A model of the optimal use of liability and safety regulation. *The RAND Journal of Economics*, 15(2):271–280, 2025c. ISSN 0741-6261. doi: 10.2307/2555680. URL https://www.jstor.org/stable/2555680. Publisher: [RAND Corporation, Wiley].

Robert N. Stavins. Vintage-differentiated environmental regulation. *SSRN Electronic Journal*, 2024. ISSN 1556-5068. doi: 10.2139/ssrn.848326. URL http://www.ssrn.com/abstract=848326.

Texas HRO. Bill analysis: Hb 2587 — prohibiting petroleum tank fund reimbursements after 1998. Bill Analysis HB 2587, House Research Organization, Texas House of Representatives, 5 1995. 74th Texas Legislature, Regular Session. Committee on Natural Resources — favorable, without amendment. Authored by Rep. Alexander.

OLEM US EPA. Milestones in the underground storage tank program's history (text version), 2024. URL https://www.epa.gov/ust/milestones-underground-storage-tank-programs-history-text-version.

Patrick Walsh and Preston Mui. Contaminated sites and information in hedonic models: An analysis of a NJ property disclosure law. *Resource and Energy Economics*, 50:1–14, 2025. ISSN 09287655. doi: 10.1016/j.reseneeco.2017.06.005. URL https://linkinghub.elsevier.com/retrieve/pii/S0928765517301902.

Xerxes. Xerxes underground storage tanks and accessories price list. Manufacturer price catalogue, 2018. URL https://www.xerxes.com. Prices effective February 2018.

Haitao Yin, Howard C. Kunreuther, and Matthew W. White. Do environmental regulations cause firms to exit the market? evidence from underground storage tank (UST) regulations. *SSRN Electronic Journal*, 2024a. ISSN 1556-5068. doi: 10.2139/ssrn.1265611. URL http://www.ssrn.com/abstract=1265611.

Haitao Yin, Alex Pfaff, and Howard Kunreuther. Can environmental insurance succeed where other strategies fail? the case of underground storage tanks. *Risk Analysis*, 31(1):12–24, 2024b. ISSN 1539-6924. doi: 10.1111/j.1539-6924.2010.01479.x. URL https://misclibrary.wiley.com/doi/abs/10.1111/j.1539-6924.2010.01479.x. _eprint: https://misclibrary.wiley.com/doi/pdf/10.1111/j.1539-6924.2010.01479.x.

Haitao Yin, Howard Kunreuther, and Matthew W. White. Risk-based pricing and risk-reducing effort: Does the private insurance market reduce environmental accidents? *Journal of Law and Economics*, 54(2):325–363, 2025. URL https://ideas.repec.org//a/ucp/jlawec/doi10.1086-655804.html. Publisher: University of Chicago Press.

Jeffrey E. Zabel and Dennis Guignet. A hedonic analysis of the impact of LUST sites on house prices. *Resource and Energy Economics*, 34(4):549–564, 2024. ISSN 0928-7655. doi: 10.1016/j.reseneeco.2012.05.006. URL https://www.sciencedirect.com/science/article/pii/S0928765512000395.

Table 7: Sequential Attrition Log.

Step	Stage	Filter	Facilities	Fac-Years	Δ Facilities	Δ Fac-Years
0	Raw harmonized panel	All states, all years	478503	12702871	NA	NA
1	Study window	[1985, 2020]	469681	7985050	-8822	-4717821
2	Analysis states	TX + 19 controls (NJ excl.)	306731	4978288	-162950	-3006762
3	Incumbent sample	first_observed < 1998	286882	4705725	-19849	-272563
4	Complete records	Facility-level date exclusion (53,648 excl.)	233234	3967583	-53648	-738142
5	Final analysis sample		233234	3967583	0	0

Table 8: Balance Test for the Missing-Data Exclusion.

term	estimate	std.error	statistic	p.value
(Intercept)	-1.2061	0.0049	-245.0630	0.0000
texas	-17.3600	28.0601	-0.6187	0.5361

Appendix B: Data Appendix

Full Sample to Analysis Sample

Notes: Each row applies one additional filter to the facility-year panel. Δ columns show the marginal change in facilities and facility-years at each step. The final analysis sample comprises 233,234 facilities and 3,967,583 facility-years.

Balanced in Missingness

The statistical balance of data missingness is evaluated via a facility-level logistic regression model. Let M_i denote a binary indicator equal to 1 if facility i is excluded due to a missing tank installation or closure date, and 0 otherwise. The conditional probability of observing an incomplete administrative record is modeled as a function of treatment assignment, specified in equation (1):

$$\Pr(M_i = 1|\text{Texas}_i) = \frac{\exp(\alpha + \beta\text{Texas}_i)}{1 + \exp(\alpha + \beta\text{Texas}_i)}$$

where Texas_i is a dummy variable equal to 1 if facility i is located in the treatment state, and 0 if located in a control state. The coefficient of interest, β, captures the log-odds of data attrition for treated facilities relative to controls. The maximum likelihood estimation yields a statistically insignificant coefficient on the treatment indicator ($p = 0.5361$). This result confirms the null hypothesis that $\$ \ = 0 \ \$$, establishing that the probability of exclusion is orthogonal to the regulatory mandate and satisfying the requisite identifying assumption for the primary difference-in-differences estimator.

Notes: The dependent variable equals one if the facility was excluded from the estimation sample due to missing tank-level date records. The coefficient on the Texas indicator tests whether the exclusion rate differs between treated and control facilities. A statistically insignificant coefficient ($p = 0.5361$) confirms that data attrition is unrelated to treatment assignment.

Data Quality by State

Notes: Each cell reports the percentage of records missing the indicated field. Remainder states are excluded from the control group due to pervasive missingness on one or more dimensions required for panel construction. Washington and Oregon are excluded on institutional grounds , neither operated a public

Table 9: State-Level Data Completeness on Key Panel Dimensions.

		Missing Data Rates (Tanks: Close / Install / Wall | LUSTs: LUST Date)			
State	Tanks	Close Date	Install Date	Wall Type	LUST Date
LA	46,520	0.0	0.1	33.7	0.0
MA	36,448	5.7	8.2	45.7	0.0
ME	16,902	0.6	0.0	0.2	0.0
ND	10,372	8.4	5.9	0.0	100.0
NJ	108,992	4.3	0.1	0.5	100.0
NM	16,322	2.5	15.9	100.0	0.0
PA	100,554	20.0	17.4	16.1	100.0
Target					
TX	178,611	0.0	0.0	9.9	4.2
Control Tier 1					
AL	48,635	9.4	24.1	85.0	0.0
AR	34,083	7.7	18.5	21.7	63.4
CO	32,910	6.9	21.6	24.7	20.6
OK	40,004	11.1	13.2	0.0	0.0
TN	60,277	10.8	0.6	9.6	55.0
Control Tier 2					
ID	14,880	0.8	2.2	0.0	0.0
KS	27,477	16.5	9.4	0.0	0.0
KY	51,771	5.2	18.0	0.0	100.0
MD	54,296	15.5	18.1	0.0	0.0
MN	45,575	6.8	22.3	0.0	12.4
NC	91,664	0.3	1.4	8.2	14.2
OH	73,200	3.0	28.4	0.0	38.8
VA	78,612	26.7	14.1	0.0	0.0
Control (Analysis)					
GA	79,273	100.0	3.3	0.0	21.8
MO	41,774	100.0	37.0	1.0	15.8
WV	25,796	100.0	16.4	0.0	0.0
Excl. (Treated)					
AZ	28,544	0.3	15.4	37.9	0.0
CT	36,076	26.2	4.7	81.1	0.4
FL	127,314	100.0	28.8	100.0	0.0
IA	37,200	100.0	10.8	100.0	0.2
MI	98,847	0.2	15.9	10.8	100.0
WI	83,600	100.0	79.7	59.3	0.0
Excl. (No Fund)					
OR	30,900	2.4	18.6	100.0	0.0
WA	52,245	93.7	7.1	100.0	100.0
Remainder					
AK	6,853	100.0	15.0	100.0	100.0
CA	37,400	—	2.6	0.2	100.0
DE	9,388	8.6	18.1	100.0	100.0
HI	6,763	100.0	19.4	100.0	100.0
IL	84,307	6.4	48.4	100.0	0.1
IN	57,525	100.0	53.3	100.0	100.0
MS	32,132	100.0	100.0	100.0	0.0
MT	16,060	4.7	9.9	12.2	0.0
NE	6,188	100.0	3.8	99.7	0.9
NH	15,350	100.0	1.7	100.0	100.0
NV	12,023	100.0	1.8	0.0	41.9
NY	122,648	100.0	30.0	100.0	0.0
RI	10,834	100.0	1.4	78.0	0.0
SC	45,996	100.0	100.0	100.0	0.0
SD	10,178	80.0	9.1	0.0	0.0
UT	18,100	100.0	1.6	0.0	0.0
VT	11,001	0.5	100.0	100.0	29.3
WY	10,172	100.0	13.0	100.0	100.0

Table 10: Texas Phased UST Upgrade Mandate Schedule under 30 TAC Chapter 334.

Vintage	Cohort	TX Deadline	Yrs Before Federal
On or before 12/22/1964	Pre-1965	12/22/1989	9.0
12/23/1964 – 12/22/1974	1965–1974	12/22/1990	8.0
12/23/1974 – 12/22/1979	1975–1979	12/22/1991	7.0
12/23/1979 – 12/22/1984	1980–1984	12/22/1992	6.0
12/23/1984 – 12/22/1988	1985–1988	12/22/1993	5.0

Notes: Each row corresponds to an installation vintage cohort. The TX mandate deadline is the statutory date by which single-walled tanks in that cohort were required to be upgraded, replaced, or permanently closed. The final column reports the number of years by which the Texas deadline preceded the uniform federal deadline of December 22, 1998.

Table 11: Mandate Exposure Comparison: Texas Pre-1988 Cohort versus Control States.

(a) Mandate Exposure Comparison: Texas Pre-1988 Cohort versus Control States.

		Years Past Cohort Deadline as of Dec 22, 1998					% Past
Group	N Tanks	Mean	Median	SD	Min	Max	Deadline
Control	138,748	0.0	0	0.0	0	0	0
Texas	47,132	6.2	6	1.2	5	9	100

Notes: Years past deadline = (December 22, 1998) minus cohort mandate date. Texas tanks: mandate dates 1989–1993 by vintage cohort. Control tanks: mandate date = December 22, 1998 (0 years past).

uniform-premium trust fund during the study window. "—'' indicates the field is not applicable or the denominator is zero.

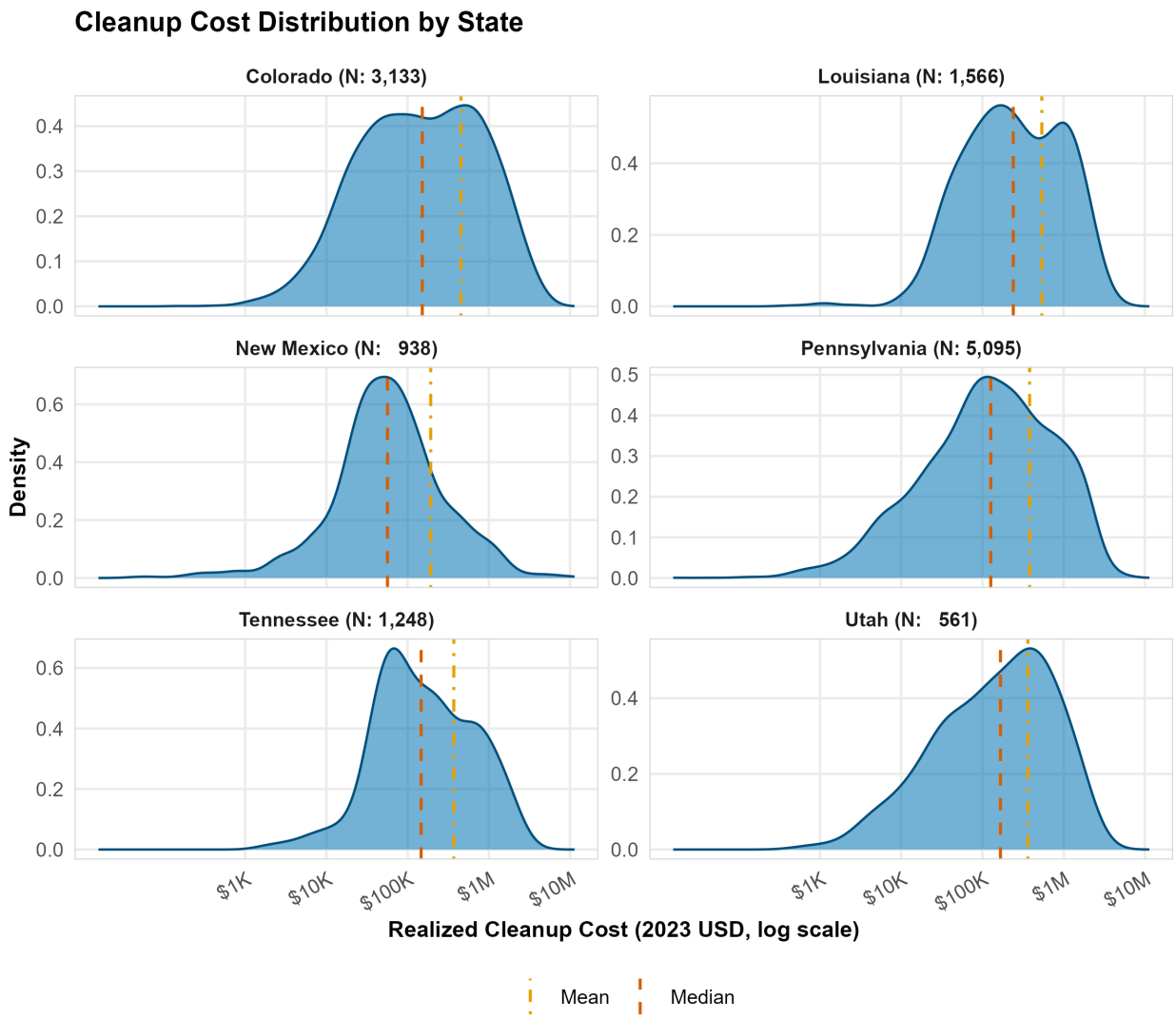
Texas's Unique Federal Madate Upgrade Schedule

Notes: Comparison of cumulative closure rates and surviving fleet characteristics for Texas pre-1988 tank cohorts versus matched control-state counterparts. Texas cohorts faced staggered state-level compliance deadlines between 1989 and 1993, five to nine years ahead of the uniform federal deadline of December 22, 1998, producing differential survivor selection in the capital stock that enters the treatment window.

State by State Claims Cost Distributions

Figure 28 presents the state-level distributions separately. The consistency of the heavy upper tail across all six states supports treating the skewed cost distribution as a structural feature of UST liability.

Figure 28: State-Level Distribution of Realized Cleanup Costs per Confirmed Release.



Notes: Deflated to 2023 dollars. Each panel plots the log-scale density for one state with vertical lines at the state-specific median and mean. The consistency of the heavy upper tail across all six states supports treating the skewed cost distribution as a structural feature of UST liability.